



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ACADEMIC REGULATIONS & SYLLABUS

Faculty of Technology & Engineering

Bachelor of Technology Programme

(Second Year Civil Engineering)

Effective from 2022-23

Vision

“To provide state of the art education in Civil Engineering guided by innovative research leading to centre of excellence in Civil Engineering education having recognition at national and international levels”

Mission

“Being a pioneering branch of Engineering, the department of Civil Engineering under the shelter of CHARUSAT is intended as a facilitator for creating a liaison between the brilliant student community and the next generation industrial needs”

Programme Educational Objectives (PEO's):

PEO 1: The graduate will possess foundation of engineering knowledge and exhibiting critical thinking and problem solving skills

PEO 2: The graduates will have trait of lifelong learning and be able to inculcate the capabilities to meet the diversified needs of industry, academia and research.

PEO 3: The graduate will exhibit the professional ethics and be supportive to the social needs

PEO 4: The graduates will possess comprehending, analyzing and designing capabilities to generate sustainable solutions

Programme Outcomes (PO's)

Engineering Graduates will be able to:

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSO's):

By the completion of Civil Engineering program, the student will attain:

PSO 1: The ability to serve the infrastructure sector with capabilities to plan, design, analyze and building civil engineering-based systems

PSO 2: The ability to adopt to the state-of-the-art practices in all sectors of Civil Engineering.

PSO 3: Employability skills with the cognizance of social and environmental necessity along with ethical responsibility to have a successful career and to become an entrepreneur.

CHARUSAT welcomes you for a Bright Future



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Accredited with Grade A by NAAC,
Accredited with Grade A by KCG

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Civil Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
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Academic Year – 2022-23

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Bachelor of Technology Programmes

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least of 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen subject of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate programme	(B. Tech.)
Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for Admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of Admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme Structure and Credits

As per Annexure – 1 attached

6. Attendance

All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

Student's attendance in a course should be 80%.

7. Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.2 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 7.1.3 Final examination by the University through modes such as; written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, is set to 70% of the marks for each the course.

7.2 Internal Evaluation

As per Annexure – I attached

7.3 University Examination

The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.4 In order to earn the credit in a course a student has to obtain grade other than FF.

7.5 Performance at Internal & University Examination

- 7.5.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows:

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

- 7.5.2 A student failing to score 40% in the final examination will get an FF grade.

- 7.5.3 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8. Grading

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	<80 ≥73	<73 ≥66	<66 ≥60	<60 ≥55	<55 ≥50	<50 ≥45	<45
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Numerical point (Grade Point) corresponding to the letter grade	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where,
 C_i = Number of credits of course i
 G_i = Grade Point for the course i
 $i = 1$ to n
 n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where,
 C_i = Number of credits of course i
 G_i = Grade Point for the course i
 $i = 1$ to n
 n = number of courses of all semesters up to which CGPA is computed
- (iii) No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

9. Award of Degree

- 9.1 Every student of the programme who fulfills the following criteria will be eligible for the award of the degree:
- 9.1.1 He/ She should have earned minimum required credits as prescribed in course structure; and
 - 9.1.2 He/ She should have cleared all internal and external evaluation components in every course; and
 - 9.1.3 He/ She should have secured a minimum CGPA of 4.5 at the end of the programme;
 - 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.
- 9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction	:	$CGPA \geq 7.5 \ \& \ \leq 10.0$
First class	:	$CGPA \geq 6.0 \ \& \ < 7.5$
Second Class	:	$CGPA \geq 5.0 \ \& \ < 6.0$
Pass	:	$CGPA < 5.0$

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM

A. Choice Based Credit System:

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/Specialization.

1.1.2 Programme Core (PC)

A ‘Core Course’ is a course which acts as a fundamental or conceptual base for Chosen Specialization of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An ‘Elective Course’ is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization

1.2.2 Programme Elective Course (PE)

A ‘Programme Elective Course’ is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time.

1.2.3 Cluster Elective Course (CE)

A 'Cluster Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (e.g. Common Courses for EC/CE/IT/EE) as offered by or decided by the Institute from time-to-time.

1.3 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non Credit Courses.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CIVIL ENGINEERING											
Level	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Level 2	MA251	Engineering Mathematics III	4		4	4	30	70			100
	CL241	Geology for Civil Engineers	4	2	6	5	30	70	25	25	150
	CL242	Surveying	4	2	6	5	30	70	25	25	150
	CL243	Mechanics of Solids	4	2	6	5	30	70	25	25	150
	HS121.02 A	Creativity, Problem Solving and Innovation	2		2	2	30	70			100
	XXXXX	University Elective - I	0	2	2	2			30	70	100
		Remedial Classes			10						
					36	23					750
	CL244	Fluid Mechanics-I	4	2	6	5	30	70	25	25	150
	CL245.01	Building Construction Technology	3	2	5	4	30	70	25	25	150
	CL246.01	Structural Analysis I	3	2	5	4	30	70	25	25	150
	ME241	Material Science & Technology	3	2	5	4	30	70	25	25	150
	HS III.02 A	Human Values and Professional Ethics	0	2	2	2			30	70	100
	MA249	Numerical Methods	3	2	5	4	30	70	25	25	150
	XXXXX	University Elective - II	0	2	2	2			30	70	100
		Remedial Classes			06						
					36	25					950

LIST OF UNIVERSITY ELECTIVE COURSES

3RD SEMESTER UNDERGRADUATE PROGRAMME (ACADEMIC YEAR 2022-23)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC281.01: Introduction to MATLAB Programming	EC / FTE
2	CE281.01: Art of Programming	CE / FTE
3	CL281.01: Environmental Sustainability and Climate Change	CL / FTE
4	CL283: SDG Handprint Laboratory	CL / FTE
5	EE284: Python Programming	EE/FTE
6	IT283: Web Designing & UI/UX	IT/FTE
7	ME281.01: Engineering Drawing	ME/FTE
8	PH233.01: Fundamentals of Packaging	RPCP/FPH
9	PD260.01: Basic Laboratory Techniques	PDPIAS/FAS
10	NR251.01: First Aid & Life Support	NURSING / FMD
11	PT191.01: Health Promotion and Fitness	ARIP / FMD
12	CA224: Introduction to Web Designing	CMPICA / FCA
13	BM231: Banking and Insurance	I2IM / FMS
14	PD261: Astrophysics, Space and Cosmos-1 (ASC-1)	PDPIAS/FAS

LIST OF UNIVERSITY ELECTIVE COURSES

4TH SEMESTER UNDERGRADUATE PROGRAMME (ACADEMIC YEAR 2022-23)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC282.01: Prototyping Electronics with Arduino	EC / FTE
2	CE282.01: Web Designing	CE / FTE
3	CL282.01: Basics of Environmental Impact Assessment	CL / FTE
4	EE287: MATLAB Programming	EE/FTE
5	EE288: Maintenance Of Household Apparatus	EE/FTE
6	IT284: Data Visualization	IT/FTE
7	ME282.01: Material Science	ME/FTE
8	PH238.01: Cosmetics in daily life	RPCP/FPH
9	NR261.01: Life Style Diseases & Management	NURSING / FMD
10	PT192.01: Occupational Health & Ergonomics	ARIP / FMD
11	CA225: Programming the Internet	CMPICA / FCA
12	BM241: Health Care Management	I2IM / FMS
13	PD262: Astrophysics, Space and Cosmos-2 (ASC-2)	PDPIAS/FAS

CONTENTS

MA251: ENGINEERING MATHEMATICS-III	18
CL241: GEOLOGY FOR CIVIL ENGINEERS	22
CL242: SURVEYING	27
CL243: MECHANICS OF SOLIDS.....	31
HS121.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION	36
CL244: FLUID MECHANICS- I	41
CL245.01: BUILDING CONSTRUCTION TECHNOLOGY	46
CL246.01: STRUCTURAL ANALYSIS -I.....	51
ME241: MATERIAL SCIENCE AND TECHNOLOGY	54
HS111.02 A: HUMAN VALUES & PROFESSIONAL ETHICS	58
MA249: NUMERICAL METHODS.....	61
ANNEXURE II - UNIVERSITY ELECTIVE SYLLABI.....	65

B. Tech. (Civil Engineering) Programme

SYLLABI (Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

MA251: ENGINEERING MATHEMATICS-III
B TECH 3rd SEMESTER (CL/EE/ME)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	-	4	4
Marks	100	-	100	

A. Outline of the course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fourier Series and Its Applications	15
2.	Laplace Transform and Its Applications	15
3.	Vector Differential Calculus	10
4.	Vector Integral Calculus	12
5.	Curve Fitting	08
	Total hours:	60

B. Detailed Syllabus:

1. **Fourier Series and Its Applications:** 15 Hours 25%
 - 1.1 Periodic functions, Dirichlets conditions, Trigonometric series.
 - 1.2 Euler formulae, Fourier series of periodic function of period p .
 - 1.3 Fourier series: Discontinuous functions, Even and odd functions.
 - 1.4 Half range Fourier series.
 - 1.5 Solutions of heat, wave, Laplace equations by method of separation of variables and Fourier series.
2. **Laplace Transform and Its Applications:** 15 Hours 25%
 - 2.1 Laplace transform as an improper integral and its existence.
 - 2.2 Laplace transform of elementary functions and its properties.
 - 2.3 Inverse Laplace transform and its properties.
 - 2.4 First and second shifting theorems. Laplace transform of derivatives and integrals.
 - 2.5 Convolution theorem and its application to obtain inverse

	Laplace transform.		
2.6	Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function).		
2.7	Solving differential equations using Laplace transform.		
2.8	Applications of ODE : Mechanical vibration system, deflection of beams.		
3	Vector Differential Calculus:	10 Hours	17%
3.1	Revision of concepts of Vector algebra, Scalar and Vector fields.		
3.2	Gradient of a scalar functions, Directional derivatives.		
3.3	Divergence and Curl of a vector field and their properties.		
3.4	Physical interpretations of gradient, divergence and curl. Irrotational and Solenoidal vector fields.		
3.5	Scalar potential function.		
4	Vector Integral Calculus:	12 Hours	20%
4.1	Introduction to line integrals and examples.		
4.2	Statement and examples of Green's theorem.		
4.3	Introduction to Surface and Volume integrals.		
4.4	Statements and examples of Stokes' and Gauss' divergence theorem.		
5	Curve Fitting:	08 Hours	13%
5.1	Normal equations.		
5.2	Fitting a straight line: $y = a + bx$ (linear regression).		
5.3	Fitting a parabola: $y = a + bx + cx^2$.		
5.4	Fitting a curve by transformation: $y = ab^x$, $xy^a = b$, $y = ax^b$.		

C. Course Outcome :

At the end of the course, the students will be able to

- CO1 Represent continuous time periodic signals using concepts of Fourier series and determine the solution of complex engineering problems heat, wave and Laplace's equations using it.
- CO2 Classify and apply the necessary Laplace transform and inverse Laplace transform techniques to solve ordinary differential equations of real world problems.

- CO3 Differentiate the dot product, cross product, length of vectors, partial derivatives, derivatives of vector-valued functions, gradient functions and these tools are employed successfully in different branches of engineering and physics.
- CO4 Analyze parameterize curves and calculate line integrals using them and Use vector operators and implement into surface integrals, Classify and examine the Green, Stokes and Divergence theorem
- CO5 Appraise the Least squares fitting method of data approximation and Discuss the concept of curve fitting of different types of curves using specific data

Course Articulation Matrix (ME/CL):

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
CO2	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
CO3	3	2	2	1	1	-	1	-	-	1	-	1	2	2	-
CO4	3	2	2	1	1	-	1	-	-	1	-	1	2	2	-
CO5	3	3	2	2	2	-	1	-	-	1	-	1	2	2	-

- Enter correlation levels 1, 2 or 3 as defined below:
- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)
- If there is no correlation, put “-”

D. Recommended Study Material:

Text Books:

1. Erwin Kreyszig; Advanced Engineering Mathematics, 9th Ed., Jhon Wiley & Sons, India, 1999.
2. Shankar S. Sastry; Introductory methods of numerical analysis. PHI Learning Pvt. Ltd., 2015.

Reference Books:

1. Ahsaan, Zafar; Differential equations and their applications. PHI Learning Pvt. Ltd., 2004.
2. Stewart James; Calculus Early Transcendental, 5th Ed., Thomson India, 2007.
3. Wylie & Barrett; Advanced Engineering Mathematics, Mc graw Hill pub., 1982.

4. M. D. Greenberg; Advanced Engineering Mathematics, 2nd ed., Pearson. 1998.
5. B. S. Grewal; Higher engineering mathematics. Khanna Publisher, New Delhi, 1996.
6. H. K. Dass; Advanced engineering mathematics. S. Chand, 2008.
7. Debnath, Lokenath, and Dambaru Bhatta. Integral transforms and their applications. CRC press, 2014.
8. Kenneth A. Stroud and Dexter J. Booth. Advanced engineering mathematics. Palgrave Macmillan, 2011.

Web Materials:

1. <http://mathworld.wolfram.com>
2. <http://en.wikipedia.org/wiki/Math>

CL241: GEOLOGY FOR CIVIL ENGINEERS
B TECH 3RD SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Physical Geology	17
2	Natural Materials	13
3	Structural Geology	12
4	Engineering Geology	18

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

1	Physical Geology	17 Hours	28%
1.1	Earth: Origin, internal structure.	3	
1.2	Work of natural agencies: Lakes, Oceans, Atmosphere, wind, streams, sea, glacier, mass movements	7	
1.3	Plate tectonics: Concepts and plate boundaries	2	
1.4	Earthquakes: Basics of earthquake, intensity and magnitude, causes, Earthquake zones of India	5	
2	Natural Materials	13 Hours	22%
2.1	Minerals: Formation, Identification and Use:	5	

Physical properties of minerals; basics of optical mineralogy, SEM, XRD

2.2 Rocks:	8	
Types of rock and origin:		
Igneous (extrusive and intrusive), sedimentary and metamorphic		
Igneous Rock:		
Agents, structure, texture, IUGG classification of intrusive and extrusive rocks		
Metamorphic Rock :		
Causes of metamorphism (stress, temperature, tectonism, pore fluid), structure and texture		
Sedimentary Rock:		
Sedimentation environments, structure, textural classification of siliciclastic and carbonate rock		
3 Structural Geology	12 Hours	20%
3.1 Introduction:	2	
Outcrop, stratification, dip and strike relation		
3.2 Study of Structural Features:	10	
Fold, Fault, Joints and Unconformities; Classification, formation and Identification		
4 Engineering Geology	18 Hours	30%
4.1 Geologic Mapping:	3	
Various methods of geological investigations, mapping, preparation of geologic sections Interpretation of reports.		
4.2 Indian geology:	2	
Geological framework of India, Geology of Gujarat		
4.3 Ground water:	2	
Basics and engineering difficulties		
4.4 Geology for Site selection:	6	
Dam, Tunnel, Reservoir and Highways		
4.5 Case studies:	5	
Important international and Indian examples of failures of civil structures due to geological constrains.		

C. Course Outcomes:

On the successful completion of this course:

- CO1 The student will have basic understanding of Geology, natural materials such as minerals and rocks, use of rocks and minerals and availability of rocks and mineral.
- CO2 The student will have basic understanding of Geological framework of India and Gujarat.
- CO3 The student will be able to analyse geological maps and process of resolving geological issues in civil engineering projects.
- CO4 The student will be able to apply the knowledge of natural dynamic process, and other geological factors while taking decision on civil engineering constructions.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	-	1	1	-	1	2	-	-	-	-	1	1	1	1
CO2	3	-	1	1	-	1	2	-	-	-	-	1	1	1-	1
CO3	3	1	1	1	-	1	2	-	-	-	-	1	1	1	1
CO4	3	1	1	1	-	1	2	-	-	-	-	1	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Prabin Singh, Engineering & General Geology, and S.K. Kataria & Sons, Katson Books, 1999, New Delhi
2. Kesavulu, C., Textbook of Engineering Geology, Macmillan India Ltd, 1993, New Delhi

Reference Books:

1. Bangar, K.M., Principles of Engineering Geology, Standard Publishers Distributors, 1995, New Delhi
2. Billings, M.P., Structural Geology, Prentice-Hall India, 1974, New Delhi
3. Blyth, F.G.H. and DE'Freits, M.H. Geology for Engineers. ELBS, 1974, London

4. Gokhale, K.V.G.K. and Rao, D.M., Experiments in Engineering Geology, Tata McGraw Hill, 1981, New Delhi
5. Reddy, V., Engineering Geology for Civil Engineers, Oxford & IBH, 1997, New Delhi
6. Parthasarathy, A., Panchapakesan, V. and Nagarajan, R. "Engineering Geology", Wiley India Pvt Ltd., 2013. ISBN-13: 9788126541829
7. Reddy, D. V. "Engineering Geology", Vikas Publishing House, 2010. ISBN13: 9788125919032
8. Krynine, D.P. and Judd, W.R. "Principles of Engineering Geology and Geotechnics", CBS Publishers & Distributors, New Delhi, 1957. ISBN-10: 812390603X.

Web Materials:

1. <http://nptel.ac.in/courses/105105106/>
2. <http://freevidelectures.com/Course/87/Engineering-Geology>
3. <http://www.cosmolearning.com/courses/engineering-geology/video-lectures/>
4. <http://geology.about.com/>

LIST OF EXPERIMENTS

Sr. No.	Name of Experiment
1	Fundamentals of Geology
2	Study of Minerals
	Study of Physical Properties of Minerals
	Identification Rock Forming Minerals by Studying Physical Properties
	Identification Ore Minerals by Studying Physical Properties
3	Study of Rocks
	Study of Igneous Rocks
	Study of Sedimentary Rocks
	Study of Metamorphic Rocks
4	Geological Mapping
	Map of layered horizontal beds
	Map of layered inclined beds
	Map of layered inclined beds
	Map of Unconformity with Inclined and Horizontal Bed Series
	Map of Faulted Sequence
	Map of Folded Sequence
	Various Combination Maps for General Practice

CL242: SURVEYING
B TECH 3RD SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Out Line of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Measurement of Elevation, Contouring	5
2	Theodolite Traversing	10
3	Plane Table Surveying	10
4	Engineering Curves	10
5	Computation of Areas and Volumes	10
6	Hydrography	5
7	Setting Out Works	5
8	Modern Surveying Instruments	5

Total hours (Theory): 60 Hours

Total hours (Lab): 30 Hours

Total hours: 90 Hours

B. Detailed Syllabus:

1	Measurement of Elevation, Contouring	05 Hours	08%
1.1	Profile levelling and cross sectioning & contouring		
1.2	Errors in levelling		
1.3	Permanent adjustment of level.		
2	Theodolite Surveying	10 Hours	17%
2.1	Introduction to vernier transit theodolite		
2.2	Temporary and permanent adjustment of theodolite		
2.3	Measuring horizontal and vertical angles		
2.4	Computation of latitudes and departure & gale's table		

2.5	Methods of traversing, closing error		
2.6	Check in closed and open traverse, balancing of traverse		
2.7	Area of traverse, omitted measurement, co-ordinate method		
3	Plane Table Surveying	10 Hours	17%
3.1	Introduction		
3.2	Principle		
3.3	Advantages & limitation		
3.4	Accessories of plane tabling		
3.5	Setting up the plane table		
3.6	Methods of plane table surveying		
3.7	Sources of errors		
4	Engineering Curves	10 Hours	17%
4.1	Introduction to engineering curves		
4.2	Classification & elements of simple circular curve		
4.3	Methods of setting out a simple circular curve		
4.4	Elements of compound curve & reverse curve		
4.5	Transition curve & vertical curve		
5	Computation of Areas and Volumes	10 Hours	17%
5.1	Different methods to compute area of traverse		
5.2	Determining areas from plans, trapezoidal rule- Simpson's rule		
5.3	Planimeter, digital planimeter		
5.4	Computation of volumes, volume from cross sections		
5.5	Trapezoidal and prismoidal formulae		
5.6	Prismoidal correction, curvature correction		
5.7	Determination of capacity of reservoir and volume of borrow pits		
6	Hydrography	05 Hours	08%
6.1	Introduction		
6.2	Purposes, control points		
6.3	Soundings		
6.4	Instruments and methods of locating soundings		
7	Setting Out Works	05 Hours	08%
7.1	Introduction		

- 7.2 Setting out the buildings
- 7.3 Setting out the sewer grades
- 7.4 Setting out the bridge & culvert

8 Modern Survey Instruments

05 Hours 08%

- 8.1 Introduction
- 8.2 Electromagnetic distance measurement
- 8.3 Electronic theodolite
- 8.4 Total station

C. Course Outcomes :

At the end of the course, the students will be able to

- CO1 Carry out elevation measurement and prepare contour maps.
- CO2 Plan theodolite survey applications.
- CO3 Measure and Plot maps with plane table.
- CO4 Set out curves, buildings, culverts and tunnels
- CO5 Compute ground areas and earthwork volumes
- CO6 Carry out basic hydrographic survey and invoke application of modern surveying instruments over conventional instruments.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	1	-	2	-	-	-	2	1	-	1	3	1	2
CO2	1	2	1	-	3	-	-	1	-	1	-	2	1	2	2
CO3	1	-	-	-	-	-	-	-	1	1	-	-	-	-	1
CO4	3	2	2	-	-	1	-	-	1	-	-	2	1	-	2
CO5	3	2	2	1	-	-	1	-	-	1	-	1	2	1	2
CO6	1	-	2	-	3	-	-	-	1	3	2	2	2	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books

- Duggal, S. K., Surveying Vol. I & II, Tata Mcgraw Hill, New Delhi
- Subramanian, R., Surveying & Levelling, Oxford University Press, New Delhi

Reference Books:

1. Punamia, B.C., Surveying Vol. I, II & III, Laxmi Publications
2. Kanetkar, T.P. and Kulkarni, S.V., Surveying and Levelling Vol. I & II, Pune Vidhyarthi Gruh
3. Arora, K.R., Surveying Vol. I, II & III, Standard Book House. New Delhi
4. Basak, N.N., Surveying and Levelling, Tata Mcgraw Hill, New Delhi
5. Agor, R., Surveying and Levelling, Khanna Publishers, New Delhi

Web Materials:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-ROORKEE/SURVEYING>
2. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITROORKEE/SURVEYING/home.htm>
3. <http://nptel.iitm.ac.in/courses.php?branch=Civil>

CL243: MECHANICS OF SOLIDS
B TECH 3RD SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Out Line of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	3
2	Simple Stresses and Strains	15
3	Principal Stresses and Strains	10
4	Shear Force and Bending Moment	10
5	Moment of Inertia	6
6	Bending and Shear Stresses in Homogeneous and Composite Beam Sections	10
7	Strain Energy	6

Total hours (Theory): 60 Hours

Total hours (Lab): 30 Hours

Total hours: 90 Hours

B. Detailed Syllabus:

1	Introduction	03 Hours	05%
1.1	Introduction to mechanics of deformable bodies		
1.2	Principle of superposition		
1.3	Classification of loaded bar		
1.4	Gradual, sudden, impact and shock loading		
1.5	Mechanical properties of materials		
2	Simple Stresses and Strains	15 Hours	22%

- 2.1 Stress and types of stress, Strain and types of strain
- 2.2 Stress strain Characteristics for ductile and brittle materials
- 2.3 Shear stresses and strains, Elasticity, Hook's law
- 2.4 Axial and shear deformations, Axial force diagram
- 2.5 Bars of varying section, Bars of uniformly varying cross section
- 2.6 Analysis of stress for statically determinate structures and indeterminate structures
- 2.7 Poisson's ratio, Volumetric strain, Biaxial and tri-axial deformations
- 2.8 Elastic constant and relation between three elastic constants
- 2.9 Stresses due to thermal effect
- 3 Principal Stresses and Strains** **10 Hours 18%**
- 3.1 Introduction, Sign convention
- 3.2 Transformation of stresses for a state of stresses under axial loading
- 3.3 Transformation of stresses for plate under biaxial loading
- 3.4 Principal plane and principal stresses
- 3.5 Maximum shear stress, Element subjected to principal stresses
- 3.6 Mohr's circle for stresses on an oblique section of a body subjected to direct stress in one plane and two plane with or without shear stress,
- 3.7 Pure shear
- 4 Shear Force and Bending Moment** **10 Hours 22%**
- 4.1 Concept of shear force and bending moment
- 4.2 Sign conventions
- 4.3 Relation between bending moment, shear force and rate of loading
- 4.4 Bending moment and shear force diagrams for statically determinate beams subjected to all different types of loading
Important points for shear force and bending moment diagram.

5	Moment of Inertia	6 Hours	12%
5.1	Introduction, Radius of gyration		
5.2	Parallel axis theorem and perpendicular axis theorem		
5.3	Polar moment of inertia		
5.4	Moment of inertia by integration		
5.5	Moment of inertia of composite areas		
6	Bending and Shear Stresses in Homogeneous and composite beam sections	10 Hours	14%
6.1	Pure bending, Theory of pure bending		
6.2	Assumptions and derivation of theory of simple bending		
6.3	Neutral axis, moment resistance, section modulus		
6.4	Strength of section		
6.5	Bending stress in symmetrical, unsymmetrical and composite sections		
6.6	Shear stresses, Shear flow		
6.7	Shear stress distribution for various cross section		
7	Strain Energy	06 Hours	07%
7.1	Elastic strain energy due to gradual loading, sudden loading, impact loading, shear and bending, Resilience		

C. Course Outcomes

At the end of the course, the students will be able to

- CO1 Classify and determine the strength parameters of materials and compute stresses & strains for structural elements due to normal, shear loads and temperature changes.
- CO2 Calculate bending moment and shear force for statically determinate beams and draw the distributions.
- CO3 Calculate the cross sectional moment of inertia using the parallel axis theorem.
- CO4 Analyse stresses in a beam under combined loadings and also Calculate bending stress, shear stress and their distribution at any desired location along the beam elements.
- CO5 Evaluate the different mechanical properties of materials and also Calculate strain energy due to different loadings.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	1	-	-	-	-	-	-	-	-	-	-	2	2	-
CO2	3	2	2	-	1	-	-	-	-	-	-	-	2	-	-
CO3	3	2	1	-	-	-	-	-	-	-	-	-	2	1	-
CO4	3	2	2	-	-	-	-	-	-	-	-	-	2	1	-
CO5	3	2	2	-	-	-	-	-	-	-	-	-	2	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I, Charotar Publishing House
2. Shah, H. J., Mechanics of Solids, Charotar Publishing House
3. Khurmi R. S., Strength of Materials, S. Chand Publications
4. S. S. Bhavikatti, Strength of Materials, Vikas Publishing House Pvt. Ltd
5. Hibbeler, R.C., “Mechanics of Materials”, 6th SI edition

Reference Books:

1. Beer and Johnston, Mechanics of Materials
2. Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
3. S. Timoshenko, Strength of Materials (Part -1), D. Van Nostrand Company, Inc.
4. R. Subramanian, Strength of Materials, Oxford University Press
5. S. S. Rattan, Strength of Materials, Tata McGraw Hill Education Pvt. Ltd.
6. R. K. Bansal, Strength of Materials, Lakshmi Publications House Pvt. Ltd.

E Books:

1. Barry Dupen, Applied Strength of Materials for Engineering Technology
2. S. Timoshenko, Strength of Materials (Part -1), D. Van Nostrand Company, Inc

Web Materials:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/Mechanics%20Of%20Solids/index.htm>
2. <http://nptel.ac.in/video.php?subjectId=112107147>
3. <https://www.youtube.com/watch?v=GkFgysZC4Vc>
4. <http://nptel.ac.in/syllabus/112106141/>
5. <http://nptel.ac.in/courses/Webcourse-contents/IIT-Delhi/Mechanics%20Of%20Solids/index.htm>

HSI21.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION
B TECH 3RD SEMESTER (UNIVERSITY ELECTIVE)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

Creative Problem Solving

<https://www.coursera.org/learn/creative-problem-solving>

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	

Total hours (Theory) : --

Total hours (Practical): 30

Total hours (Lab) : --

Total hours : 30

B. Detailed Syllabus:

1. Introduction to Creativity, Problem Solving and Innovation 06 Hours 20%
Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills
2. Questioning, Learning and Visualization 6 Hours 20%

Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking

- | | | | |
|----|--|---------|-----|
| 3. | Creative Thinking and Problem Solving
Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving | 6 Hours | 20% |
| 4. | Logic, Language and Reasoning
Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning | 6 Hours | 20% |
| 5. | Contemporary Issues and Practices in Creativity and Problem Solving
Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving | 6 Hours | 20% |

C. Course Outcome :

At the end of the course, the students will be able to

- CO1 Demonstrate creativity in their day to day activities and academic output.
- CO2 Solve personal, social and professional problems with a positive and an objective mindset.
- CO3 Think creatively and work towards problem solving in a strategic way.
- CO4 Initiate new and innovative practices in their chosen field of profession.
- CO5 Give logical ideas, opinions, and solutions to problems.
- CO6 Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	-	3	-	-	1	1	-	-	-	2	-	1	1	-	-
CO2	-	3	2	1	-	-	-	1	1	1	-	1	-	1	-
CO3	-	2	2	1	-	-	-	-	-	-	-	-	-	-	-
CO4	1	-	1	-	-	-	2	-	-	-	-	-	1	-	-
CO5	-	1	-	1	-	-	-	-	1	3	-	1	-	1	-
CO6	-	-	-	2	-	-	-	-	-	-	2	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

❖ Text book:

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossey-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genius, Ten Speed Press 2001

❖ Reference book:

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map, Thorsons Publication 2002
5. Scott Berkun, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

❖ Additional Readings

1. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
9. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993
14. Mihaly Csikszentmihalyi, Creativity-Flow and Psychology of Discovery and Invention, Harper Publication 1996
15. Martin Gardner, W. H., Aha! Insight, Freeman Publication 1978
16. Paul Slovic, Test Your Lateral Thinking IQ, Sterling Publication 1994
17. Paul Slovic & Des Macale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

❖ Web material:

1. Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

B. Tech. (Civil Engineering) Programme

SYLLABI (Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CL244: FLUID MECHANICS- I
B TECH 4TH SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Properties of Fluid	4
2	Fluid Statics	10
3	Fluid Kinematics	9
4	Fluid Dynamics	8
5	Measurement of Flow	8
6	Flow Through Pipes	9
7	Open Channel Flow	10
8	Introduction to the Hydraulic machinery	2

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

1	Properties of Fluid	04 Hours	7%
1.1	Types of Fluid		
1.2	Fluid properties		
1.3	Fluids as a continuum		
1.4	Control volume concept		
1.5	Viscometers		
2	Fluid Statics	10 Hours	17%
2.1	Pressure and its measurement		

2.2	Pascal's law, hydrostatic pressure, atmospheric, absolute, gauge and vacuum pressure		
2.3	Pressure measurement through piezometer and manometer		
2.4	Total pressure, intensity of pressure, centre of pressure, Pressure on horizontal, vertical, inclined and curved surface		
2.5	Floating Bodies, Buoyancy and centre of buoyancy, meta centre and meta centre height		
2.6	Condition of equilibrium of floating and submerged body, determination of meta-centric height by experimental and analytical method, stable and unstable equilibrium, pressure in case of accelerated rigid body motion		
3	Fluid Kinematics	09 Hours	15%
3.1	Types of fluid motion, methods of describing fluid flow - Lagrangian and Eulerian method		
3.2	Inviscid flows, velocity and acceleration, flow rate		
3.3	Continuity equation		
3.4	Potential flows, flow lines, velocity potential and stream function		
3.5	Flownet its characteristic and utility, circulation and vorticity		
4	Fluid Dynamics	08 Hours	13%
4.1	Energy possessed by a fluid body, Types of forces, Forces influencing fluid motion, head-energy correction factor		
4.2	Euler and Bernoulli's equations, application of Bernoulli's equation		
4.3	Flow measurement, momentum of fluid in motion, momentum equation and momentum correction factor		
4.4	Application of momentum equation, forces on a pipe bend, free jets		
5	Measurement of Flow	08 Hours	13%
5.1	Orifice & Mouth piece Classification, hydraulic coefficients, experimental determination of hydraulic coefficient, discharge through all types of orifice & mouthpiece, time of emptying the tank through orifice and mouthpiece		

5.2	Notches and Weirs Classification, discharge through various types of Notches and weirs, time of emptying a reservoir or a tank with notches & weirs		
5.3	Venturimeter		
5.4	Nozzles and Bendmeter		
6	Flow Through Pipes	09 Hours	15%
6.1	Introduction		
6.2	Major and minor losses of energy in pipes, hydraulic gradient, total energy line		
6.3	Pipes in series and parallel, flow through branched pipes		
6.4	Hydraulic transmission of power		
6.5	Water hammer and its effects		
7	Open Channel flow	10 Hours	17%
7.1	Comparison between pipe flow and open channel flow		
7.2	Classification of open channel flow		
7.3	Uniform flow – Chezy's formula		
7.4	Manning's formula- numerical		
7.5	Hydraulically efficient channel cross section		
7.6	Rectangular section		
7.7	Trapezoidal section and circular section		
8	Introduction to the Hydraulic machinery	2 Hours	3%
8.1	Introduction to various types of Turbines and hydraulic pumps, Hydraulic press - hydraulic accumulator - Hydraulic ram		
8.2	Working principle, discharge calculations and use of and machines Centrifugal pumps		

C. Course Outcomes (COs):

At the end of the course, the students will be able to

- CO1 Determine the properties of fluid and pressure and their measurement.
- CO2 Apply continuity equation and energy equation in solving problems on flow through conduits.
- CO3 Compute the frictional loss in laminar and turbulent flows.

CO4 Design open channels for most economical sections like rectangular, trapezoidal and circular sections.

CO5 Select the type of pumps and turbine required with reference to available head of water and discharge.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	-	2	-	-	-	-	-	-	-	2	1	-	-
CO2	3	3	2	-	-	-	-	-	-	1	-	2	1	-	-
CO3	2	2	1	-	-	-	-	-	-	-	-	1	1	-	-
CO4	3	3	2	1	-	-	-	-	-	1	-	2	2	-	2
CO5	3	1	-	-	-	1	-	-	-	-	-	2	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Jain, A.K., Fluid Mechanics, Khanna Publishers, New Delhi
2. Bansal, R.K., Fluid Mechanics, Laxmi Publications

Reference Books:

1. Streeter, V.L. and Wylie, E.B., Fluid Mechanics, McGraw Hill, 1985, New York
2. Subramanya, K., Theory and Applications of Fluid Mechanics, Tata-McGraw Hill Publishing Co., 1993, New Delhi
3. Shaughnessy, E.J., Katz, I.M. and Schaffer, J.P., Introduction to Fluid Mechanics, SI edition, 2005, Oxford University Press, New Delhi
4. White, F.M. Fluid Mechanics, McGraw Hill, New York
5. Kumar, D.S., Fluid Mechanics, S.K.Kataria & Sons
6. Rajput, R.K., Fluid Mechanics, S. Chand & Co. publications
7. Modi, P.N. and Sheth, Fluid Mechanics & Hydraulic Machines, Standard Book House
8. Ramamurtham, S., Hydraulic Fluid Mechanics & Fluid Machines, Dhanpatrai Publishing Co.
9. Garde, R.J. and Mirajgaoker, A.C., Engineering Fluid Mechanics, New Chand & Sons

Web materials:

1. <http://www.msubbu.in/ln/fm/>
2. <http://nptel.ac.in/courses/105103095/>

CL245.01: BUILDING CONSTRUCTION TECHNOLOGY
B TECH 4TH SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Masonry Works	09
2	Foundations	07
3	Doors, Windows & Ventilators	05
4	Stairs and Staircases	05
5	Floors and Floorings	05
6	Roofs and Roof Coverings	05
7	Wall Finishes	04
8	Temporary Works	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- 1 Masonry Works** 09 Hours 20%
- 1.1 Stone masonry:**
Technical terms; materials used; classification – random rubble, square rubble, Ashlar fine tooled, Ashlar chamfered; joints
- 1.2 Brick masonry:** Technical terms; bonds in brick work – stretcher, header, English, Flemish, Raking; pointing – method and types

1.3	Composite masonry: Brick-stone composite masonry - properties and uses; Concrete masonry – hollow concrete blocks, Autoclaved aerated concrete (AAC), Advantages and Disadvantages over other cement construction materials; Reinforced brick masonry		
2	Foundations	07 Hours	16%
2.1	Classification, Necessity, essential requirements		
2.2	methods of site exploration		
2.3	Settlement, causes of failures of foundation and remedial measures		
2.4	Bearing capacity of soils		
2.5	Shallow foundations: Depth of footing, types and construction, design of strip footing		
3	Doors, Windows & Ventilators	05 Hours	11%
3.1	Doors: Location, technical terms, door frames, types of doors – construction, suitability - panelled, glazed, flushed doors, collapsible steel doors		
3.2	Windows: Types of windows–construction, suitability - Casement, Sash, and Skylight windows		
3.2	Ventilators: Ventilators combined with window; fan light fixtures and fastenings		
4	Stairs, Staircases and escalators	05 Hours	11%
4.1	Technical terms, requirements of good stair		
4.2	Dimensions of a step, types of steps, classification of stairs		
4.3	Design of stair, example – stair planning, elevators, escalators		
5	Floors and Floorings	05 Hours	11%
5.1	Components of a floor		
5.2	Flooring material and factors affecting selection of flooring material		
5.2	Types of ground floors –cement concrete, tiles, marble, timber etc., Recent Developments in Flooring Technology		
5.3	Types of upper floors –Conventional floors (timber, steel joist), reinforced cement concrete & pre-cast concrete floors		
6	Roofs and Roof Coverings	05Hours	11%
6.1	Requirements of a good roof, technical terms		

6.2	Types of roofs – single, trussed, steel roof trusses		
6.3	Types of roof covering –G.I. Sheets, light weight roofing, Flat terraced roof – advantages, disadvantages, types, Water proofing techniques in flat roofs.		
7	Wall Finishes	04 Hours	09%
7.1	Plastering: Objectives, requirements, terms, tools, cement plaster, special materials used in plastering, defects		
7.2	Painting: Painting on different surfaces, defects, Recent Developments in painting technology		
8	Temporary Works	05 Hours	11%
8.1	Timbering in trenches		
8.2	Component and types of scaffolding		
8.3	Formwork		

C. Course Outcomes:

On the successful completion of this course, the students will be able to:

- CO1 Understand the types, dimensions and method of construction of various building components along with various checks required for their construction.
- CO2 Identify, select and recommend appropriate type of building elements suitable for a particular situation.
- CO3 Select and recommend appropriate temporary structures suitable for various building construction needs.
- CO4 Select and recommend appropriate mode of vertical movements and design of staircase
- CO5 Design strip footing type of shallow foundation.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	-	-	-	1	-	-	-	-	-	-	-	1	-	2
CO2	1	2	-	1	-	-	-	-	-	-	-	1	1	-	2
CO3	1	1	-	1	1	-	-	-	-	-	-	-	1	-	1
CO4	1	1	3	-	1	-	-	-	-	-	-	-	1	-	1
CO5	1	-	3	-	-	-	-	-	-	-	-	-	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Punamia, B.C., Building Construction, Laxmi Publication, New Delhi
2. Sushil Kumar, Building Construction, Standard Publishers Distributors, New Delhi
3. Rangwala, S.C., Building Construction, Charotar Publishing House Pvt. Ltd., Anand

Reference Books:

1. Varghese, P.C., Building Construction, Prentice-Hall of India (PHI) Learning Pvt. Ltd., New Delhi
2. Roy Chudley & Roger Greeno, Construction Technology, Prentice Hall
3. Roy Chudley & Roger Greeno, Building Construction Handbook, Routledge Publications
4. Francis D. K. Ching, Building Construction Illustrated, Wiley Publications
5. Edward Allen & Joseph Iano, Fundamentals of Building Construction: Materials and Methods, Wiley Publications
6. National Building Code of India, Indian Standard Institution (ISI), 2005, New Delhi

Web Materials:

1. <http://nptel.ac.in/courses/105102088/>
2. https://www.youtube.com/watch?v=fDUD36VPD_U
3. <https://www.youtube.com/watch?v=wmRYKrfQjuk>

LIST OF TUTORIALS

Sr. No.	Topic
1.	Masonry work A) Stone masonry work
2.	Masonry work B) Brick masonry work
3.	Foundations -Shallow Foundation
4.	Doors, Windows & Ventilators
5.	Stairs and Staircases
6.	Floors and Flooring
7.	Roofs and Roof Coverings
8.	Temporary Works

CL246.01: STRUCTURAL ANALYSIS -I
B TECH 4TH SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	04
2	Slope and Deflection of Statically Determinate Beams	09
3	Combined Direct and Bending Stresses	06
4	Columns and Struts	06
5	Influence Line for Determinate Beams	10
6	Torsion in Circular Shafts	04
7	Arches	06

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1. Introduction	04 Hours	10%
1.1 Concepts of Structural Analysis		
1.2 Idealization of structures		
1.3 Static indeterminacy and kinematic indeterminacy		
2 Slope and Deflection of Statically Determinate Beams	09 Hours	20%
2.1 Differential Equation of the Elastic Curve, Relationship of slope deflection with radius of curvature, bending moment, shear force and load intensity		

2.2	Relation between Moment, Slope and Deflection using fundamental, Macaulay's Method (Method of Singularity Function), Moment Area Method, Conjugate Beam Method		
3	Combined Direct and Bending Stresses	06 Hours	13%
3.1	Eccentric Loading along One Principle Axis		
3.2	Middle Third Rule, Kernal (Core) of Section		
3.3	Column Subjected to Biaxial Bending		
3.4	Structures Subjected to Lateral Pressure		
4	Columns and Struts	06 Hours	13%
4.1	Buckling of Columns, different end conditions		
4.2	Euler's Theory, Equivalent Length		
4.3	Applicability and limitations of Euler's Formula		
4.4	Rankine's Formula		
5	Influence Line for Determinate Beams	10 Hours	22%
5.1	Uses of influence lines, Various Types of Loading		
5.2	Influence lines for statically determinate beams under moving loads		
5.3	Influence lines for support reactions, shear force & bending moment for uniformly distributed load and several point loads		
5.4	Criteria for maximum effects		
6	Torsion in Circular Shafts	04 Hours	9%
6.1	Assumption for shear stress in a circular shaft subjected to torsion		
6.2	Torsion Stress and Strain, Torsion Formula		
6.3	Power Transmitted by Shaft		
6.4	Design of Shaft: Shafts of Uniform Section, Shaft of Varying Section, Coupling and Keys		
7	Arches	06 Hours	13%
7.1	Arches as structural forms – Examples of arch structures, Types of arches		
7.2	Analysis of three hinged - parabolic and circular arches		

C. Course Outcomes :

On the successful completion of this course students will be able to

- CO1 Identify the stability and determinacy of planar structures.
- CO2 Calculate deflections for statically determinate beams using different methods of analysis.
- CO3 Analyses the columns with different end conditions and understand the effect of same.
- CO4 Construct influence lines for statically determinate beams and to use influence lines to find maximum load effects in beams.
- CO5 Calculate the combined stresses and torsion in the circular shaft.
- CO6 To analyze the determinate arches.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	1	-	1	-	-	-	-	-	-	-	-	1	-	-
CO3	2	2	1	1	-	-	-	-	-	-	-	-	1	-	-
CO4	2	1	1	1	1	-	-	-	-	-	-	-	1	-	-
CO5	1	-	1	-	-	-	-	-	-	-	-	-	1	-	-
CO6	2	2	1	1	-	-	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

- 1) Junarkar, S.B. and Shah, H.J., Mechanics of Structures Vol. I, Charotar Publishing House
- 2) Negi, L.S. and Jangid, R.S., Structural Analysis, Tata McGraw Hill
- 3) Reddy, C.S., Basic Structural Analysis, Tata McGraw Hill

Reference Books:

- 1) Dupen Barry, Applied Strength of Materials for Engineering Technology
- 2) Gere and Timoshenko, Mechanics of Materials, CBS Publishers
- 3) Hibbler, R.C., Mechanics of Materials, Pearson Education
- 4) Wang, C.K., Intermediate Structural Analysis, Tata McGraw Hill

ME241: MATERIAL SCIENCE AND TECHNOLOGY
B TECH 4TH SEMESTER (CIVIL / MECHANICAL ENGINEERING)

Credit and Hours

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction to Material Science	04
2	Crystal Structure and Properties	10
3	Mechanism of crystallization & Phase Transformation	06
4	Engineering Steels and Alloys	07
5	Corrosion and degradation of metals	04
6	Fiber Reinforced Composites	08
7	Engineered Wood & Wood Products	06

Total Hours (Theory): 45

Total Hours (Lab): 20

Total Hours: 65

B. Detailed Syllabus

- | | | |
|-----|---|-----------------|
| 1.0 | Introduction to Material Science | 04Hours 9% |
| 1.1 | Classification of Engineering Materials | |
| 1.2 | Engineering requirements of materials | |
| 1.3 | Properties of engineering materials | |
| 1.4 | Criteria for selection of materials for engineering applications | |
| 1.5 | Structure property relationship | |
| 2.0 | Crystal Structure and Properties | 10 Hours 22% |
| 2.1 | Atomic structure & bonding in solids | |
| 2.2 | Bravais lattices & Crystal Structure, Crystal planes & directions, Atomic packing | |

2.3	Bragg Law & Diffraction		
2.4	Imperfection in solids, Linear defects, Slip, Dislocation & Plastic deformation, Planar defects, Volume defects		
2.5	Mechanical properties of metals, Strain hardening, Thermal, Magnetic, Electrical & Optical properties of materials		
3.0	Mechanism of crystallization & Phase Transformation	06 Hours	13%
3.1	Nucleation- Homogeneous and Heterogeneous Nucleation- Growth - Single crystal - Polycrystalline Materials - Basic principles of solidification of metals and alloys		
3.2	Phase diagram & phase transformation		
4.0	Engineering Steels and Alloys	07 Hours	16%
4.1	Iron carbon diagram and phase diagrams		
4.2	Grades of CI, Alloyed Cast Iron, Malleable Iron and S. G. Iron		
4.3	Classification, Properties and Use of Structural Steels, High Carbon and Mild Steels, TMT Steel		
4.4	Effects of different alloying metals		
5.0	Corrosion and degradation of metals	04 Hours	9%
5.1	Causes and nature of corrosion		
5.2	Inter-granular corrosion (IGC), Hydrogen embrittlement		
5.3	Measures of counteracting corrosion,		
5.4	Metal coatings, Organic coatings, Lining and cladding, Use of Corrosion inhibitors, Cathodic protection against corrosion		
6.0	Fiber Reinforced Composites	08 Hours	18%
6.1	Types and application of composites		
6.2	Particle reinforced composites, Influence of fiber orientation and concentration		
6.3	Fiber phase and the matrix phase, Polymer matrix composite		
6.4	Metal matrix composite, Ceramic matrix composite and carbon-carbon composites		
7.0	Engineered Wood and Wood Products	06 Hours	13%
7.1	Laminated veneer lumber (LVL): Laminated		

strand lumber (LSL), Parallel strand lumber (PSL),
Wood I-joists, Glue-laminated beams

- 7.2 Reconstituted products: Particle boards,
Medium Density Fibre Board (MDF), Hardboard
and Block Board

C. Course Outcomes :

On the successful completion of this course:

- CO1 Familiar with various atomic bonding and crystal structure, and also their characteristics and influence on engineering properties
- CO2 Understand basic engineering properties of materials and the underlying structural features which are governing such properties
- CO3 Know the mechanism of crystallization and phase transformation in major engineering materials
- CO4 Be able to understand fundamentals of various engineering steels and alloys and should have ability to select suitable materials based on their structure-property relations
- CO5 Be able to understand the corrosion and other degradation in metals and suggest suitable remedial measures.
- CO6 Know the structure, properties and application of various fiber reinforced composites used in engineering, Know process, structure, properties and application of various processed wood products used in engineering

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	1	-	-	-	1	2	-	2	-	-	-	1	-	-
CO2	3	2	2	2	2	1	1	-	1	-	-	-	2	1	1
CO3	3	3	2	3	2	1	2	-	-	-	-	-	1	-	-
CO4	3	2	3	2	2	2	2	-	1	-	-	-	2	3	3
CO5	2	2	3	3	3	2	1	-	-	-	-	3	1	-	3
CO6	2	2	2	2	2	1	1	-	-	-	-	3	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material

Text Books:

1. Materials Science and Engineering: A First Course, Raghvan V, Prentice Hall of India
2. Callister W. D. Jr, "Materials science and engineering : An Introduction", Edition:-2006, Wiley India, New Delhi, India
3. Dharmendra K umar and Jain S. K., "Material science and manufacturing process", Vikas Pub House, New Delhi, India
4. Van Vlack, Elements of Materials Science and Engineering, 1989

Reference books

1. Narula& Gupta, Material Science, Tata McGraw-Hill Education
2. Avner Sidney H., "Physical Metallurgy", Tata McGraw Hill Education
3. Khanna O. P., "Material Science - A Text Book of Material Science & Metallurgy", Dhanpat Rai Pub
4. Narang G. B. S. and Manchanedy K., "Materials and Metallurgy", Khanna Pub New Delhi, India
5. Thomas G. Williamson, APA Engineered Wood Handbook, McGraw Hill Professional, 2002

Reading Materials, web materials with full citations

1. <http://ocw.mit.edu/OcwWeb/web/courses/courses/index.htm#MaterialsScienceandEngineering>
2. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIScBANG/Material%20Science/New_index1.html

Credits and Hours:

Pre-requisite courses:

- ### A. Outline of the Course:

Total hours (Theory):	--
-----------------------	----

Total hours (Practical): 30

Total hours (Lab) : --

Total hours : 30

B. Detailed Syllabus:

- | | | | |
|----|---|---------|-----|
| 1. | Introduction to Values and Ethics
Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics | 5 Hours | 17% |
| 2. | Elements and Principles of Values
Universal & Personal Values, Social, Civic & Democratic Value | 8 Hours | 26% |

- | | | | |
|----|--|---------|-----|
| 3. | Applied Ethics | 8 Hours | 26% |
| | Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics | | |
| 4. | Value, Ethics & Global Issues | 5 Hours | 17% |
| | Cross-Cultural Issues, Role of Ethics & Values in Sustainability | | |
| 5. | Contemporary Issues in Values and Ethics | 4 Hours | 14% |
| | Case Studies, Presentations, Projects | | |

C. Course Outcome :

At the end of the course, the students will be able to

- CO1 Understand the concepts and mechanics of values and ethics.
- CO2 Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
- CO3 Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
- CO4 Develop their responsibility towards society.
- CO5 Comprehend their own core values and adhere to those values at their workplace.
- CO6 Practice Ethical Leadership.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	-	-	-	-	-	-	1	3	2	1	-	-	-	-	-
CO2	-	-	-	-	-	1	-	-	1	-	-	2	-	2	-
CO3	-	-	-	-	-	3	1	-	-	-	-	-	-	1	-
CO4	-	-	-	-	-	3	1	1	-	-	-	-	-	-	-
CO5	-	-	1	1	-	-	-	-	2	-	-	-	-	1	-
CO6	-	-	-	-	-	-	-	3	-	2	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

❖ Reference book:

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
3. Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

❖ Web material:

- www.ethics.org

MA249: NUMERICAL METHODS
B TECH 4TH SEMESTER (CL/ME)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/Week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No	Title of Unit	Minimum number of Hours
1	Errors	05
2	Approximate solutions of nonlinear equations and system of linear equations	08
3	Interpolation	12
4	Numerical Integration and Differentiation	08
5	Numerical Solution of Ordinary and Partial Differential equations	12

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Errors	05 Hours	(12%)
1.1	Exact and Approximate Numbers and Significant figures		
1.2	Rules of Rounding of numbers, Correct digit		
1.3	Types of errors: Absolute, Relative and Percentage		
1.4	Rounding error, Inherent error and Truncation error		
1.5	Propagation of errors		
2	Approximate solutions of nonlinear equations and System of linear equations	08 Hours	(18%)
2.1	Bisection method, Method of False position, Method of Iteration.		

2.2	Newton-Raphson method for single variable and their convergenc		
2.3	Gauss Jordan method		
2.4	Gauss Seidel method and Gauss Jacobi method		
3	Interpolation	12 Hours	(26%)
3.1	Finite differences and associated operators		
3.2	Newton's Interpolation formulas		
3.3	Lagrange interpolation, Hermite interpolation		
3.4	Error estimates of these formulae		
3.5	Piecewise Interpolation : Linear Spline, Cubic Spline		
4	Numerical Integration and Differentiation	08 Hours	(18%)
4.1	Newton-Cotes Quadrature formula		
4.2	Trapezoidal rule with error		
4.3	Simpson's 1/3 rule and Simpson's 3/8 rule with respective errors		
4.4	Romberg Integration and Gaussian Quadrature		
4.5	Numerical Differentiation		
5	Numerical Solution of Ordinary and Partial Differential Equations	12 Hours	(26%)
5.1	Overview of ordinary differential equations and partial differential equations		
5.2	Euler method and Modified Euler method		
5.3	Runge -Kutta 2 nd and 4 th order methods		
5.4	Finite difference approximations to derivatives		
5.5	Solution of Laplace Equation by Gauss Jacobi and Gauss Seidel Method		
5.6	Solution of Heat Equation in One dimension		
5.7	Solution of Wave Equation in One dimension		

C. Course Outcome:

After learning the course, students will able to

- CO1 Employ the knowledge of various types of numerical methods to Engineering and real world problems.
- CO2 Determine approximate solution of non-linear equations and system of linear equations and examine the error using numerical methods.
- CO3 Use numerical techniques for differentiation and definite integration.
- CO4 Interpolate or extrapolate approximate value of dependent variable for any value of independent variable, employing only finitely many tabulated values.
- CO5 Interpret the solution of ordinary and partial differential equation numerically.
- CO6 Construct the algorithm and perform the various commands in programming language.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	1	2	-	-	-	-	-	-	-	1	1	1	1
CO2	3	3	1	2	-	-	-	-	-	-	-	-	1	1	1
CO3	3	3	1	2	-	-	-	-	-	-	-	-	1	1	1
CO4	3	3	1	2	-	-	-	-	-	-	-	-	1	1	1
CO5	3	3	1	2	-	-	-	-	-	-	-	-	1	1	1
CO6	2	3	-	2	1	-	-	-	-	-	-	1	1	2	1

- Correlation levels 1, 2 or 3 as defined below:
- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

D. Recommended Study Material:

Text Books:

1. S. S. Sastry,, Introductory methods of numerical analysis. PHI Learning Pvt. Ltd.,2012.
2. Steven C. Chapra and Raymond P. Canale; Numerical methods for engineers. Vol. 2. New York: McGraw-Hill, 2012.

Reference Books:

1. S. Grewal and J. S. Grewal; Numerical Methods in Engineering & Science: With Programs in C, C++ & MATLAB. Khanna, 2014.
2. Jain, M. K., Iyengar, S. R. K. and Jain, R. K., "Numerical Methods for Scientific and
3. Engineering Computation", New Age Pvt. Pub, New Delhi, 2000
4. K. E. Atkinson, Introduction to Numerical Analysis, 2nd Edn, John Wiley, 1989

Web Materials:

1. <http://numericalmethods.eng.usf.edu>
2. <http://mathworld.wolfram.com/>
3. <http://nptel.ac.in>

B. Tech. (Civil Engineering) Programme

ANNEXURE II - UNIVERSITY ELECTIVE SYLLABI

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering
Chandubhai S Patel Institute of Technology
M. S. Patel Department of Civil Engineering
Bachelor of Technology Programme
(Third Year Civil Engineering)

Effective From 2022-23

Vision

“To provide state of the art education in Civil Engineering guided by innovative research leading to centre of excellence in Civil Engineering education having recognition at national and international levels”

Mission

“Being a pioneering branch of Engineering, the department of Civil Engineering under the shelter of CHARUSAT is intended as a facilitator for creating a liaison between the brilliant student community and the next generation industrial needs”

Programme Educational Objectives (PEO's):

PEO 1: The graduate will possess foundation of engineering knowledge and exhibiting critical thinking and problem solving skills

PEO 2: The graduates will have trait of lifelong learning and be able to inculcate the capabilities to meet the diversified needs of industry, academia and research.

PEO 3: The graduate will exhibit the professional ethics and be supportive to the social needs

PEO 4: The graduates will possess comprehending, analyzing and designing capabilities to generate sustainable solutions

Programme Outcomes (PO's)

Engineering Graduates will be able to:

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSO's):

By the completion of Civil Engineering program, the student will attain:

PSO 1: The ability to serve the infrastructure sector with capabilities to plan, design, analyze and building civil engineering-based systems

PSO 2: The ability to adopt to the state-of-the-art practices in all sectors of Civil Engineering.

PSO 3: Employability skills with the cognizance of social and environmental necessity along with ethical responsibility to have a successful career and to become an entrepreneur.

CHARUSAT welcomes you for a Bright Future



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Accredited with Grade A by NAAC,
Accredited with Grade A by KCG

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Civil Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Academic Year – 2022-23

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Bachelor of Technology Programmes

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least of 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen subject of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate programme (B. Tech.)

Minimum 8 semesters (4 academic years)

Maximum 12 semesters (6 academic years)

3. Eligibility for Admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of Admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme Structure and Credits

As per Annexure – 1 attached

6. Attendance

All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

Student's attendance in a course should be 80%.

7. Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.2 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 7.1.3 Final examination by the University through modes such as; written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, is set to 70% of the marks for each the course.

7.2 Internal Evaluation

As per Annexure – I attached

7.3 University Examination

The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.4 In order to earn the credit in a course a student has to obtain grade other than FF.

7.5 Performance at Internal & University Examination

- 7.5.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows:

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

- 7.5.2 A student failing to score 40% in the final examination will get an FF grade.
- 7.5.3 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8. Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	<80 ≥73	<73 ≥66	<66 ≥60	<60 ≥55	<55 ≥50	<50 ≥45	<45
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Numerical point (Grade Point) corresponding to the letter grade	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where,
 C_i = Number of credits of course i
 G_i = Grade Point for the course i
 $i = 1$ to n
 n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where,
 C_i = Number of credits of course i
 G_i = Grade Point for the course i
 $i = 1$ to n
 n = number of courses of all semesters up to which CGPA is computed
- (iii) No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9. Award of Degree

- 9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:
- 9.1.1 He/ She should have earned minimum required credits as prescribed in course structure; and
 - 9.1.2 He/ She should have cleared all internal and external evaluation components in every course; and
 - 9.1.3 He/ She should have secured a minimum CGPA of 4.5 at the end of the programme;
 - 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.
- 9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction	:	CGPA ≥ 7.5 & ≤ 10.0
First class	:	CGPA ≥ 6.0 & < 7.5
Second Class	:	CGPA ≥ 5.0 & < 6.0
Pass	:	CGPA < 5.0

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM

A. Choice Based Credit System:

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/Specialization.

1.1.2 Programme Core (PC)

A 'Core Course' is a course which acts as a fundamental or conceptual base for Chosen Specialization of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An 'Elective Course' is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization

1.2.2 Programme Elective Course (PE)

A 'Programme Elective Course' is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time.

1.2.3 Cluster Elective Course (CE)

A 'Cluster Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (e.g. Common Courses for EC/CE/IT/EE) as offered by or decided by the Institute from time-to-time.

1.3 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non Credit Courses.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CIVIL ENGINEERING (CBCS)

Level	Course Code	Course Title	Teaching Scheme								Examination Scheme				
			Contact Hours				Credit				Theory		Practical		Total
			Theory	Practical	Tutorial	Total	Theory	Practical	Project	Total	Internal	External	Internal	External	
Level 3	CL341	Structural Analysis-II	4	2	0	6	4	1		5	30	70	25	25	150
	CL344	Summer Internship - I	0	3	0	3	0	0	3	3	0	0	75	75	150
	CL349	Concrete Technology	3	2	1	6	3	1		4	30	70	25	25	150
	CL350	Building Planning	3	2	1	6	3	1		4	30	70	25	25	150
	HS131.02 A	Communication And Soft Skills	0	2	0	2	0	2		2	0	0	30	70	100
	CLXXX	Programme Elective-I	3	2	0	5	3	1		4	30	70	25	25	150
						28				22					850
	CL351	Environmental Engineering-I	4	2	0	6	4	1		5	30	70	25	25	150
	CL346	Geotechnical Engineering-I	3	2	0	5	3	1		4	30	70	25	25	150
	CL352	Water Resources Engineering-I	4	2	1	7	4	1		5	30	70	25	25	150
	CL353	Transportation Engineering-I	4	2	1	7	4	1		5	30	70	25	25	150
	HS132.0 A	Contributory Personality Development	0	2	0	2	0	2		2			30	70	100
	CLXXX	Programme Elective-II	3	2	0	5	3	1		4	30	70	25	25	150
						32				25					850

CONTENTS

CL341: STRUCTURAL ANALYSIS-II.....	16
CL344: SUMMER INTERNSHIP-I.....	20
CL349: CONCRETE TECHNOLOGY.....	24
CL350: BUILDING PLANNING	29
HS131.02 A: COMMUNICATION AND SOFT SKILLS	34
CL370: CONSTRUCTION ENGINEERING MATERIALS	38
CL372: FLUID MECHANICS-II.....	42
CL373: AIR POLLUTION AND CONTROL	46
CL374: ARCHITECTURAL DESIGN OF FACILITIES.....	51
CL375: BUILDING SERVICES.....	55
CL381: INFRASTRUCTURE MANAGEMENT	60
CL382: DISASTER PREPAREDNESS & PLANNING MANAGEMENT	64
CL351: ENVIRONMENTAL ENGINEERING- I.....	70
CL346: GEOTECHNICAL ENGINEERING-I	75
CL352: WATER RESOURCES ENGINEERING - I.....	79
CL353: TRANSPORTATION ENGINEERING- I	84
HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT	90
CL376: BUILDING REPAIR AND REHABILITATION	93
CL377: ADVANCED SURVEY.....	99
CL378: STRUCTURAL ANALYSIS-III.....	104
CL379: TOWN & URBAN PLANNING	108
CL380: CONSTRUCTION TECHNOLOGY	112
CL383: INSTRUMENTATION & SENSOR TECHNOLOGIES FOR CIVIL ENGINEERING APPLICATIONS.....	116

B. Tech. (Civil Engineering) Programme

SYLLABI (Semester – 5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CL341: STRUCTURAL ANALYSIS-II
B. TECH 5th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Energy Principles	10
2	Fixed and Continuous Beam	08
3	Influence Line for Indeterminate Structures	10
4	Consistent Deformation	06
5	Slope-Deflection Method	10
6	Moment-Distribution Method	10
7	Introduction to Matrix- Beam Element	06

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

1	Energy Principles	10 Hours	17%
1.1	Strain energy- Introduction		
1.2	Strain energy stored in linear elastic systems		
1.3	Castigliano's theorems		
1.4	Computation of displacements of statically determinate beams and frames by unit load method		
1.6	Analysis of statically indeterminate structures-beams and frames by unit load method		

2	Fixed and Continuous Beam	08 Hours	13%
2.1	Statically determinate and indeterminate structures		
2.2	Degree of static indeterminacy		
2.3	Advantages and disadvantages of indeterminate structures		
2.4	Analysis of fixed beams for various types of loads and secondary effects		
2.5	Claypeyron's three moment theorem and its application		
2.6	Analysis of continuous beams		
2.7	Analysis of propped cantilever beams		
3	Influence Line for Indeterminate Structure	10 Hours	17%
3.1	Muller-Breslau's principle		
3.2	Quantitative I.L. for reaction and internal forces in propped cantilever and continuous beams		
3.3	Qualitative I.L. for rigid jointed structures having higher degree of static indeterminacy		
4	Consistent Deformation	06 Hours	10%
4.1	Introduction		
4.2	Method of consistent deformation		
4.3	Analysis of Statically indeterminate beams		
5	Slope-Deflection Method	10 Hours	17%
5.1	Introduction		
5.2	Basic concepts, Definitions and Sign conventions		
5.3	Development of Slope-deflection equations		
5.4	Analysis of indeterminate beams		
5.5	Analysis of portal frames with no lateral translation of joints		
5.6	Analysis of frames with lateral translation of joints		
6	Moment-Distribution Method	10 Hours	17%
6.1	Introduction		
6.2	Absolute and relative stiffness of members, Carry over factor (COF), Distribution factor (DF)		
6.3	Sign conventions		
6.4	Applications of method		
6.5	Analysis of continuous beam with secondary effects		

6.6 Symmetrical frames

6.7 Analysis of frames with no lateral translation of joints

6.8 Analysis of frames with lateral translation of joints

7 Introduction to Matrix- Beam Element

06 Hours 09%

7.1 Introduction to flexibility and stiffness method

7.2 Application of stiffness method system approach to analysis of beam

C. Course Outcomes:

On the completion of the course the students will be able to:

CO1 Identify the fundamentals of structure and various methods of Analysis

CO2 Recognize the role of professional societies in developing new structural software and updating current knowledge

CO3 Identify and formulate an engineering problem and to develop a solution

CO4 Use various classical methods for analysis of indeterminate structures

CO5 Demonstrate the concepts of qualitative influence line diagram for rigid jointed structures having higher degree of statically indeterminacy

CO6 Apply the basic concepts of matrix methods in structural analysis

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	-	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2	-	1	3	1	2	1	-	-	-	-	-	-	1	-	1
CO3	2	2	3	1	1	-	-	-	-	-	-	-	2	-	-
CO4	2	2	2	1	2	-	-	-	-	-	-	-	1	-	-
CO5	3	1	2	1	1	-	-	-	-	-	-	-	1	-	-
CO6	1	1	-	-	-	-	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I & II, Charotar Publishing House
2. Negi, L.S. and Jangid, R.S., Structural Analysis, Tata McGraw Hill
3. Vazirani, V.N. and Ratwani, N.M., Analysis of Structures, Khanna Publishers

4. Weaver William and Gere James, Matrix Analysis of Framed Structures, CBS Publishers
5. R.S. Khurmi and N. Khurmi, Theory of structures, S. Chand

Reference Books:

1. Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
2. Hibbler, R.C., Structural Analysis, Pearson Education
3. Wang, C.K., Intermediate Structural Analysis, Tata McGraw Hill
4. Reddy, C.S., Basic Structural Analysis, Tata McGraw Hill
5. Jangid, R.S., Structural Analysis, Tata McGraw Hill
6. Devdas Menon., Structural Analysis, Alpha Science
7. William M.C. McKenzie, Examples in Structural Analysis, Second Edition, CRC Press

Web Materials:

1. <http://nptel.ac.in/downloads/105101085/>
2. <http://nptel.ac.in/courses/105101086/>
3. http://nptel.ac.in/courses/Webcourse-contents/IIT%20Kharagpur/Structural%20Analysis/New_index1.html
4. <http://www.nptelvideos.in/2012/11/structural-analysis-ii.html>
5. <http://nptel.ac.in/courses/105106050/2>
6. [http://nptel.iitg.ernet.in/Civil_Eng/IIT%20Delhi/Structural%20Analysis%201%20\(Video\).htm](http://nptel.iitg.ernet.in/Civil_Eng/IIT%20Delhi/Structural%20Analysis%201%20(Video).htm)
7. <http://freevideolectures.com/Subject/Civil-Engineering>
8. <http://freevideolectures.com/Course/97/Structural-Analysis-II>

LIST OF EXPERIMENTS

Experiment No.	Name of Experiment
1	Deflections of beams and cantilevers
2	Continuous and indeterminate beam
3	Pin jointed frameworks
4	Frame deflections and reactions
5	Bending moments in a portal frame

CL344: SUMMER INTERNSHIP-I
B TECH 5th SEMESTER (CIVIL ENGINEERING)

Credit and Hours:

Teaching Scheme	Project	Total	Credit
Hours/week	30	90	3
Marks	150	150	

A. Instructional Method and Pedagogy:

- Summer internship shall be for at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for their summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit following two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.

- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

B. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and main conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.

- **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
- **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, computer programs, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

C. Course Outcomes:

After completion of the course students will able:

- CO1 To apply knowledge and skills gained in company/industry/organization to real-world problems and to solve engineering problems.
- CO2 To learn to work as a team and to work with teammates from other disciplines.
- CO3 To use experience related to professional and ethical issues in the actual work environment.
- CO4 To explain the impact of engineering solutions employed in a project, in a global, economic, environmental, and societal context.
- CO5 To find relevant sources (e.g., library, Internet, experts) and gathers information and to demonstrate knowledge of contemporary issues related with engineering in general and to use new tools and technologies.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	3	2	1	1	1	3	3	3	3	3	3
CO2	-	-	-	-	-	-	-	-	3	2	-	-	-	-	-
CO3	-	-	-	-	-	1	-	3	-	-	-	-	-	-	-
CO4	3	3	2	2	2	2	3	1	-	1	1	2	1	1	1
CO5	2	3	2	3	3	2	2	1	-	2	1	2	2	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CL349: CONCRETE TECHNOLOGY
B TECH 5th SEMESTER (CIVIL ENGINEERING)

Credits and hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	1	2	6	4
Marks	100	-	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	01
2	Cement	07
3	Aggregate	07
4	Admixtures	05
5	Fresh Concrete	06
6	Harden Concrete	06
7	Selection of Concrete Mix Proportion (Mix Design)	07
8	Special Concrete	04
9	Non-Destructive test of Concrete	02

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction	01 Hours	02%
1.1	Ingredients of concrete		
1.2	Strength development		
1.3	Different types of concrete		
1.4	New developments and future trends		

2	Cement	07 Hours	15%
2.1	History of portland cement		
2.2	Manufacture process of portland cement		
2.3	Chemical composition		
2.4	Hydration and heat of hydration		
2.5	Structure of hydrated cement and product of hydration		
2.6	Setting of cement		
2.7	Field testing of cement, standard consistency test, initial and final setting time, strength of cement, soundness test		
2.8	Storage of cement		
2.9	Types of cement		
3	Aggregate	07 Hours	15%
3.1	Source and classification of aggregates		
3.2	Shape, size and texture of aggregates		
3.3	Mechanical properties of aggregates, strength of aggregates, specific gravity, bulk density, porosity and absorption of aggregates, moisture content of aggregates, bulking of fine aggregate		
3.4	Deleterious substance in aggregate		
3.5	Soundness of aggregates		
3.6	Alkali aggregates reaction		
3.7	Grading of aggregates		
3.8	Test on aggregate: Aggregates crushing value, aggregates impact value, aggregates abrasion value, sieve analysis and fineness modulus.		
3.9	Quality of mixing water		
3.10	Use of Sea water in concrete		
4	Admixtures	05 Hours	11%
4.1	Advantages of admixture		
4.2	Types of admixture: Accelerators, retarders, water reducing admixture, plasticizers and air entraining, super plasticizer, water proofing admixture, antibacterial admixture		
4.3	Cementitious materials: Fly ash, silica fume, rice husk ash, ground granulated blast furnace slag, metakaolin		

5	Fresh Concrete	06 Hours	14%
5.1	Rheology of fresh concrete		
5.2	Workability and factors affecting workability		
5.3	Measurement of workability		
5.4	Segregation		
5.5	Bleeding		
5.6	Process of manufacture of concrete: batching, mixing, transportation, placing, compaction, curing, re-tampering, finishing		
6	Hardened Concrete	06 Hours	14%
6.1	Gain of strength of concrete		
6.2	Factors affecting strength of concrete: water cement ration, gel space ratio, age of concrete		
6.3	Durability and permeability		
6.4	Nature of strength of concrete		
6.5	Microcracking , Creep and Shrinkage		
6.6	Aggregate cement paste interface		
6.7	Maturity of concrete		
7	Selection of Concrete Mix Proportions (Mix Design)	07 Hours	15%
7.1	Concept of concrete mix design: variables in proportioning, various methods of proportioning, statistical quality control of concrete, calculation of standard deviation, relation between average design strength and specified minimum strength		
7.2	Principle of mix proportioning		
7.3	Factors governing the selection of mix proportion		
7.4	Mix proportions and quantities per batch		
7.5	ACI, IS Method, concrete mix design using admixture, mix design for pumpable concrete		
7.6	Variability of test result		
7.7	Acceptance criteria and IS code provisions		

8	Special Concrete	04 Hours	9%
8.1	High performance concrete		
8.2	Lightweight aggregate		
8.3	High strength concrete		
8.4	Polymer concrete		
8.5	Fibre reinforce concrete		
8.6	Ready mix concrete		
8.7	Mass concrete		
8.8	Self-Compacting Concrete		
9	Non Destructive testing of Concrete	02 Hours	5%
9.1	Introduction to Destructive, Semi destructive and Non-destructive testing,		
9.2	Problem faced during Non-destructive evaluation.		
9.3	Rebound Hammer test,		
9.4	Ultrasonic Pulse Velocity test		

C. Course Outcomes:

At the end of the course, the students will be able to

- CO1 The students will be able to: test all the concrete materials as per IS code.
- CO2 The students will understand concrete mix design using ACI and IS code methods.
- CO3 The students able to determine the properties of fresh and hardened concrete.
- CO4 The students have knowledge of special concretes and their specific applications

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	-	-	-	-	-	-	3	-	2	-	-	1	-	1
CO2	1	-	-	-	-	-	-	3	-	2	-	-	1	-	1
CO3	-	-	-	-	1	-	-	-	-	-	-	-	1	-	-
CO4	-	-	-	-	-	-	2	-	-	-	1	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text books:

1. Neville, A.M., Concrete Technology, Pearson Education.
2. Santakumar, A.R., Concrete Technology, Oxford University Press.
3. Shetty, M.S., Concrete Technology, S. Chand Publication.
4. Gambhir, M.L., Concrete Technology, Tata McGraw Hill Publication.

Reference books:

1. Paulo, Mehta P. Kumar and Monteiro, J.M., Concrete Micro-structure, Properties and Materials, Prentice Hall INC and McGraw Hill, USA.
2. IS: 10262- 2009 Recommended guidelines for Concrete Mix Design.
3. IS: 456- 2000, Indian Standard Plain and Reinforced Concrete Code of Practice.

Web Materials:

1. <http://nptel.iitm.ac.in/video.php?courseId=1059>
2. <http://www.aboutcivil.com/concrete-technology.html>
3. www.understanding-cement.com
4. <http://www.engineeringcivil.com/theory/concrete-engineering/>

CL350: BUILDING PLANNING
B. TECH 5th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	1	6	4
Marks	100	50		150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Min. No. of Hours
1	Building Drawing	03
2	Building Byelaws	06
3	Planning of Residential Buildings	12
4	Planning of Public Buildings	04
5	Building Services	06
6	Perspective Drawing	08
7	Introduction to Town Planning	06

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | |
|--|-----------------|------------|
| 1 Building Drawing | 03 Hours | 07% |
| 1.1 Classification of buildings | | |
| 1.2 Basic principles of planning | | |
| 1.3 Principles of Architecture | | |
| 1.4 Standard conventional signs, symbols and abbreviations | | |
| 1.5 ISI nomenclature: Size of scale, standard method of Dimensioning | | |
| 2 Building Byelaws | 06 Hours | 13% |
| 2.1 Objectives, Scope, Applicability and Principles | | |
| 2.2 Basic terminology | | |
| 2.3 Various forms of building coverage | | |
| 2.4 Bye-laws for Open spaces, Area limitations, Height limitations, | | |

	Plinth Area, Means of Access, Light and Ventilation, Parking spaces		
2.4	Minimum standard dimensions of building elements		
2.5	Examples on FSI or FAR		
2.6	Submission of drawings for issue of permission for construction		
3	Planning of Residential Buildings	12 Hours	28%
3.1	Different types of Residential Buildings		
3.2	Selection of site for residential buildings		
3.3	Orientation of Buildings		
3.4	Guidelines for making wall thickness plan from line diagram		
3.5	Detailed drawing–plan, elevation, sections		
3.6	Design of Individual rooms with particulars attention to anthropometric data, functional and furniture requirements.		
3.7	Design of buildings with particular attention to orientations, FSI and allowable built up area.		
3.8	Furniture layout, Kitchen layout with emphasis on kitchen work triangle		
4	Planning of Public Buildings	04 Hours	08%
4.1	School – Principles of planning a school building, Components, Design of a classroom, Sanitary Requirements		
4.2	Hospital – Site selection, Principles of planning a hospital building, Wards, Different departments, Operation theatre, Sanitary Requirements		
4.3	Hostel – Site selection, Principles of planning a hostel building, wardens office, residential area, recreation room, special requirements, Sanitary Requirements, Kitchen and Dining Hall		
5	Building Services	06 Hours	13%
5.1	Water supply and Drainage – water distribution systems, materials used for plumbing, drainage systems		
5.2	Electrification – distribution of electrical energy, accessories of electrical installation, purpose of earthing		
5.3	Working drawing of a building – Electrical layout, Water supply and drainage layout		

6	Perspective Drawing	08 Hours	18%
6.1	Introduction & technical terms		
6.2	Classifications, elements of perspective		
6.3	Examples		
7	Introduction to Town Planning	06 Hours	13%
7.1	History, ancient planning in India		
7.2	Objects & importance of town planning		
7.3	Principles of town planning		
7.4	Horizontal growth, vertical growth		
7.5	Satellite town, ribbon development, concentric growth		

C. Course Outcomes:

On the successful completion of this course, the students will be able to:

- CO1 Discern various aspects of principles of planning and architecture in building design and read the building drawings.
- CO2 Understand local building bye-laws and provisions of National Building Code in respect of building and town planning.
- CO3 Utilise basic principle of planning and architecture in design of residential buildings and public buildings.
- CO4 Utilise the knowledge of different types of plumbing and electric fitting and laying procedure to create working drawings.
- CO5 Understand perspective drawing and represent various forms in perspective view.
- CO6 Have basic understanding of town planning and its history.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	1	-	2	-	-	-	1	-	-	-	-	-	3	-	-
CO4	1	-	2	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-
CO6	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Singh Gurcharan, Building Planning, Designing and Scheduling, Standard Publishers and Distributors.
2. Dr. N. Kumara Swamy & Rao A. Kameswara, Building Planning and Drawing, Charotar Publishing House.
3. Wagh Sajjan V., Building and Town Planning, Tech-Max Publications.
4. Khasiya R. B, Motiani A. T and Khasiya K. R, Building & Town Planning, Mahajan Publishing House.

Reference Books:

1. Rangwala S. C., Town Planning, Charotar Publishing House, Anand.
2. Hiraskar G. K., Town Planning, Dhanpatrai & Sons, New Delhi.
3. Sikka V. B., Civil Engineering Drawing, S. K. Kataria & Sons Publication.
4. Shah M. G., Kale C. M and Patki S. Y., Building Drawing, Tata McGraw Hill Publication, New Delhi.
5. General Development Control Regulations published by AUDA and GICEA.
6. National Building Code of India, Indian Standard Institution (ISI), New Delhi, 2005.

Web Materials:

1. <http://www.vastu-design.com/vastu-video-workshop/>
2. https://www.youtube.com/watch?v=EIDXE28_8eQ
3. <https://www.youtube.com/watch?v=VYiVjVulnm4>
4. <https://www.youtube.com/watch?v=bCn0X9RRjN0&list=PL060E3166E87E1FD5>
5. https://www.designingbuildings.co.uk/wiki/Types_of_drawings_for_building_design
6. <http://freevidelectures.com/Course/86/Building-Materials-and-Construction/31>

List of Drawings for Lab Sessions

Sr. No.	Details
1	Conventional Signs & Symbols
2	Plan, Sectional Elevation, Front Elevation and site plan for a small house
3	Plan, Sectional Elevation, Front Elevation and site plan for Duplex Type House

4	Working Drawings
5	Design of houses with given functional requirements and climatic data and Model Making
6	Perspective Drawing: One Point Perspective
7	Perspective Drawing: Two Point Perspective

HSI31.02 A: COMMUNICATION AND SOFT SKILLS
B. TECH 5th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communicative English

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

B. Detailed Syllabus:

1. **An Introduction to Communication** 06 Hours 20%
- 1.1 Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers;
- 1.2 Applications of Communication;
- 1.3 Rhetoric in Professional Communication;
- 1.4 Importance of Ethos, Logos, and Pathos in Communication

2. Cross-cultural Communication and Globalization	03 Hours	10%
2.1 Basic Concepts: Culture, Globalization and Cross-cultural Communication;		
2.2 Social and People Skills;		
2.3 Communicating with People of Different Cultures;		
2.4 Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them;		
2.5 Persuasive Communication		
3. Communication for Career Building	10 Hours	33%
3.1 Cover Letters and Resume;		
3.2 E-mail and Report;		
3.3 Types of Resume;		
3.4 Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion;		
3.5 Concept and Rationale of Personal Interview;		
3.6 Types of Personal Interview;		
3.7 Writing Statement of Purpose		
4. Group Dynamics and Soft Skills	05 Hours	17%
4.1 An Introduction to Group Dynamics and Soft Skills;		
4.2 Groups and their Structures;		
4.3 Roles and Functions of Members in Groups;		
4.4 Conflict Management;		
4.5 Aptitude and Attitude;		
4.6 Various Intelligences;		
4.7 Developing an Open Mindset		
5. Effective Presentation Strategies	04 Hours	14%
5.1 Designing Appealing Presentation;		
5.2 Audience Analysis and Supporting Material;		
5.3 Presentation Mechanics and Presentation Process;		
5.4 Managing Yourself during Q and A Session;		
5.5 Fundamentals of Persuasion		
6. Contemporary Issues in Communication and Soft Skills	02 Hours	06%
6.1 Trends and Practices in Communication, Case Studies		

C. Learning Outcomes:

At the end of the course, the students will be able to

- CO1 Gain thorough understanding and proficiency in various Professional Communication Skills.
- CO2 Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
- CO3 Develop business writing and presentation skills to succeed in career.
- CO4 Develop soft skills to stand out and take their career to the next level.
- CO5 Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.
- CO6 Practice new trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	-	-	-	-	-	2	2	2	-	2	-	-	-	2	-
CO2	-	-	2	-	-	3	-	-	3	-	-	-	-	3	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	1	-	-	2	1	3	2	-	-	-	2	-
CO5	-	-	-	-	-	2	2	-	-	-	2	-	-	2	-
CO6	-	-	-	-	-	-	-	-	-	3	-	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) no correlation, put “-”

Recommended Study Material:

- Text book:
 1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
 2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
 3. Raman, M & Singh, P. Business Communication, Oxford University Press
- Reference book:
 1. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education

2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited

- Web material:

1. <https://www.coursera.org/learn/careerdevelopment>
2. <https://www.futurelearn.com/courses/writing-applications>
3. <https://www.futurelearn.com/courses/workplace-englis>

CL370: CONSTRUCTION ENGINEERING MATERIALS
B. TECH 5th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical / Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Construction Materials	3
2	Timber	7
3	Masonry Units	8
4	Plastics	4
5	Glass	4
6	Ceramic Materials	4
7	Metals	7
8	Paint & Varnishes	3
9	Advanced Construction Materials	5

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction to Construction Materials	03 Hours	07%
1.1	General Introduction		
1.2	Physical Properties		
1.3	Mechanical Properties		
1.4	Classification of materials based upon their uses		
2	Timber	07 Hours	15%
2.1	General Introduction		

2.2	Classification of trees		
2.3	Structure of a tree		
2.4	Seasoning of Timber – Meaning, Objects, Methods, Advantages and Disadvantages of different methods of seasoning		
2.5	Defects in Timber		
2.6	Industrial Timber		
3	Masonry Units	08 Hours	18%
3.1	Bricks – general, composition of good brick earth, qualities of good bricks, factors affecting qualities of bricks, tests for bricks		
3.2	Stone - Qualities of good building stone, tests for stones, dressing of stones		
3.3	AAC blocks - General introduction, Composition, Manufacturing process, Advantages and Disadvantages, properties		
4	Plastics	06 Hours	09%
4.1	General Introduction		
4.2	Composition of Plastic, Polymerization		
4.3	Classification of Plastics		
4.4	Properties of Plastics, Uses of Plastic		
4.5	Biodegradable plastic		
4.6	Fibre Glass Reinforced Plastic		
5	Glass	04 Hours	09%
5.1	General		
5.2	Classification of glass and composition of glass		
5.3	Properties of glass , Types of glass		
5.4	Coloured Glass and special varieties of glass		
6	Ceramic Materials	04 Hours	09%
6.1	General Introduction, Properties, Importance		
6.2	Terracota, Stoneware		
6.3	Tiles - Characteristics, types		
6.4	Glazing and refractories		

7	Metals	07 Hours	15%
7.1	Ferrous - Introduction to different types of ferrous metals		
7.2	Steel - General Information, Manufacture of steel, Uses of Steel, Factors affecting physical properties of Steel, Defects in steel, Properties of Mild Steel, Properties of Hard Steel, Mechanical Treatment of Steel, Market forms of steel		
7.3	Non-Ferrous Metals - Different types Aluminium - Manufacture, Properties, Evaluation as building material, Economics of using aluminium, Uses		
8	Paint & Varnishes	03 Hours	07%
8.1	Paints - Characteristics, Pigment Volume Concentration Number, Ingredients, Types		
8.2	Varnishing - Characteristics, Ingredients, Types		
9	Advanced Construction Materials	05 Hours	11%
9.1	Application of Nano Technology in construction materials		
9.2	Advanced Composite Reinforcement		
9.3	Ground granulated blast furnace slag		
9.4	Hydrophobic concrete for waterproofing		
9.5	Insulated concrete forms		

C. Course Outcomes:

On the successful completion of this course, the students will:

- CO1 Understand the constituents of various materials like timber, glass, plastic, ceramic material, metal, varnish and paint.
- CO2 Be familiar with the varieties of construction material available in market.
- CO3 Understand the application of various construction materials with their properties like timber, glass, plastic, ceramic material, metal, varnish and paint.
- CO4 Be familiar about various advanced construction materials available in market.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	-	-	-	3	2	-	-	-	-	-	-	-	-	2
CO2	1	2	-	-	-	-	1	-	-	1	-	-	1	-	-
CO3	1	2	-	-	2	-	1	-	1	1	-	-	-	-	-
CO4	2	-	-	-	3	2	-	-	-	-	-	-	1	-	2

D. Recommended Study Material:

Text Books:

1. Duggal S. K., Building Materials, New Age International Publication.
2. Rangwala S.C., Engineering Materials, Charotar Publishing House
3. Soni Saurabhkumar, Building Materials & Construction, S.K.Kataria publication

Reference Books:

1. Dr. Janardan Jha, Engineering Materials, Khanna Publication
2. A K Roy, Materials of Construction, Chaudhary Publication
3. Vazirani and Chandola, Engineering Materials
4. D.N. Ghose, Construction Materials, TATA Mc Graw Hill
5. TTTI, Chandigarh, Civil Engineering materials TTTI
6. Rangwala Ketki, Essential of Civil Engineering, Charotar Publishing House

Web Materials:

1. https://onlinecourses.nptel.ac.in/m?desturl=cHJvZmlsZQ==#/course/noc19_ce42
2. https://onlinecourses.nptel.ac.in/m?desturl=cHJvZmlsZQ==#/course/noc19_ce40
3. https://en.m.wikipedia.org/wiki/Insulating_concrete_form
4. <https://www.nbmcw.com/tech-articles/concrete/3725-new-construction-materials-for-modern-projects.html>
5. <https://nptel.ac.in/courses/105106053/>
6. https://geniebelt-com.cdn.ampproject.org/v/s/geniebelt.com/blog/10-innovative-construction-materials/amp?usqp=mq33IAQQKAFwAZgB47zlkNnQu6WrAQ%3D%3D&_js_v=a2&am_gsa=l#referrer=https%3A%2F%2Fwww.google.com&share=https%3A%2F%2Fgeniebelt.com%2Fblog%2F10-innovative-construction-materials

CL372: FLUID MECHANICS-II
B. TECH 5th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1	Dimensional analysis and Model similitude	8
2	Viscous fluid flow	8
3	Turbulent flow in pipes	8
4	Flow around a body- Forces on submerged body	6
5	Non-uniform flow in open channel	7
6	Hydraulic machinery	8

Total Hours (Theory): 45
Total Hours (Practical): 30
Total Hours: 75

B. Detailed Syllabus:

1	Dimensional analysis and Model similitude	08 Hours	18%
1.1	Fundamental dimensions		
1.2	Rayleigh's and Buckingham's method		
1.3	Dimension less numbers and their significance		
1.4	Hydraulic similitude		
1.5	Type of models		
2	Viscous fluid flow	08 Hours	18%
2.1	N-S equations in Cartesian Coordinates and Cylindrical coordinates.		
2.2	Exact Solutions of N-S equations, Poiseuille flow		
2.3	Couette flow		

2.4	Laminar flow in pipe		
	Boundary Layer, Developments of boundary layer along flat plate,		
2.5	laminar boundary layer, turbulent boundary layer, boundary layer separation		
2.6	Boundary Layer thicknesses		
3	Turbulent flow in pipes	08 Hours	18%
3.1	Laminar flow stability, causes and mechanism of turbulence, shear stress due to turbulence		
3.2	Prandtl's mixing length theory, Laws of turbulent flow		
3.3	Turbulent flow in pipes, pipe resistance factors smooth and rough pipes		
3.4	Resistance to commercial pipes, Moody's diagram, Explicit equation for friction factor		
3.5	Pipe Network, Aging of pipes		
3.6	Water Hammer, Causes, Rapid closure, slow closure, Basic equation of water hammer		
3.7	Surge tanks, Types of surge tanks		
4	Flow around a body- Forces on submerged body	06 Hours	13%
4.1	Regime of external flow, Drag and Lift , Pressure drag, Friction drag, stream lined body, Blunt body		
4.2	Drag coefficient and similarity considerations		
4.3	Flow past a cylinder, sphere		
4.4	Flow past airfoil of infinite length and finite length		
5	Non-uniform flow in open channel	07 Hours	15%
5.1	Specific energy, specific energy diagram, Discharge Diagram		
5.2	Application specific energy diagram and discharge diagram, Critical depths, critical flow, critical slope		
5.3	Momentum principle to open channel flow, specific force, small waves in open channel		
5.4	Gradual Varied flow, Derivation of water surface slopes, classification of water surface profiles		
5.5	Hydraulic Jump, Types, elements and characteristics of hydraulic jump		
6	Hydraulic machinery	08 Hours	18%
6.1	Jet striking a stationary plate and moving plate,		

- 6.2 Force exerted by jet on curved vane
- 6.3 Turbo machineries, Pumps and turbines
- 6.4 Classification of hydraulic turbines, impulse turbine
- 6.5 Reaction turbines, specific of turbines
- 6.6 Centrifugal pump, classification, definitions for CF,
- 6.7 Installation and priming of pump, NPSH, Cavitation in pump,

C. Course Outcomes:

On the completion of the course one should be able to:

- CO1 Develop an ability to analyze laminar flow and turbulent flow for pipe
- CO2 Will be able to design the pipe system (Network) and process for the required need of whether discharge or head.
- CO3 Develop the skill to perform experiments as well as analyze and interpret the results.
- CO4 Design open channels for non-uniform flow water surface profiles and dissipation of energy in hydraulic jump.
- CO5 Select the type of pumps and turbine required with reference to available head of water and discharge
- CO6 Able to define the relationship between dimensional magnitude contained in mass, force and acceleration and empirical constants linking various dimensionless parameters.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	1	1	1	-	1	1	-	-	-	1	2	1	1	-
CO2	-	-	1	1	-	2	1	-	-	-	1	2	1	1	-
CO3	2	2	2	2	2	2	2	-	-	-	1	2	1	2	1
CO4	1	-	-	2	-	-	1	-	1	-	3	2	2	1	-
CO5	-	2	-	-	2	1	-	1	1	1	-	-	1	-	2
CO6	-	1	-	-	2	-	-	1	-	-	1	-	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Jain, A.K., Fluid Mechanics, Khanna Publishers, New Delhi.
2. Bansal, R.K., Fluid Mechanics, Laxmi Publication, New Delhi.

Reference Books:

1. Streeter, V.L. and Wylie, E.B., Fluid Mechanics, McGraw Hill, 1985, New York.
2. Subramanya, K., Theory and Applications of Fluid Mechanics, Tata-McGraw Hill Publishing Co., New Delhi, 1993.
3. Shaughnessy, E.J., Katz, I.M. and Schaffer, J.P., Introduction to Fluid Mechanics, SI edition, Oxford University Press, New Delhi, 2005.
4. White, F.M. Fluid Mechanics, McGraw Hill Publication, New York.
5. Kumar, D.S., Fluid Mechanics, S.K.Kataria & Sons Publication.
6. Rajput, R.K., Fluid Mechanics, S. Chand & Co. Publication.
7. Modi, P.N. and Sheth, Fluid Mechanics & Hydraulic Machines, Standard Book House.
8. Ramamurtham, S., Hydraulic Fluid Mechanics & Fluid Machines, Dhanpatrai Publishing Co. Publication.
9. Garde, R.J. and Mirajgaoker, A.C., Engineering Fluid Mechanics, New Chand & Sons Publication.

Web Materials:

1. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT%20Guwahati/fluid_mechanics/index.htm
2. <http://nptel.iitm.ac.in/video.php?subjectId=105101082>
3. <http://www.mvsengineering.com>
4. <http://nptel.iitm.ac.in/courses/IIT-MADRAS/Hydraulics/index.php>
5. <http://nptel.iitm.ac.in/video.php?subjectId=105103096>

CL373: AIR POLLUTION AND CONTROL
B. TECH 5th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours / week	03	02	05	04
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Air Quality: History, Definitions, Characteristics and Perspectives	04
2	Sources and Effects of Air Pollution	06
3	Meteorological Aspects of Air pollution	06
4	Transport of Pollution in Atmosphere	07
5	Sampling and analysis of Air Pollutants	05
6	Air Pollution Control Systems	12
7	Air Pollution Legislation	02
8	Current Global Issues of Air Pollution	03

Theory Hours: 45
Laboratory Hours: 30
Total Hours: 75

B. Detailed Syllabus:

- 1 Air Quality: History, Definitions, Characteristics and Perspectives 04 Hours 08%
 - 1.1 Composition of dry ambient air properties of air, function of air, definition of pollution
 - 1.2 Air pollution – historical overview
 - 1.3 Air pollutants – classification
 - 1.4 Global and national scope of the problem- general, urban, rural and specific
 - 1.5 Air quality management concept

2	Sources and Effects of Air Pollution	06 Hours	10%
2.1	Air pollutants – sources		
2.2	Units for measurement of air pollution, Emission factors		
2.3	Major pollutants from different sources in Gujarat		
2.4	Effects of air pollution on human health, plants , animals, properties and visibility		
2.5	Economic effects of air pollution		
2.6	Indoor air pollution and personal exposure to air pollution		
2.7	Simple numerical problems based on COH, CoHb		
3	Meteorological Aspects of Air Pollution	06 Hours	16%
3.1	Introduction to meteorology: Global winds, Hadley cells, wind rose terrestrial wind profile		
3.2	Effects of terrain and topography on winds		
3.3	Atmospheric stability and instability		
3.4	Theory and application of acoustic sounding (SODAR) technique		
4	Transport of Pollution in Atmosphere	07 Hours	16%
4.1	Lapse rate		
4.2	Plume behaviour under different atmospheric conditions		
4.3	Plume behaviour in valley and terrains		
4.4	Maximum mixing depths		
4.5	Mathematical models of dispersion of air pollutants		
4.6	Stack design		
5	Sampling and analysis of Air Pollutants	05 Hours	12%
5.1	Methods and Instruments for sampling and analysis of air for stack and ambient air monitoring		
6	Air Pollution Control Systems	12 Hours	25%
6.1	Air Pollution control at source		
6.2	Equipment for control of air pollution : For particulate matter - Settling chambers - Fabric filters - Scrubbers - Cyclones - Electrostatic precipitators		
6.3	For Gaseous pollutants : control by absorption - adsorption - scrubbers - secondary combustion after burners		

6.4	Working principles advantages and disadvantages		
6.5	Design criteria and examples		
6.6	Fugitive Emissions – sources and control , odour control		
7	Air Pollution Legislation	02 Hours	06%
7.1	Government of India: air pollution laws		
7.2	Indian standards- emission and air quality standards		
7.3	Role of pollution control boards		
8	Current Global Issues of Air Pollution	03 Hours	07%
8.1	Greenhouse effect, Ozone depletion, heat island, dust storms		
8.2	Automobile pollution sources and control		
8.3	Photochemical smog		
8.4	Future engines and fuels		

C. Course Outcomes:

On the successful completion of this course,

- CO1 The students will be able to classify the sources of air pollution, describe effects, various methods and equipment available for controlling air pollution.
- CO2 The students will recognize the diverse knowledge of air pollution control practices applied to real life problems.
- CO3 The students will be able to review the theories and practical aspects of air pollution control along with the applications and further, they shall have an ability to plan air pollution control systems.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	1	-	2	-	-	-	-	-	-	-	1	-	-	-
CO2	3	2	-	3	-	3	-	1	-	1	1	3	2	2	2
CO3	1	3	3	3	2	3	3	3	3	3	2	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

- Peavy, Howard S., Donald R. Rowe, and George Tchobanoglous. 1985. *Environmental Engineering*. McGraw-Hill, New York (Latest edition)

2. Rao, C.S. 2007. *Environmental pollution control Engineering*. New Age International, New Delhi (Latest edition)
3. Rao. M. N. and Rao, H. V. N. 2007. *Air Pollution*. Tata McGraw Hill Publication, New Delhi (Latest edition)

Reference Books:

1. Government of India's Publication of laws related to air pollution, Gujarat Pollution Control Board's (GPCB) Publication of standards. Indian Standards relevant to Air Pollution Monitoring.
2. David H.F. Liu, Bela G. Liptak. 2016 Special Indian Edition. *Environmental Engineers' Handbook* Part 1 and Part 2, CRC Press LLC, Florida
3. Doren, K. L. 2016. *Air Pollution*, CBS Publishers and Distributors, New Delhi
4. Wark, K., & Warner, C. F. 1981. *Air pollution: its origin and control*. Harper & Row, New York (Latest edition)
5. Parker, A. 1978. *Industrial Air Pollution Handbook*. Tata McGraw Hills Publication, New Delhi (Latest edition)
6. Wang, Lawrence K., Pereira, Norman C., Hung, Yung-Tse. 1979. *Handbook of Environmental Engineering*. Vol. 1, Air and Noise Pollution Control, The Humana Press, Clifton, New Jersey (Latest edition)
7. Brauer, H. and Verma, Y. B. G. 1981. *Air Pollution Control Equipment*. Berlin Heidelberg, New York (Latest edition)
8. Richard C. Flagan & John H. Seinfeld. *Fundamentals of Air Pollution engineering*. 1988. Prentice Hall, Englewood Cliffs, N.J (Latest edition)
9. Roy M. Harrison. *Pollution: Causes, Effects and Control*. 2000. Royal Society of Chemistry, Cambridge (Latest edition)

Web Material:

1. <<http://cpcbenvi.nic.in/airpollution/database.htm>>
2. <<http://cpcb.nic.in/>>
3. <<http://www.gpcb.gov.in/>>
4. <<http://nptel.ac.in/courses/105104099/>>
5. <<http://nptel.ac.in/courses/105102089/8>>
6. <<http://www.envfor.nic.in/legis/air/air1.html>>

7. <<http://www.indiacode.nic.in/acts-in-pdf/2015/201503.pdf>>
8. <<http://envfor.nic.in/>>

LIST OF EXPERIMENTS

Sr. No.	Title of Experiment
1	Introduction to Air Quality Monitoring and Air Quality Standards
2	Determination of Wind Direction and Speed
3	Monitoring of Suspended Particulate Matter
4	Monitoring of Respirable Particulate Matter (PM ₁₀ , PM _{2.5} and PM _{1.0})
5	Monitoring of Gaseous Air Pollutants in Ambient Air: Oxides of Nitrogen
6	Monitoring of Gaseous Air Pollutants in Ambient Air: Sulphur Dioxide
7	Monitoring of Gaseous Air Pollutants in Ambient Air: Carbon Monoxide
8	Monitoring of Stack for Particulate Matter
9	Monitoring of Stack for Gaseous Pollutants
10	Automobile Exhaust Monitoring

CL374: ARCHITECTURAL DESIGN OF FACILITIES
B. TECH 5th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Architectural Design	12
2	Architectural Detailing	15
3	Architectural Facilities	18

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Architectural Design	12 Hours	25 %
1.1	The Site: Challenges and Opportunities		
1.2	Site Design and Environmental Analysis		
1.3	Sustainable Infrastructure		
1.4	The Skin: Materials and Techniques		
1.5	Evolving a Sustainable Design Practice		
1.6	Sustainable Design and Existing Buildings		
2	Architectural Detailing	15 Hours	40%
2.1	General Planning and Design Data		
2.2	Site work		
2.3	Masonry		
2.4	Metals		
2.5	Wood and Plastics		

- 2.6 Thermal and Moisture Protection
- 2.7 Doors and Windows
- 2.8 Interior Stairs and Stairwells
- 2.9 Specialties
- 2.10 Conveying Systems
- 2.11 Electrical
- 2.12 Energy and Environmental Design
- 2.13 Layout of Plates
- 2.14 Exterior Details

3 Architectural Facilities 18 Hours 35%

- 3.1 Controlling Water
- 3.2 Controlling Air
- 3.3 Controlling Heat Flow
- 3.4 Controlling Water Vapor
- 3.5 Controlling Sound
- 3.6 Accommodating Movement
- 3.7 Providing Structural Support
- 3.8 Providing Passages For Mechanical and Electrical Services
- 3.9 Health and Safety

C. Course Outcomes:

At the end of the course, the students will be able to

- CO1 Use basic architectural principles in the design of buildings, interior spaces and sites
Understand the western architectural canons and traditions in architecture, landscape and urban design as well as the climatic, technological, socioeconomic and other cultural factors that have shaped and sustained them
- CO2
- CO3 Read architectural drawings and make technically precise drawings.

Course Articulation Matrix:

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	-	2	-	1	2	1	2	2	2	1	1
CO2	-	-	3	-	-	3	3	2	2	-	1	2	-	1	3
CO3	3	2	3	2	2	1	-	-	2	1	2	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Daniel E. Williams., Sustainable Design: Ecology, Architecture, and Planning, John Wiley & Sons, 2007.
2. Charles George Ramsey, Harold Reeve Sleeper, Architectural Details: Classic Pages from Architectural Graphic Standards 1940 – 1980, Wiley, 2001.
3. Louis Rouillion, Charles George Ramsey, Architectural Details: Classic Pages from Architectural Graphic Standards 1940 – 1980, J. Wiley, 1924.
4. Edward Allen, Patrick Rand, Architectural Detailing: Function, Constructibility, Aesthetics, John Wiley & Sons, 2016.

Reference Books:

1. Rangwala S. C., Building Construction, Charotar Publishing House, Anand.

LIST OF TUTORIALS

Sr. No.	Details
1	Internet Case studies of sustainable buildings like Suzlon One Earth, Pune, ZED housing, Bangalore etc. (Group work)
2	Study of green building rating systems like GRIHA, TERI, LEED etc. (Individual Work)
3	Study of materials that can bring sustainability like solar panels, wind mills etc. (Group work)
4	Case study and model making on: (Group Work) A. DOORS: Types of doors, its use, materials and joinery details B. EXTERIOR DETAILS: Type of paints finishes, alco panels, aluminum panels, glass etc. C. MASONRY: Brick masonry detail and types of brick bonds D. METALS: Types of metals and its use in building construction E. INTERIOR STAIRCASES: Study and analysis of staircases used in interior spaces F. EXTERIOR STAIRCASES: Study and analysis of staircases used in exterior spaces F. ELECTRICAL SYSTEMS: Electric conducting systems, types of conduits used in construction and interior designs
5	To make a detail report on one building/mall that depicts all the Architectural facilities like; HVAC (Heating Ventilation and Air Conditioning), AHU (Air Handling Units), Fire safety

	measure and fire Exits, Plumbing and Sanitation, Structural layout, Mechanical Services, Water supply services etc. (Group work)
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CL375: BUILDING SERVICES
B. TECH 5th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Building Services	5
2	Plumbing Services in Buildings	8
3	Mechanical Services in Buildings	10
4	Illumination, Fire Protection, Acoustic and Sound Insulations	12
5	Heating, Ventilation and Air Conditioning (HVAC)	10

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

B. Detailed Syllabus:

- | | | | |
|-----------|---|-----------------|------------|
| 1. | Introduction to Building Services | 05 Hours | 11% |
| 1.1 | Definitions, Objective and uses of services | | |
| 1.2 | Applications of services for different types building considering | | |
| 1.3 | Classification of building services | | |
| 1.4 | Types of services and selection of services | | |
|
 | | | |
| 2 | Plumbing Services in Buildings | 08 Hours | 18% |
| 2.1 | Elements of plumbing | | |
| | Objectives of plumbing, purpose of plumbing, role of plumber, licensing of plumbers their functions, sewer Air, supply pipes, drainage & vent pipes application for obtaining supply connection | | |

2.2 Pipes joints & fittings

Introduction. Types of Pipe – G.I. Pipes, PVC Pipes, Copper pipes, C.I. Pipes, A.C. Pipes, prestressed concrete pipes, joints in pipes, method of fixing pipes such as G.I. fitting C.I. fitting

2.3 Valves & Terminal Fittings

Types of valves & its purpose, sluice valve, reflux valve, scour valve, Air relief valve, pressure relief valve, gate valves, Bio-taps & stop valve self-closing valve. Flush valve, mixing valve

2.4 Sanitary fixture & Building drainage system

Building sanitary fittings – water closet, flushing appliances, urinals, washbasins, flushing cisterns, principles of building drainage, siphonic action, traps & its types, capacity & sizing of pipe, soil pipe, waste pipe, rain water pipe, system of plumbing, Installation of pipes, testing of pipes

3. Mechanical Services in Buildings

10 Hours

22%

3.1 Introduction of mechanical services

3.2 Lifts: Definition, Types of Lifts, Design Considerations, Location, Sizes, Component parts- Lift Well, Travel, Pit, Hoist Way, Machine, Buffer, Door Locks, Suspended Rope, Lift Car, Landing Door, Call Indicators, Call Push

3.3 Elevators & Escalators: Different types of elevators and Escalators, Freight elevators, Passenger elevators, Hospital elevators, Uses of different types of elevators Escalators

3.4 Pumps: Reciprocating, Centrifugal, Deep well, Submersible, Automatic pumps, Sewerage pumps, Compressors, Vacuum pump – their selection, installation and maintenance

3.5 Conveyors: Different types of Conveyors, Uses of different types of Conveyors

4. Illumination, Fire Protection, Acoustic and Sound Insulations

12 Hours

27%

4.1 Natural and artificial lighting- principles and factors, Arrangement of luminaries, Distribution of illumination, Utilization factors

4.2 Causes of fire and Effects of fire, General Requirements of Fire

Resisting building as per IS and NBC 2005., Characteristics of Fire resisting materials, Maximum Travel Distance, Fire Fighting Installations for Horizontal Exit, Roof Exit / Fire Lifts, External Stairs

- 4.3 Requirement of good Acoustic, Various sound absorbers, Factors to be followed for noise control in residential building

5. Heating, Ventilation and Air Conditioning (HVAC) 10 Hours 22%

- 5.1 Behaviour of heat propagation, thermal insulating materials and their co-efficient of thermal conductivity. General methods of thermal insulation: Thermal insulation of roofs, exposed walls
- 5.2 Ventilation: Definition and necessity, Principles, Temperature Control, Air Velocity Control, Humidity Control, Air Distribution system, Cleaners, Filters, Spray washers, Electric preceptors, Types of Air Conditioners, (Central type, Window Type, Split Unit)

C. Course Outcomes:

On successful completion of this course

- CO1 The students will acquire competencies to plan various types of services required for different types of building.
- CO2 Students will be able to manage building services provisions in big construction sites.
- CO3 Students will be able to synchronize the construction activities with installation of building services.
- CO4 Students will learn the needs of a building occupant in an enclosed built environment regarding; thermal comfort, indoor air quality, fire safety, electric usage and wet area usage.
- CO5 Students will gain technological solution alternatives and knowledge of equipment in the market for ventilation, heating, cooling techniques, building service systems.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	-	-	2	-	-	-	-	-	-	-	-	-	2	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	1	2	-	1	-	-	-	-	-	-	-	-	-	-	1
CO5	1	-	-	-	-	-	-	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Building Services by S. M. Patil, Seema Publication, Mumbai, Revised edition.
2. Building Construction by Dr. B. C. Punmia, Laxmi Publications (P) Ltd., New Delhi.
3. Water supply and Sanitary Engineering by S. K. Garg, Delhi.

Reference books

1. A text book on Building Services by R. Uday kumar. Eswar Press, Chennai.
2. Building Construction by P. C. Varghese PHI Learning (P) Ltd., New Delhi.
3. Elements of Public Health Engineering, by K. N. Duggal, New age international.
4. Building repair and Maintenance Management by P. S. Gahlot CBS Publishers & Distribution (P) Ltd.
5. Building Construction by Charotar publishing House, Anand.
6. A to Z of practical building and its management by Sandeep Mantri, Mantri Institute of Development & Research, Pune.
7. Building Systems for Interior Designers by Corky Binggeli, John Wiley & Sons, 2003.

Web Material:

1. www.academia.edu
2. www.nptel.iitm.ac.in
3. [http://en.wikipedia.org/w/index.php?title=Dumbwaiter \(elevator\)&oldid=62176813](http://en.wikipedia.org/w/index.php?title=Dumbwaiter_(elevator)&oldid=62176813)
4. www.bis.org.in/sf/nbc.htm
5. cpwd.gov.in/Units/handbook.pdf
6. <http://www.civilengineeringnews.tk/2014/07/methods-of-demolition-of-building.html>
thecontractor.org

LIST OF PRACTICALS / EXERCISE

Tutorials	Tutorial 1	Introduction to building services
	Tutorial 2	Draw diagrams of various plumbing fixtures.
	Tutorial 3	Identify proper locations for Lift/ Escalator/ Elevator in a given commercial complex
	Tutorial 4	Prepare a case study for the firefighting services for commercial building.

	Tutorial 5	Suggestions for noise control methods for a given commercial complex
Seminar	Seminar on course related topic to be given to a group of students not more than three. Students have to present/ defend the seminar in front of teachers and students. This work should be accompanied in the submission	
Site visit	Visit a residential building & commercial building under construction and study the services like mechanical, plumbing, HVAC, Fire safety, acoustics and related allied services of civil engineering and prepare site visit detailed report	

CL381: INFRASTRUCTURE MANAGEMENT
B. TECH 5th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction and Infrastructure scenario	04
2	An overview of Basic Concepts Related to Infrastructure: Urban Infrastructure	10
3	Strategies for Successful Infrastructure Project Implementation	09
4	Private Involvement in Infrastructure	06
5	Infrastructure Economics and Finance	05
6	Infrastructure Risk Management	06
7	Infrastructure Maintenance	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction and Infrastructure scenario	04 Hours	09%
1.1	Types of Infrastructure		
1.2	Role of Infrastructure		
1.3	Need and scenario of infrastructure		
1.4	Infrastructure crisis		
2	An overview of Basic Concepts Related to Infrastructure: Urban Infrastructure	10 Hours	22%

2.1	An overview of the Power Sector in India, Water Supply and Sanitation Sector in India, Road, Rail, Air and Port Transportation Sectors in India and Telecommunications Sector in India.		
2.2	An overview of the urban infrastructure in India, rural infrastructure in India		
2.3	Legislations pertaining to urban infrastructure		
2.4	State of urban and rural infrastructure		
2.5	Strategies to improve infrastructure in rural areas		
2.6	Stages of an Infrastructure Project Lifecycle		
3	Strategies for Successful Infrastructure Project Implementation	09 Hours	20%
3.1	Shaping the Planning Phase of Infrastructure Projects to mitigate risks		
3.2	Sustainable Contracts		
3.3	Introduction to Fair Process and Negotiation, Negotiating with multiple Stakeholders on Infrastructure Projects		
3.4	Sustainable Development of Infrastructure, Information Technology and Systems for Successful Infrastructure Management		
3.5	Capacity Building and Improving the Governments Role in Infrastructure Implementation		
4	Private Involvement in Infrastructure	06 Hours	11%
4.1	Overview		
4.2	Benefits		
4.3	Problems and challenges of infrastructure privatization		
4.4	Privatization of Infrastructure in India – Case Study		
5	Infrastructure Economics and Finance	05 Hours	18%
5.1	Principles of finance		
5.2	Infrastructure economics		
5.3	Introduction to project finance		
6	Infrastructure Risk Management	06 Hours	11%
6.1	Risks in infrastructure		
6.2	Quantitative risk analysis		

6.3 Qualitative risk management

6.4 Risk management strategies

7 Infrastructure Maintenance

05 Hours

9%

7.1 Introduction to Infrastructure maintenance

7.2 Need and requirement for Infrastructure maintenance

7.3 Preventive techniques for maintenance

C. Course Outcome (COs) :

On the successful completion of this course

CO1 Students will be able to understand an overview of basic concepts of infrastructure and its scenario.

CO2 Students will be able to describe the strategies for successful infrastructure project implementation.

CO3 Students will be able to explain the role of private sector in infrastructure growth.

CO4 Students will be able to know about infrastructure economics and finance.

CO5 Students will be able to identify and manage risks involved in infrastructure.

CO6 Students will be able to execute maintenance of infrastructure.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	-	-	-	-	1	-	-	-	-	-	-	1	-	-
CO2	2	2	1	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	1	-	-	-	1	-	-	-	-	-	-	1	-	-
CO4	-	-	-	-	-	-	-	-	-	-	2	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	-	2	-	2	-	-
CO6	1	-	1	-	-	-	-	-	-	-	-	-	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. J. Parkin and D. Sharma, Infrastructure planning, Thomas Telford
2. P. Chandra, Projects: Planning, analysis, selection, financing, implementation, and review, Tata McGraw-Hill
3. S. Goodman and M. Hastak, Infrastructure planning handbook: Planning, engineering, and economics, McGraw-Hill

Reference Books:

1. Ronald W Hudson, "Infrastructure Management: integrating design, Construction, maintenance, rehabilitation and renovation", MGH, 1st Edition, 1997
2. L. Squire and H. G. van der Tak, Economic analysis of projects, John Hopkins University Press, London, 1975.
3. J. D. Finnerty, Project financing - Asset-based financial engineering, John Wiley & Sons, New York, 1996.

Web Materials:

1. Ministry of Urban Development: <http://moud.gov.in/>
2. Indian Infrastructure Finance Company Limited: <http://www.iifcl.co.in/>
3. Ministry of Shipping: <http://shipping.nic.in/index.php>
4. Indian Ports Association: <http://www.ipa.nic.in/about.htm>
5. National Maritime Foundation: <http://www.maritimeindia.org>
6. India Maritime 2015: <http://www.indiamaritime.in>
7. <http://nptel.ac.in/courses/105106115/>

Term Work

Term work will be based on above syllabus with seminar/group project to be incorporated.

CL382: DISASTER PREPAREDNESS & PLANNING MANAGEMENT
B. TECH 5th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Disaster Management	6
2	Role of Remote sensing, Geographical Information System and Information Technology in Disaster Management	10
3	Risk Assessment and Vulnerability Analysis	10
4	Disaster Preparedness and Response	8
5	E-Governance and Emerging Technology in Disaster Management	6
6	Community Linkage in Disaster Management	5

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- 1 Introduction to Disaster Management** **6 Hours 13%**
- 1.1 Concepts of : Hazard, Risk, Vulnerability, Disaster
- 1.2 Natural and Manmade Disasters – Meaning, Types and effects
- Hydrological Disasters – Flood, Flash flood, Drought, Cloud burst, Cyclone, Storm, Storm surge, Tidal waves
- Geological Disasters – Earthquakes, Tsunamis, Landslides, Avalanches, Volcanic eruptions, Mudflow
- Climatic Change, Global Warming, Sea Level Rise, Ozone Depletion, Deforestation, Industrial Waste, Heat and Cold Waves

CBRN – Chemical disasters, Biological disasters, Radiological disasters, Nuclear disasters, Road accidents, Rail accidents, Air accidents, Sea accidents, Armed conflicts, Technological disasters

Fire – Building, Coal, Forest, Oil fire

Pollution – Air, Water, Soil pollution

1.3 Importance, Dimensions & Scope of Disaster Management

1.4 Disaster Management Cycle

1.5 Disaster and Development

2 Role of Remote Sensing, Geographical Information System and Information Technology in Disaster Management 10 Hours 22%

2.1 Definition of GIS, Concept of Space and Time, Spatial data, Map Projection and Datum, Domains of Spatial information system, Components of GIS (Hardware, Software, Data, People and Process), GIS Functionalities for end user / system (Data Acquisition, Data Input, Data Management, Data Analysis, Data Modeling and Data Output), Web based GIS Technology

2.2 Introduction to Remote Sensing, Fundamentals of Remote Sensing, Electromagnetic Radiation, Electromagnetic Spectrum, Energy interaction with Atmosphere, Energy interaction with Earth Surface, Platform and Sensors, Characteristics of Image, Image Interpretation and Analysis – Visual Image Interpretation & Digital Image Processing, Microwave Remote Sensing, Indian Remote Sensing Satellites

2.3 Importance of Information in Disasters :

Methods of collecting relevant information – libraries, internet, interviews questionnaires, survey, observation, Mass media, Meetings, Role of Information from disaster affected community

2.4 Organizing an effective dissemination of information: feedback for improving information, Role of Communication in Disasters, Types of communication in case of disasters – HAM radio, Satellite, Video Conferencing, Electronics devices

3 Risk Assessment and Vulnerability Analysis 10 Hours 22%

3.1 Risk Concepts, Elements Of Risk, Perception of Risk, Acceptable risk, Requirements in Risk assessment

3.2	Risk analysis techniques; Process of Risk assessment, Analytical systems for risk assessment, Natural hazard/ risk assessment, Understanding climate risk, Mapping of risk assessment, Decision making for risk reduction, Problems in risk assessment		
3.3	Observation and perception of vulnerability – Vulnerability Identification, Vulnerability types and dimensions, Vulnerability – Social factors and economic factors		
3.4	Strategic development for Vulnerability reduction : Physical & Social infrastructure for vulnerability reduction, Hazard resistant designs and construction, Systematic management and Strategic planning for vulnerability reduction		
4	Disaster Preparedness and Response	8 Hours	18%
4.1	Disaster Preparedness: concept and significance		
4.2	Essentials of Disaster Preparedness Plan		
4.3	Disaster Response Plan – Communication, Participation, and Activation of Emergency Preparedness Plan, Search, Rescue, Evacuation and Logistic Management		
4.4	Relief and Recovery, Psychological Response and Management (Trauma, Stress, Rumor and Panic)		
5	E-Governance and Emerging technology in Disaster Management	6 Hours	13%
5.1	E-Governance : Concept and Significance		
5.2	E-Governance in Urban and Rural Development		
5.3	ICT Implementation in Governance: Issues and Challenges		
6	Community Linkage in Disaster Management	5 Hours	12%
6.1	Community Based Disaster Management, Human Behavior and Response: Individual, Community, Institutional		
6.2	Community Participation and Awareness		
6.3	Community Health During Disasters : Drinking Water, Food and Nutrition, Hygiene and Sanitation		
6.4	Soft skills for Disaster Manager : Leadership and Coordination, Life skills, Time Management Skills		

C. Course Outcomes (COs):

On the completion of the course one should be able to:

- CO1 To become familiar with definitions and terms used within the disciplines of disaster preparedness and emergency management.
- CO2 To understand the scope and extent to which disasters and emergencies influence the social and political environment.
- CO3 To know the scope of modern technologies in disaster management
- CO4 To understand planning, mitigation, in-crisis decision making, evacuation, response, and recovery within the leadership during an event.
- CO5 To illustrate knowledge about development of viable plans and appropriately consider management options for disaster planning and emergency management.

Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	-	1	1	1	-	1	1	-	-	1	1	2	1	1	-
CO2	-	1	1	1	-	2	1	-	1	-	1	3	1	-	-
CO3	-	1	1	2	2	2	2	-	-	-	1	2	1	1	1
CO4	1	3	2	1	2	2	3	-	1	3	-	2	1	1	2
CO5	2	2	2	1	-	3	2	1	1	1	3	3	2	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Singh Jagbir, Disaster Management Future challenges and opportunities, I K International Pub.
2. Pandey Mrinalini, Disaster Management 1st ed. Wiley India
3. Joseph George, Fundamentals of Remote Sensing, Universities Press.

Reference Books:

1. Natural Disaster Hotspots A Global Risk Analysis, The World Bank
2. Sahni Pardeep, Disaster Risk Reduction in South Asia, Prentice Hall of India.
3. Gupta Harsh K, Disaster Management, Universities Press
4. Modh Satish, Introduction to Disaster Management, Macmillan Pub
5. Sensarma Suman Ranjan, Disaster Risk Management Conflict And Cooperation, Concept Publishing Company
6. Modh Satish, Citizens Guide to Disaster Management, Macmillan Pub.
7. Panchal Priyank K., State Disaster Resource Network, Macmillan Pub.

8. Shah Riddhi, State Disaster Resource Network (UNDP), Macmillan Pub.
9. Sulphrey M M. Disaster Management, PHI Learning
10. Sharma Sanjay K., Environment Engineering and Disaster Management, University Science

Web Materials:

1. Geneva: Sphere Project. <https://www.spherestandards.org/handbook-2018/>
2. <https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030>
3. <https://gidm.gujarat.gov.in/>
4. <https://ndma.gov.in/en/>
5. <https://www.undrr.org/>

List of Tutorials

Experiment No.	Name of Tutorial
1	Role of Engineers in Disaster Management
2	Study of Recent Disasters : at state level, preparation of Disaster Risk Management Plan of an Area or Sector
3	Study of Recent Disasters : at national level, preparation of Disaster Risk Management Plan of an Area or Sector
4	Study of Recent Disasters : at international level, preparation of Disaster Risk Management Plan of an Area or Sector
Global-National-State level Case Studies in Disaster Management	
1	Japan's Tohoku Earth Quake (2011) and Nepal Earthquake (2015)
2	East Africa Drought (2011)
3	Indian Ocean Earthquake (2004)(Tsunami) and Gujarat Earthquake (2001)
4	Uttarakhand Flash Floods and Kashmir Floods
5	Drought Management in Gujarat & Rajasthan
6	Landslides in Shiwalik Hills Case Study
7	Avalanches in Jammu and Kashmir: Case Studies
8	Pukhrayan train derailment 20 November 2016 and Bhopal Gas Tragedy
9	Case studies as per the interest and suggestions from the students
10	Case studies as per the interest and suggestions from the students

B. Tech. (Civil Engineering) Programme

SYLLABI (Semester – 6)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CL35I: ENVIRONMENTAL ENGINEERING- I
B. TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	1	7	5
Marks	100	50		150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Environmental Engineering	02
2	Water Demands and Sources of Water	06
3	Water Conveyance and Distribution Systems	08
4	Quality and Treatment of water	14
5	Sewage: Quality, Collection and estimation of discharge	06
6	Unit operations/ processes for sewage treatment	10
7	Design of sewage treatment units	10
8	Plumbing and House Drainage	04

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

1	Introduction to Environmental Engineering	02 Hours	04%
1.1	Scope of Environmental Engineering, Environmental Ethics and Role of Environmental Engineer		
1.2	Necessity of planned water supply and Sewerage System		
2	Water Demands and Sources of Water	06 Hours	10%
2.1	Various types of water demands & factors affecting per capitademand, Sources of water, Impurities in water		
2.2	Variations in Demand & Design Periods		

2.3	Population Forecasting Methods, Examples		
3	Water Conveyance and Distribution Systems	08 Hours	13%
3.1	Water Intake Structures-Types and design		
3.2	Types of Conduits, Flow in pipe systems and types of pressure pipes		
3.3	Distribution System-Requirements, layouts and methods		
3.4	Distribution Reservoirs-functions, types and determination of storage capacity		
4	Quality and Treatment of water	14 Hours	23%
4.1	Water Quality Parameters and Standards for drinking water		
4.2	Layout of water treatment plant, Design period, and factors considered for selection of design period. Treatment plant site selection, factors considered, selection of treatment train		
4.3	Plain sedimentation, aeration, sedimentation tank & its design, sedimentation with coagulation, types of coagulants, optimum dose of coagulants, mixing devices, design of flocculation unit. theory of filtration, types of filters and their comparison, design of rapid sand filter, washing of filter, methods of disinfection, methods of removing hardness Computation of dose of chemicals for removal of hardness		
5	Sewage: Quality, Collection and estimation of discharge	06 Hours	10%
5.1	Characteristics - physical, chemical, biological		
5.2	Standards for effluent disposal & receiving water body		
5.3	Different types of sewers, sewerage systems, variation in sewage flow, sewer appurtenance, estimation of wastewater discharge in a sewer, separate and combined sewerage systems and design of sewers		
6	Unit operations/ processes for sewage treatment	10 Hours	17%
6.1	Layout plan and section of sewage treatment plant		
6.2	Physical unit operation screening, flow equalization, mixing, flocculation, sedimentation		
6.3	Chemical unit processes: chemical precipitation		

6.4 Biological unit processes: Aerobic attached growth and aerobic suspended growth treatment processes, anaerobic suspended growth treatment processes low cost sanitation systems, septic tanks, soak pit, stabilization ponds

7 Design of sewage treatment units

10Hours

17%

7.1 Design of racks, screens, grit chamber, aeration units, primary & secondary clarifiers, activated sludge plant and trickling filter units, rotating biological contactors

7.2 Sludge dewatering units, sludge digesters and drying beds

8 Plumbing and House Drainage

04Hours

06%

8.1 Principles of Plumbing and House Drainage, House drainage plan for residential building

8.2 Types of Traps

C. Course Outcomes:

On the completion of the course one should be able to understand

CO1 The water quality criteria and standards with their relation to public health and environment.

CO2 Water and sewage characteristics and their relation with treatment process selection.

CO3 To understand the design principles involved in treatment of water and sewage

CO4 To apply appropriate breadth and depth of skills in identification of engineering problems designed with realistic constraints and contribute to sustaining and improving community.

CO5 To improve written communication and design skills by preparing a preliminary design report of water and sewage treatment plant units.

CO6 The fundamentals of plumbing and house drainage system.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-		1
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-		1
CO3	3	3	3	-	-	-	2	-	-	-	3	-	3	2	1
CO4	-	3	3	2	1	3	3	3	2	-	3	3	3	2	3

CO5	-	-	-	2	1	1	2	2	3	3	-	3	-	-	3
CO6	3	-	-	-	-	-	1	1	-	-	1	-	3	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Book:

1. Garg, S.K., Environmental Engg. Vol. – I & II, Khanna Publications.
2. Punmia, B.C., Environmental Engg. Vol. - I & II, Laxmi Publications.
3. Peavy, Rowe and Tchobanoglous, Environmental Engg, Tata McGraw Hill.

Reference Books:

1. Bhole, A.G., Water Treatment Plant Design, Indian Water Works Association.
2. Dix, H.M., Environmental Pollution, Edward Arnold Publishers Ltd.
3. Mackenzie L. Davis, David, A. Cornwell, Introduction to Environmental Engineering, TataMcgraw Hill Publications.
4. Sanitary Engg. and Sewage Treatment, Manual, Ministry of Works & Housing, New Delhi
5. Chaterjee, A.K., Environmental Engg, Khanna Publishers.
6. Steel, E.W. and McGhee, T.J., Water Supply & Sewerage, McGraw-Hill College.
7. Water Supply and Treatment, Manual, Ministry of Works and Housing, New Delhi.
8. Wastewater Engineering Treatment, Disposal, Refuse: Metcalf and Eddy, Tata McGrawHill Publishers, New Delhi, 1995.
9. Introduction to Environmental Engineering :P. Aarne Vesilind, Susan M. Morgan, Thompson /Brooks/Cole; Second Edition
10. CPHEEO Manual on Sewage and Treatment

Web Materials:

1. <http://nptel.iitm.ac.in>
2. <http://www.epa.gov>
3. <http://www.nesc.vu>
4. <http://www.indiaenvironmentportal.org.in>
5. <http://www.filtersource.com>
6. <https://dgserver.dgsnd.gov.in>

List of Experiments

Experiment No.	Name of Experiment
A	Analysis of Water and Sewage Samples
1.	Determination of pH, Turbidity and Conductivity
2.	Determination of Solids
3.	Determination of Acidity and Alkalinity
4.	Determination of Hardness
5.	Determination of Chlorides and Residual Chlorine
6.	Determination of Dissolved Oxygen (DO) and Bio chemical Oxygen Demand (BOD)
7.	Determination of Chemical Oxygen Demand (COD)
8.	Determination of Oil and Grease
9.	Determination of Optimum Coagulant Dose by Jar Test
10.	Determination of Nitrate and Sulphate
11.	Determination of Most Probable Number (MPN)
B.	Design Problems on Water and Sewage Treatment Plant Units

CL346: GEOTECHNICAL ENGINEERING-I
B. TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	02
2	Origin & Evolution of Soil	03
3	Index Properties & Plasticity Characteristics	07
4	Soil water	06
5	Compaction	08
6	Shear Strength	10
7	Consolidation	09

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction	02 Hours	06%
1.1	Historical Development of Soil Engineering		
1.2	Importance of Soil Engineering		
1.3	Major soil deposit in India		
2	Origin & Evolution of Soil	03 Hours	10%
2.1	Origin and General types of soils		
2.2	Geological cycle, physical and chemical agencies for soil		
2.3	Soil structure & Clay minerals		
3	Index Properties & Plasticity Characteristics	07 Hours	15%
3.1	Three phase diagram of soil		

3.2	Volumetric relationship and Weight – Volume relationships		
3.3	Determination of different index properties		
3.4	Grain size distribution by sieve analysis, Hydrometer analysis		
3.5	Atterberg's Limits and indices		
3.6	Activity, sensitivity & thixotropy of soil		
3.7	Different classification systems: Textural Classification, AASHTO classification, Unified soil classification, IS classification		
3.8	Field Identification of soil		
4	Soil water	06 Hours	14%
4.1	Types of soil water, capillary phenomena		
4.2	Concept of effective and neutral stresses		
4.3	Permeability of soil, Darcy's law, Factors affecting permeability		
4.4	Seepage velocity, Permeability of Stratified soil		
4.5	Tests for determination of coefficient of permeability in field and laboratory		
4.6	Seepage pressure, quick sand phenomenon and piping		
4.7	Introduction of flow net		
5	Compaction	08 Hours	16%
5.1	Definition and importance, factors affecting compaction, Influence of compaction on soil properties		
5.2	Laboratory compaction test, relative density and its determination		
5.3	Field compaction and its control,		
6	Shear Strength	10 Hours	20%
6.1	Definition, Mohr strength theory, Mohr-coulomb's strength theory		
6.2	Shear test based on drainage condition		
6.3	Factors affecting shear strength of granular and cohesive soil		
6.4	Direct shear test, Triaxial compression test, unconfined compression test, vane shear test		
7	Consolidation	09 Hours	19%
7.1	Compressibility of soil & its type, mechanism of consolidation		
7.2	Assumptions and Terzaghi's one dimensional consolidation theoretical equation		
7.3	One dimensional consolidation test, Co-efficient of consolidation and		

its determination, determination of pre-consolidation pressure

7.4 Consolidation settlement, rate of settlement for uniform pressure increment in a clay layer

7.5 Introduction to secondary compression

C. Course Outcomes (COs):

At the end of the course, the students will be able to

- CO1 students will be able to evaluate soil properties by performing various experiments
- CO2 students will be able to carry out soil classification
- CO3 students will be able to solve practical problems related to permeability, seepage and consolidation settlement,
- CO4 Students will be able to relate soil properties to workout theoretical soil strength with confidence.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	2	2	-	-	-	-	-	-	-	-	-	1	1	-
CO2	1	2	1	-	-	-	-	-	-	-	-	-	1	-	-
CO3	2	2	1	-	-	-	-	-	-	-	-	-	1	1	-
CO4	2	2	1	-	-	-	-	-	-	-	-	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Arora, K.R., Soil Mechanics & Foundation Engineering, Standard Publication, New Delhi.
2. Punamia, B.C., Soil Mechanics & Foundation Engineering; Laxmi Publication Pvt. Ltd., Delhi.
3. Murthy, V.N.S., Soil Mechanics & Foundation Engineering, Sai Kripa Technical Consultants, Bangalore.
4. Shroff, A. V., Shah D. L., Soil Mechanics & Geotechnical Engineering, Oxford & IBH, Delhi.

Reference Books:

1. Singh Alam, Soil Engineering, Agion Publishers, Jodhpur.
2. Purshottam Raj, Geotechnical Engineering, Tata McGraw Hill Publication.
3. Purushothama, P. Raj, Soil Mechanics and Foundation Engineering, Pearson Education.

4. Singh, Alam, Soil Mechanics & Foundation Engineering, CBS Publishers & Distributors, New Delhi.
5. Taylor, D.W., Fundamentals of Soil Mechanics, Asia Publishing House, Mumbai.
6. Ranjan Gopal and Rao, A.S.R., Basic and Applied Soil Mechanics, New Age International Pvt. Ltd.
7. Braja Das, M., Principles of Geotechnical Engineering, Thomson Asia Pvt. Ltd.

Web Materials:

1. <http://edudel.nic.in>
2. <http://bis.org.in/other/quake.htm>
3. <http://www.thepeninsulaneighborhood.com/ThePlan.html>
4. http://www.historytution.com/indus_valley_civilization/town_planning.html

LIST OF EXPERIMENTS

Experiment No.	<u>Name of Experiment</u>
1	Moisture Content by Oven-drying Method
2	Specific Gravity by Pycnometer Method
3	Sieve Analysis
4	Hydrometer Analysis
5	Liquid Limit & Plastic Limit Tests
6	Shrinkage Limit Test
7	In Situ Density by Core Cutter Method
8	In Situ Density by Sand Replacement Method
9	Direct Shear Test

CL352: WATER RESOURCES ENGINEERING - I
B. TECH 6TH SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	1	7	5
Marks	100	50		150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Water Resources Engineering	01
2	Surface water Hydrology	12
3	Groundwater Hydrology	08
4	Irrigation & Crop Water Requirement	03
5.	Dams	12
6.	Reservoir	04
7.	Diversion Head Works	06
8.	Spillways & Energy Dissipators	05
9.	Canal Irrigation System	09

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1 | Introduction to Water Resources Engineering | 01 Hours | 2% |
| 1.1 | Need of water resources projects | | |
| 1.2 | Hydrologic cycle - scope and application, Hydrologic Equation (Water Balance Equation) | | |
| 2 | Surface water Hydrology | 12Hours | 20% |
| 2.1 | Precipitation: Types, forms, measurement, estimation of missing data, mean rainfall computation | | |

2.2	Evaporation: Process, factors affecting, measurement of evaporation		
2.3	Evapotranspiration: Transpiration, evapotranspiration, factors affecting, measurement of evapotranspiration		
2.4	Infiltration: Process, infiltration rate, infiltration capacity, infiltration indices, measurement - infiltrometers		
2.5	Runoff: Types, factors affecting, estimating volume of runoff (yield) – rainfall runoff correlation		
2.6	Hydrograph: Components, unit hydrograph, S-Hydrograph, computation of flood		
3	Groundwater Hydrology	08 Hours	13%
3.1	Occurrence and movement of groundwater, Darcy's law, Types of aquifers, Aquifer properties		
3.2	Hydraulics of wells under steady & introduction to unsteady condition in confined and unconfined aquifers yield of wells, pumping and recuperation tests ⁴		
3.3	Needs of ground water recharge, ground water recharge methods, Rain water harvesting		
3.4	Occurrence and causes of saline water intrusion, Ghyben-Herzberg Relation, Upconing of saline water, Control of saline water intrusion		
4	Irrigation & Crop Water Requirement:	03 Hours	5%
4.1	Irrigation: Definition, merits, demerits		
4.2	Irrigation methods: Detailed classification		
4.3	Duty of water and delta, factors affecting on duty and delta, relation between duty-delta & examples		
4.4	Irrigation efficiencies and calculation of efficiencies		
5	Dams	12 Hours	20%
5.1	Definition, classification, factors affecting selection of type of dam, Selection & investigation of site for a dam		
5.2	Earth dams: Types, foundation requirements, causes of failure, design criteria, suitable preliminary sections, seepage line determination & analysis, stability analysis - side slope, foundation, checking of stability, seepage control measures		
5.3	Rockfill dams: Foundation requirements, typical sections, design		

consideration for a rockfill dam		
5.4	Gravity dam: Definition, forces acting on dams, load combination for design, modes of failure & criteria for structural stability, stability analysis, elementary & practical profile, design of gravity dam, openings in dams	
6	Reservoir	04 Hours 7%
6.1	Definition, types, Storage zones	
6.2	Capacity-elevation and area elevation curves of a reservoir site	
6.3	Reservoir capacity: Catchment yield and reservoir yield, fixing the reservoir capacity for the computed value of the dependable yield of the reservoir catchment, relation between the inflow, outflow and storage data for a reservoir, fixing the reservoir capacity from the annual inflow and outflow data, mass curve and demand curve	
6.4	Reservoir losses, Reservoir sedimentation and control	
7	Diversion Head Works	06 Hours 10%
7.1	Types & components of diversion head works	
7.2	Location & typical layout of diversion headworks	
7.3	Difference between dam, weir and barrage, Types of weirs	
7.4	Criteria for designing weir and barrage	
7.5	Theories of subsurface flow: Bligh's creep theory, Lane's weighted creep theory, theory of seepage flow (critical gradient), Khosla's theory	
8	Spillways & Energy Dissipators	05 Hours 8%
8.1	Spillways: Definition, location, essential requirements, components, classification	
8.2	Energy Dissipators: Formation & types of hydraulic jump; Jump Height Curve (JHC), Tail Water Rating Curve (TWRC) and alternatives of JHC & TWRC, types of energy dissipators	
9	Canal Irrigation System	09 Hours 15%
9.1	Classification of canals, alignment	
9.2	Distribution system	
9.3	Cross section and longitudinal sections of canals	
9.4	Design of unlined canal in alluvial & non-alluvial soils	

- 9.5 Design of channel using Kennedy's Garrat's diagram & Lacey's regime diagram, Canal lining
- 9.6 Cross Drainage Works: Definition, types, selection criteria
- 9.7 Outlets: Definition, requirements, types
- 9.8 Falls: Definition, necessity, location, types
- 9.9 HR & CR: Definition, functions
- 9.10 Escapes: Definition, necessity, types

C. Course Outcomes:

On the completion of the course one should be able to:

- CO1 Understand Concepts of different hydraulic structures
- CO2 Plan and design irrigation projects.
- CO3 Design channels and other irrigation structures required for irrigation, drainage, soil etc.
- CO4 Estimate the quantity of water required by crops.
- CO5 Understand types of irrigation systems and different hydraulic structures
- CO6 Develop Conservation, flood control and other water-management projects.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2	-	-	3	3	2	3	3	3	-	3	3	3	3	2	3
CO3	-	-	3	3	2	3	3	3	-	3	2	3	3	2	3
CO4	3	2	-	-	-	3	-	-	-	-	-	2	2	-	1
CO5	-	3	-	-	-	-	-	-	-	-	-	-	1	-	-
CO6	-	-	3	3	2	3	3	3	2	3	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Subramanya, K., Engineering Hydrology, Tata McGraw Hill, New Delhi.
2. Raghunath, H.M., Groundwater, 1987, Wiley Eastern Ltd., New Delhi.
3. Garg, S.K., Irrigation Engineering and Hydraulic Structures, Khanna Publishers, New Delhi.

Reference Books:

1. Garg, S.P., Groundwater and Tube Wells, 1993, Oxford & IBH Publishing Co.
2. Modi, P.N., Irrigation Water Resources and Water Power Engineering, Standard Book House, New Delhi.
3. Raghunath, H.M., Hydrology – Principles, Analysis and Design, 1986, Wiley Eastern Ltd.
4. Todd, D.K., Groundwater Hydrology, 1993 John Wiley & Sons.
5. Karanth, K.R., Ground Water Assessment Development and Management, Tata McGraw Hill, New Delhi.
7. Patel, A.S. and Shah, D.L., Water Management – Conservation, Harvesting & Artificial Recharge, New Age International Publishers.
9. Punmia and Pando, Lal, B.B., Irrigation and Water Engineering, Standard Publishers Distributors, New Delhi.
10. Michael, A.M., Irrigation - Theory and Practice, Vikas Publishing House, New Delhi.
11. Arora, K.R. Irrigation, Water Power and Water Resources Engineering, Standard Publishers Distributors, New Delhi.

Web Materials:

1. <http://nptel.iitm.ac.in>
2. http://www.groundwatermanagement.org/module2_000.pps
3. <http://www.uiowa.edu/~c012003a/14.%20Groundwater.pdf>
4. <http://www.authorstream.com/presentation/brod-17752-lect-18-groundwater->
5. <http://www.ngwa.org/public/gwbasics/index.aspx>

CL353: TRANSPORTATION ENGINEERING- I
B TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	1	7	5
Marks	100	50	0	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	04
2	Traffic Engineering	10
3	Highway Geometric Design	10
4	Highway Materials and Construction	09
5	Highway Pavement Design	13
6	Highway Drainage, Maintenance, Economics and Arboriculture	10
7	Overlay Design and Construction	04

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

- 1 Introduction 04 Hours 06%**
- 1.1 Role and modes of Transportation, Scope of Transportation Engineering
 - 1.2 Highway Planning and Development in India
 - 1.3 Classification of Roads, Highway Alignment and Surveys: Introduction, Ideal Requirement of Alignment, Factors controlling Alignment, Engineering Surveys, Drawing and Reports, Highway Project

2	Traffic Engineering	10 Hours	18%
2.1	Introduction to Traffic Engineering		
2.2	Traffic Surveys: Traffic Volume Studies, Speed Studies, Traffic Density & Capacity, Relations between traffic parameters, Level of service, Passenger car unit, Parking studies, Accident Studies		
2.3	Traffic Operations: Traffic Regulations, Traffic Control Devices (Signs, Signals and Markings) with its design		
2.4	Highway Intersection: At grade & Grade separated intersections, Rotary intersection, IRC Recommendations , Highway Lightning		
3	Highway Geometric Design	10 Hours	16%
3.1	Introduction		
3.2	Highway Cross-section Elements		
3.3	Sight Distance		
3.4	Design of Horizontal Alignment		
3.5	Design of Vertical Alignment		
4	Highway Materials and Construction	04 Hours	14%
4.1	Introduction		
4.2	Aggregates and Test on Aggregates		
4.3	Bitumen and Test on Bitumen		
4.4	Test on Soil: CBR Test		
4.5	Bituminous Materials: conventional and modified binders, production, types and grade , physical and chemical properties and uses Modified bitumen: Crumb Rubber Modified bitumen, Natural rubber modified bitumen, polymer modified bitumen Introduction to emulsified bitumen and its characterization; Long term and short term ageing and its effect on bitumen performance		
4.6	Bituminous Mix design ,Desirable properties of bituminous mixes, Modified Marshall"s specifications, Introduction to super pave mix design procedure		
4.7	Highway Construction: Construction of Earth Roads, Gravel Roads, Water Bound Macadam Roads, Bituminous Pavement and Cement		

	Concrete Pavement		
5	Highway Pavement Design	13 Hours	22%
5.1	Introduction		
5.2	Types of Highway Pavement: Flexible pavement & rigid pavement, their components & its functions.		
5.3	Design Factors		
5.4	Design of Flexible Pavement: CBR Method, IRC Guidelines		
5.5	Design of Rigid Pavement: IRC method, Westergaard theory, loads & temperature stresses.		
6	Highway Drainage, Maintenance, Economics and Arboriculture	10 Hours	16%
6.1	Highway Drainage: Introduction, Importance of Highway Drainage, Surface Drainage, Sub-surface Drainage, Drainage of slopes and erosion controls		
6.2	Highway Maintenance: Pavement Failures, Maintenance Techniques, Strengthening of existing pavements.		
6.3	Highway Economics and Finance: Introduction, Methods of Economic Evaluation of highway projects: Annual cost method, Rate of return method & benefit- cost ratio method, Highway Project administration and Finance: Public Private Partnership		
6.4	Highway Arboriculture: Environmental Factors affecting highway planning, Road Side Development, Planning Plantation of trees, Species and their selection, Care of trees		
7	Overlay Design and Construction	04 Hours	08%
7.1	Need, Types, Overlay design methods for flexible pavements by conventional design method, Benkelman beam method, Falling Weight Deflection Method, I.R.C. Guidelines & Asphalt Institute Method.		
7.2	Overlay Design Methods for Rigid Pavements, Flexible overlays over rigid slabs		

C. Course Outcomes:

The course content should be taught and learning imparted with the aim to develop required knowledge and skills so that they are able to acquire following competency:

CO1	To design highway pavement geometrics
CO2	To understand the procedure to collect the traffic data for design and the need of traffic management
CO3	Test the highway material as per IS recommendations
CO4	Carryout preliminary design of flexible and rigid pavement as per IRC
CO5	To understand the concept for providing highway drainage, maintenance and arboriculture.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	1	2	2	2	1	-	-	2	1	-	-	2	1	1
CO2	2	1	1	3	2	1	-	-	2	2	1	-	2	1	2
CO3	1	-	-	3	1	1	1	1	2	2	-	-	1	1	1
CO 4	3	1	1	3	2	-	1	1	2	1	-	1	2	1	3
CO5	-	-	1	1	-	1	2	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Khanna, S.K. & Justo, C.E.G., Highway Engineering, NemChand & Bros, Roorkee (U.A).
2. Kadiyali, L.R., Traffic Engineering & Transport Planning, Khanna Publishers, New Delhi.
3. Kadiyali, L.R. & Lal, N.B., Principles & Practices of Highway Engineering, Khanna Publishers, New Delhi.

Reference Books:

1. Sharma, S.K., Principles, Practice and Design of Highway Engineering, S. Chand & Co., New Delhi.

Web Materials:

1. <http://www.cdeep.iitb.ac.in/nptel/Civil%20Engineering/Transportation%20Engg%20I/TOC.htm>

Other Materials:

1. IRC – 37 “Guidelines for Design of flexible Pavements”, IRC, New Delhi, 2001.
2. IRC – 67 “Code of Practice for Road Signs”, IRC, New Delhi – 2001.
3. IRC: 58, 2002: “Guidelines for the Design of Plain Jointed Rigid Pavements for Highways”, IRC, N. Delhi, December, 2002.
4. IRC:70, 1977: “Guidelines on Regulation and Control of Mixed Traffic in Urban Areas”
5. IRC:106, 1990: “Guidelines for Capacity of Urban Roads in Plain Areas”
6. IRC SP 30: “Manual of Economic Evaluation of Highway Projects in India”, IRC, New Delhi, 2009.
7. IRC SP 41: “Guidelines for the Design of At-Grade Intersections in Rural and Urban Areas”, IRC, New Delhi, 1994.
8. Khanna S. K. & Justo C. E. G., “Highway Material Testing (Laboratory Manual), Nem Chand & Bros, Roorkee
9. IRC 81: Guidelines for strengthening of flexible Road Pavement using Benkleman Beam Deflection Technique.

LIST OF EXPERIMENTS

Experiment No.	Name of Experiment
Test on Subgrade Soil	
1	California Bearing Ratio Test
Tests on Aggregates	
2	Aggregate Crushing Test
3	Aggregate Impact Test
4	Los Angeles Abrasion Test
5	Shape Tests
6	Specific Gravity & Water Absorption Test

Tests on Bitumen	
7	Penetration Test
8	Ductility Test
9	Flash & Fire Point Test
10	Softening Point Test
11	Specific Gravity Test
12	Viscosity Test

HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT
B.TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communication and Soft Skills

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

B. Detailed Syllabus:

- | | | | |
|-----|--|----------|-----|
| 1. | Concept of Personality | 06 Hours | 20% |
| 1.1 | Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling | | |
| 2. | Soft Skills and Personality Development | 08 Hours | 26% |
| 2.1 | Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player | | |

3. Developing Contributory Personality 06 Hours 20%

- 3.1 Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities

4. Life skills and Personality Development 06 Hours 20%

- 4.1 Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving

5. Contemporary Issues in CPD 04 Hours 14%

- 5.1 Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations

C. Course Outcome (COs):

At the end of the course, the students will be able to

CO1 Identify one's individual personality strengths and challenges.

CO2 Develop more assertive and optimist attitude towards work and life.

CO3 Develop quintessential soft skills to groom one's personality.

CO4 Identify traits of contributor personality.

CO5 Contribute to self, society, nation, and globe.

CO6 Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	1	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	1	-	-	-	-
CO3	-	-	-	-	-	1	-	-	3	1	2	1	1	-
CO4	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO5	-	-	2	-	-	2	2	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	2	2	-	-	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation“-”

D. Recommended Study Material:

Text book:

1. Personality Development & Soft Skills, Oxford University Press
2. Soft Skills, Bookboon
3. Personality Development, Swami Vivekananda; Advaita Ashrama

Reference book:

1. Contributor Personality Program Workbook (Volume 1,2),
2. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

Web material:

1. <https://www.coursera.org/learn/wharton-succcess>
2. <https://www.coursera.org/learn/personality-types-at-work>
3. <https://www.coursera.org/learn/self-awareness>

CL376: BUILDING REPAIR AND REHABILITATION
B. TECH 6th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - II

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Maintenance of Building	04
2	Deterioration and Durability Aspects	10
3	Conditional/damage assessment & Evaluation of structures	08
4	Materials and Techniques for Repair	08
5	Rehabilitation and Retrofitting Methods	10
6	Demolition and Dismantling Techniques	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|-----|--|----------|-----|
| 1 | Maintenance of Building | 04 Hours | 09% |
| 1.1 | Introduction | | |
| 1.2 | Objectives | | |
| 1.3 | Phases of Maintenance | | |
| 1.4 | Common Defects In Buildings And Measures To Prevent And Control The Same | | |
| 1.5 | Building Failures- Causes And Effects | | |
| 1.6 | Cracks In Buildings | | |

2	Deterioration and Durability Aspects	10 Hours	22%
2.1	<u>Deterioration:</u> Types of deterioration – Signs, causes & symptoms, Mechanism of deterioration, contributing factors like permeability, inadequate durability & micro-structure of concrete		
2.2	Physical deterioration due to moisture, temperature, shrinkage, freeze-thaw, abrasion, erosion, cavitation, crystallization of salts, Efflorescence, exposure to severe environment like marine exposure		
2.3	Chemical deterioration due to corrosion of reinforcement (chloride induced, carbonation induced), Alkali-silica reaction, sulphate attack, Acid attack		
2.4	Deterioration due to water leakage, fire – detection & mitigation of the same. Deterioration due to ageing, inadequate maintenance, Design & construction deficiencies, overloading etc.		
2.5	Visual deterioration of structures- Types of cracks, causes & characteristics of cracking in various structural components like beam, column, slab, masonry walls. Measurement of cracks, interpretation of the cause of particular type of crack		
2.6	<u>Durability:</u> Life Expectancy of Different Types of Buildings – Influence of Environmental Elements Such as Heat, Moisture, Precipitation and Frost on Buildings- Design and Construction Errors, Corrosion Mechanism		
2.7	Effect of Biological Agents like fungus, moss, plants, trees, algae, - Termite Control and Prevention		
2.8	Chemical Attack on Building Materials and Components		
2.9	Aspects of Fire and Fire Prevention on Buildings		
2.10	Impact of Pollution on Buildings		
3	Conditional/damage assessment & Evaluation	08 Hours	18%
3.1	Conditional evaluation – Importance, objective & stages, Conditional/damage assessment procedure, Preliminary & Detailed investigation – Scope, Objectives, Methodology & Rapid visual inspection of structures		

3.2	Damage Assessment allied Tests (Destructive, Semi-destructive, Non-destructive)		
3.3	Non-destructive Evaluation Tests - Concrete Strength Assessment Chemical Tests, Fire Damage Assessment, Structural Integrity/Soundness Assessment		
3.4	Interpretation & Evaluation of Test Result Data		
4	Materials and Techniques for Repair	08 Hours	18%
4.1	Essential Parameters for Repair Materials		
4.2	Materials for Repair		
	4.2.1 Premixed cement concrete/mortars		
	4.2.2 Polymer Modified Mortars and Concrete (PMM/PMC)		
	4.2.3 Epoxies and Epoxy Mortars/Concretes		
	4.2.4 Polyester Resins		
	4.2.5 Sulphur infiltrated concrete		
	4.2.6 Ferro cement		
	4.2.7 Fibre reinforced concrete		
	4.2.8 Foamed concrete		
	4.2.9 Dry pack		
	4.2.10 Vacuum concrete		
	4.2.11 Surface Coatings		
4.3	Gunniting, Grouting, Shotcrete and Epoxy injection		
5	Rehabilitation and Retrofitting Methods	10 Hours	22%
5.1	Important factors to be considered for selection of repair methods.		
5.2	Repair Stages		
5.3	Repair Methods		
5.4	Repair of stone, brick and block masonry (Cracks, dampness, efflorescence, joint separation, etc.), Flooring, Roofs (sloping, flat, pitched, etc.)		
5.5	Repair of Concrete members due to (i) Steel Corrosion (ii) Lack of Bond (iii) shear, tension, torsion, compression failure		
5.6	Repair of leakage due to rain water		

5.7 Strengthening of Earthquake Damage Buildings

6 Demolition and Dismantling techniques

05 Hours

11%

6.1 Engineered demolition techniques for Dilapidated structures

6.2 Safety measures during demolition operation

6.3 Dismantling of buildings and reuse of materials/fittings from environmental and financial point of view

6.4 Case studies

C. Course Outcomes (COs):

At the end of the course, the students will be able to

- CO1 Understand the types of requirement of maintenance in building and able to assess the quality aspects of existing building.
- CO2 Identify and define concepts associated with deterioration of concrete structures, damage assessment or inspection of a building showing signs of deterioration and should be able to detect the possible cause/source of deterioration with appropriate techniques.
- CO3 Develop a knowledge of the concrete repair process with variety of repair materials and techniques. Understand and apply rehabilitation and retrofitting processes.
- CO4 Gain knowledge regarding dismantle and demolish structures which cannot be repaired in an environment friendly, with maximum saving of materials and in a safe way.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	2	-	-	-	-	-	-	-	-	-	-	-	-	1
CO2	-	-	1	-	-	-	-	-	-	-	-	-	-	-	1
CO3	-	-	-	2	-	-	-	-	-	-	-	-	1	-	1
CO4	-	-	-	-	-	-	1	-	-	-	-	-	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. P.C.Varghese , Maintenance, Repair & Rehabilitation and Minor Works of Buildings, PHI learning.

2. R.T.Allen and S.C.Edwards, " Repair of Concrete Structures ", Blakie and Sons, UK, 1987.
3. A.R. Santhakumar,, "Concrete Technology", 2012 Oxford University Press ,2006.
4. V. M. Malhotra, Nicholas J. Carino 2004 "Handbook on Nondestructive Testing of Concrete".
5. Handbook on Repairs and Rehabilitation of RCC buildings – CPWD, Government of India.
6. R.Dodge Woodson., Concrete Structures: Protection, Repair and Rehabilitation, Elsevier, 2009.
7. Xilin Lu ,"Retrofitting Design of Building Structures",Science Press,2010.
8. Structural Condition assessment by Robert T. Ratay.
9. IS 13935 : 2009, Seismic Evaluation, Repair And Strengthening of Masonry Buildings — Guidelines

Reference Books:

1. A.C. Panchdhari, "Maintenance of Buildings", New Age International.
2. Santhakumar, A.R., Training Course notes on Damage Assessment and repairs in Low Cost Housing, "RHDC – NBO" Anna University, July 1992.
3. Concrete Technology by M.L.Gambhir, Tata McGraw-Hill Education.
4. R.T.Allen and S.C.Edwards, Repair of Concrete structures, Blakie and Sons, UK, 1987
5. M.S.Shetty, Concrete Technology – Theory and Practice, S.Chand and Company, New Delhi, 1992.
6. Concrete Structures, Protection, Repair and Rehabilitation by R.Dodge Woodson.
7. Repairs and rehabilitation of concrete structures by P. I. Modi & C. N. Patel, PHI Publication.

Web Materials:

1. cpwd.gov.in/Units/handbook.pdf
2. www.bis.org.in/sf/nbc.htm
3. <http://www.civilengineeringnews.tk/2014/07/methods%20of-demolition-of-%20building.html>
4. <http://www.indianconcreteinstitute.org/repair-and-rehabilitation.html>

TERM WORK

- 1) Study of rehabilitation/retrofitting of RCC/Masonry buildings covering (a) damage assessment by visual inspection and using various techniques including NDT (b) one/two alternatives for rehabilitation/retrofitting (considering strength criteria & serviceability criteria).
- 2) Case study of construction and design failures.
- 3) Market survey for material for repairs.
- 4) Case study of repairing/rehabilitation structures and retrofitting of EQ damage/deficit structure.
- 5) Presentations / finding engineering applications /preparation of learning material based on the syllabus.

CL377: ADVANCED SURVEY
B. TECH 6th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - II

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Tacheometric Surveying	09
2	Geodetic Surveying	07
3	Trigonometric Levelling	07
4	Theory of Errors	06
5	Field Astronomy	04
6	Aerial photogrammetry	06
7	Total Station and GPS	06

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | |
|----------|---|------------------------|
| 1 | Tacheometric Surveying : | 09 Hours 19% |
| 1.1 | Introduction | |
| 1.2 | Tacheometric Systems – Tangential, Stadia and subtense methods | |
| 1.3 | Stadia systems – horizontal and inclined sights – vertical and normal staff – fixed and movable hair – stadia constants | |
| 1.4 | Field work in tacheometry, anallatic lens – subtense bar – Self reducing tacheometers | |
| 1.5 | Errors and precisions | |
| 2 | Geodetic Surveying | 07 Hours 15% |
| 2.1 | Principle and Classification of triangulation system | |

2.2	Selection of base line and stations		
2.3	Orders of triangulation		
2.4	Triangulation figures- Horizontal and vertical control		
2.5	Station marks and signals- marking signals		
2.6	Extension of base, Reduction of Centre, Selection and marking of stations		
3	Trigonometric Levelling	07 Hours	15%
3.1	Introduction		
3.2	Methods of trigonometric levelling: Base of object accessible, base of object inaccessible – same vertical plane, base of object inaccessible – not in same vertical plane		
3.3	Determination of height of elevated object above ground when its base & top are visible but not accessible		
4	Theory of Errors	06 Hours	13%
4.1	Introduction		
4.2	Types and sources of errors & definitions		
4.3	Laws of accidental errors, laws of weights, theory of least squares		
4.4	Rules for giving weights and distribution of errors to the field observations		
5	Field Astronomy	04 Hours	10%
5.1	Introduction & purposes, astronomical terms		
5.2	Determination of azimuth, latitude, longitude and time corrections to the observations		
6	Aerial photogrammetry	06 Hours	14%
6.1	Photogrammetric terms, Applications, Type of photographs, floating marks		
6.2	Scale of a vertical and oblique photographs, heights and tilt distortions		
6.3	Flight planning, Stereoscopy, base lining		
6.4	Developments in photogrammetry, Photomaps and mosaics		
6.5	Photographic interpretations, Parallax bar		
7	Total Station and GPS	06 Hours	14%
7.1	Digital theodolite, Electronic Distance Measurement, Total		

Station: Introduction, principle and uses

- 7.2 Remote sensing: Principle of Remote sensing, EMR, types, resolutions, types of satellites, type of sensors, visual and digital image processing and its applications
- 7.3 Global Positioning System: Definition, Principles of GPS and applications.
- 7.4 Introduction and principle of GIS, Applications of GIS in Civil Engineering

C. Course Outcomes (COs):

At the end of the course, the students will be able to

- CO1 Conduct tacheometry and geodetic survey.
- CO2 Apply principles of theory of errors for correction of measurements.
- CO3 Apply knowledge of astronomy for solving civil engineering problems.
- CO4 Explain use of aerial camera, aerial photographs and procedure of aerial survey.
- CO5 Utilize stereoscope and parallax bars, total station and other modern survey instruments.
- CO6 Apply GIS in solving engineering problems.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	1	-	2	2	1	1	2	2	2	1	3	2	2
CO2	3	1	-	-	1	-	-	-	-	-	-	1	1	-	-
CO3	3	2	1	-	2	1	1	-	2	1	1	2	1	1	2
CO4	3	3	2	2	3	2	1	1	1	2	1	2	2	3	2
CO5	2	1	2	2	3	3	2	2	2	2	2	3	3	3	3
CO6	2	3	3	3	3	3	3	3	3	2	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Duggal, S. K., Surveying Vol. I & II, Tata McGraw Hill Publication, New Delhi.
2. Subramanian, R., Surveying & Levelling, Oxford University Press, New Delhi.

Reference Books:

1. Surveying Vol. I, II and III by Dr. B.C. Punamia, Laxmi Publishers. New Delhi
2. Surveying and Levelling Vol. I and II by T.P. Kanetkar and S.V. Kulkarni, Pune Vidhyarthi Gruh
3. Surveying Vol. I, II and III by Dr. K.R. Arora, Standard Book House. New Delhi
4. Surveying Vol. I and II by S. K. Duggal, Tata McGraw Hill, New Delhi
5. Surveying and Levelling by N.N. Basak, Tata McGraw Hill, New Delhi
6. Surveying and Levelling by R. Agor, Khanna Publishers, New Delhi
7. Advanced Surveying by R. Agor, Khanna Publishers, New Delhi
8. Fundamentals of Surveying by Roy, S.K., Prentice Hall India, New Delhi
9. Surveying and Leveling by Subramanian, R., Oxford University Press, New Delhi
10. Remote Sensing and GIS by B Bhatia, Oxford University Press, New Delhi.
11. Remote sensing and Image interpretation by T.M. Lillesand, R.W. Kiefer, and J.W. Chipman, 5th edition, John Wiley and Sons India
12. Surveying theory and practice 7th Edition by James M Anderson and Edward M. Mikhail Tata McGraw Hill Publication.
13. Gopi, Advanced Surveying: Total Station, GIS and Remote Sensing, Pearson Education India, 2007.

Web Materials:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITROORKEE/SURVEYING/home.htm>
2. <http://nptel.iitm.ac.in/video.php?subjectId=105104101>
3. http://en.wikipedia.org/wiki/Geographic_information_system
4. <http://www.naicc.org/meeting/2009/GPSRemoteSensing.pdf>

LIST OF EXPERIMENTS

Experiment No.	Name of Experiment
1	Introduction to Tacheometry
2	Determination of multiplying and additive constants of a Tacheometer
3	Use of (i) Tacheometer, (ii) Total Station for determination of Reduced levels and Horizontal distances of various points in the field
4	Project: Tacheometric Survey Project
5	Use of different Softwares for surveying
6	Extension of Baseline using Theodolite
7	Remote Sensing
8	Applications of GIS Software's
9	Total Station Surveying – Measurements of Distances and angles, Slope distances, Height, Traversing.
10	Use of Stereoscope for 3-D Viewing, Height determination from a Stereo pair using the Parallax bar

CL378: STRUCTURAL ANALYSIS-III
B TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Beams Curved in Plan	06
2	Cable and Suspension Bridge	04
3	Plastic Theory of Structures	08
4	Column Analogy	06
5	Approximate analysis of Indeterminate Structures	06
6	Matrix Method of Structural Analysis – System Approach	15

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Beams Curved in Plan	06 Hours	13 %
1.1	Uses of curved beam		
1.2	Types of internal forces		
1.3	Analysis of curved beam fixed at ends for point load, uniformly distributed load		
1.4	Analysis of closed circular beam supported symmetrically		
2	Cable and Suspension Bridge	04 Hours	09%
2.1	Introduction: Cables and cable bridge		
2.2	General cable theorem		
2.3	Cable under uniformly distributed loads		
2.4	Suspension bridge		
2.5	Suspension bridge with three-hinged stiffening girder		

3	Plastic Theory of Structures	08 Hours	18%
3.1	Concept, assumptions, upper and lower bound theorems		
3.2	Shape factor for different cross sections		
3.3	Collapse load, load factor, plastic modulus of section, plastic moment of resistance		
3.4	Computation of collapse load for fixed beam, continuous beam and plane frame subjected to various load cases		
4	Column Analogy	06 Hours	13%
4.1	introduction		
4.2	Sign convention		
4.3	Analysis of fixed beam and propped cantilever beam		
4.4	Stiffness and carry-over factors for non-prismatic members		
5	Approximate analysis of Indeterminate Structures	06 Hours	13%
5.1	Introduction		
5.2	Portal method		
5.3	Cantilever method		
5.4	Substitute Frame Method		
6	Matrix Method of Structural Analysis – System Approach	15 Hours	34 %
6.1	Introduction to flexibility and stiffness method		
6.2	Flexibility and stiffness coefficient		
6.3	Application of flexibility and stiffness method system approach to analysis of beams, plane frame and plane truss		

C. Course Outcomes:

On the completion of the course the students will be able to:

- CO1 Recognize the importance of structural analysis and the tools available to determine the response of a structural system to external loads.
- CO2 Identify and formulate an engineering problem and to develop a solution.
- CO3 Recognize the need for technical updating on a continuing basis, since the course emphasizes on the changing nature of software.
- CO4 Use various approximate methods for analysis of indeterminate structures

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	1	-	1	1	-	-	-	-	-	-	-	1	-	-
CO2	3	2	3	-	1	-	-	-	-	-	-	-	1	-	-
CO3	-	-	1	2	2	1	-	-	-	-	1	-	2	-	1
CO4	2	2	1	1	2	-	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I & II, Charotar Publishing House
2. Negi, L.S. and Jangid, R.S., Structural Analysis, Tata McGraw Hill
3. Vazirani, V.N. and Ratwani, N.M., Analysis of Structures, Khanna Publishers
4. Weaver William and Gere James, Matrix Analysis of Framed Structures, CBS Publishers
5. R.S. Khurmi and N. Khurmi, Theory of structures, S. Chand

Reference Books:

1. Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
2. Hibbler, R.C., Structural Analysis, Pearson Education
3. Wang, C.K., Intermediate Structural Analysis, Tata McGraw Hill
4. Reddy, C.S., Basic Structural Analysis, Tata McGraw Hill
5. Jangid, R.S., Structural Analysis, Tata McGraw Hill
6. Devdas Menon., Structural Analysis, Alpha Science
7. William M.C. McKenzie, Examples in Structural Analysis, Second Edition, CRC Press

Web Materials:

1. http://nptel.ac.in/courses/Webcourse-contents/IIT%20Kharagpur/Structural%20Analysis/New_index1.html
2. <http://www.nptelvideos.in/2012/11/structural-analysis-ii.html>

3. <http://nptel.ac.in/courses/105106050/2>
4. [http://nptel.iitg.ernet.in/Civil_Eng/IIT%20Delhi/Structural%20Analysis%20I%20\(Video\).htm](http://nptel.iitg.ernet.in/Civil_Eng/IIT%20Delhi/Structural%20Analysis%20I%20(Video).htm)
5. <http://freevidelectures.com/Subject/Civil-Engineering>
6. <http://freevidelectures.com/Course/3015/Advanced-Structural-Analysis>

LIST OF EXPERIMENTS

Experiment No.	Name of Experiment
1	Simple suspension bridge
2	Plastic bending of beams
3	Plastic bending of portals
4	Redundant truss
5	Beam Apparatus

CL379: TOWN & URBAN PLANNING
B TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Town Planning	06
2	Ancient System of Town Planning	04
3	Surveys	04
4	Zoning	05
5	Housing and Slums	08
6	Development Plan	05
7	Planning Concepts	08
8	Planned Cities of India	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction to Town Planning	06 Hours	13%
1.1	Objects & importance of town planning		
1.2	Principles of town planning		
1.3	Origin of Towns		
1.4	Growth of Towns		
1.5	Stages in Town Development		
1.6	Distribution of land uses		
1.7	Forms of planning		

2	Ancient System of Town Planning	04 Hours	9%
2.1	Town Planning in Ancient India		
2.2	Planning thought behind Fatehpur Sikri, Shahjahanabad, Jaipur and Delhi		
2.3	Indus Valley Civilization		
3	Surveys	04 Hours	9%
3.1	Necessity of surveys		
3.2	Collection of Data and Methods Adopted to Collect Data		
3.3	Types and Uses		
4	Zoning	05 Hours	11%
4.1	Objects and Principles of Zoning		
4.2	Advantages and Importance of Zoning		
4.3	Aspects of Zoning, Transition Zone		
4.4	Maps for Zoning		
5	Housing and Slums	08 Hours	18%
5.1	Importance of housing, its demand		
5.2	Requirements of residential buildings		
5.3	Design of residential areas		
5.4	Low cost housing, Laurie Baker's work and ideology		
5.5	Slums – Causes, Characteristics, Effects		
5.6	Slum clearance, Works of Improvement, Re-housing		
5.7	Prevention of slum formation		
6	Development Plan	05 Hours	11%
6.1	Objects and Necessity of development plan		
6.2	Stages of preparation of development plan, Data to be collected		
6.3	Features of Development Plan		
7	Planning Concepts	08 Hours	18%
7.1	Land use planning, Neighbourhood planning		
7.2	Protective strips, green belt		
7.3	Radburn Layout		
7.4	Urban Roads – Requirements, Classification, Types of Street Systems, Ring Roads		
7.5	Geddesian Triad, Ekistics		

8 Planned Cities of India

05 Hours 11%

8.1 Chandigarh, Contribution of Le Corbusier to Town Planning

8.2 Gandhinagar

C. Course Outcomes:

On the successful completion of this course, the students will be able to:

- CO1 Have a proper understanding of history of town planning, ideas developed in the past and its relevance in today's world.
- CO2 Understand procedure involved in making development plans and various concepts, and issues of town planning.
- CO3 Understand the importance of housing and its cost effectiveness.
- CO4 Understand the causes of slum formation and measures needed to prevent it.
- CO5 Understand the ideology behind different planned cities.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	2	1	1	-	-	-	-	-	-	-	-	1	-	-
CO3	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	1	-	-	-	1	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Rangwala, S.C., Town Planning, Charotar Publishing House, Anand.
2. Hiraskar, G.K., Town Planning, Dhanpatrai & Sons, New Delhi.
3. A. Bandopadhyay, Text book of Town Planning, Books and Allied, Calcutta 2000

Reference Books:

1. Mathur, G.C., Low Cost Housing in Developing Countries, South Asia Books
2. K. S. Rame Gowda, Urban and regional planning: principles and case studies, Prasaranga, University of Mysore, 1972

3. John Ratcliffe, An Introduction to Town and Country Planning, Hutchinson 1981
4. Kevin A. Lynch, The Image of the City, MIT Press
5. Alex Krieger and William S. Saunders, Urban Design, University of Minnesota Press

Web Materials:

1. <https://townplanning.gujarat.gov.in/>
2. <http://www.udd.gujarat.gov.in/ctp.php>
3. http://hcp.co.in/file_manager/publications/Town-Planning-of-Gujarat_Research-Paper.pdf
4. <https://www.planetizen.com/>
5. <http://www.itpi.org.in/>

LIST OF TUTORIALS

Experiment No.	Name of Tutorials
1	Origin and Growth of Towns
2	Ancient System of Town Planning
3	Surveys
4	Zoning
5	Housing and Slums
6	Development Plan – Case study of Different cities
7	Planning Concepts
8	Planned Cities of India
9	City planning case studies from around the world

CL380: CONSTRUCTION TECHNOLOGY
B. TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Min. No. of Hours
1	Introduction to Construction Equipment & Fundamental of Moving Earth	09
2	Excavating Equipment	06
3	Handling and Hauling Equipment	06
4	Overview of Other Construction Equipment	06
5	Procurement and Planning of Construction Equipment	04
6	Ground Water Control During Excavation	06
7	Latest Building Material and Technology	08

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1 | Introduction to Construction Equipment & Fundamental of Moving Earth | 09 Hours | 21% |
| 1.1 | Contribution and Importance of construction equipment | | |
| 1.2 | Classification of Equipment, Selection of construction equipment | | |
| 1.3 | Soil Fundamentals, Weight-Volume Relations of Materials | | |
| 1.4 | Machine power, coefficient of traction, rim pull and available pull and drawbar pull, Effect of grade & altitude on power | | |
| 2 | Excavating Equipment | 06 Hours | 13% |
| 2.1 | Features of excavating equipment | | |

2.2	Production/Output analysis and Suitability of Power Shovel, Dragline, Backhoe, Clamshells		
3	Handling and Hauling Equipment	06 Hours	13%
3.1	General		
3.2	Tractor, Dumper, Bulldozer, Ripper, Scraper, Truck		
4	Overview of Other Construction Equipment	06 Hours	13%
4.1	Hosting Equipment		
4.2	Conveying Equipment		
4.3	Drilling Equipment		
4.4	Pumping Equipment		
4.5	Compacting Equipment		
5	Procurement and Planning of Construction Equipment	04 Hours	09%
5.1	Procurement criteria		
5.2	Equipment cost		
5.3	Replacement analysis		
6	Ground Water Control During Excavation	06 Hours	13%
6.1	Importance, selection of method for ground water control during excavation		
6.2	Well point system, Dewatering Systems etc.		
6.3	Grouting-preparation of grouting, material and equipment used.		
7	Latest Building Material and Technology	08 Hours	18%
7.1	Introduction		
7.2	Building Material from Waste: Recycle Concrete Aggregates, Paper as building material, Composite: Matrices and Reinforcement		
7.3	Introduction to Advance Techniques used in Construction: Slip		

Formwork, Jump Formwork, Tunnel Formwork, Trenchless
Technology for excavation, Shotcreting

C. Course Outcomes (COs):

On the successful completion of this course:

- CO1 The students will be able to estimate earthwork for various construction projects.
- CO2 The students will be able to plan and select suitable construction equipment for different construction activities.
- CO3 The students will acquire ability to do planning for construction operations based on equipment productivity.
- CO4 The students will be able to analyse various financial aspects involved in the use of construction equipment.
- CO5 Students will have an understanding on various systems and techniques used in ground water control during excavation.
- CO6 Students will be familiar with some of the latest materials and systems used in construction.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	1	3	3	1	-	-	-	2	3	2	3	3	2	2
CO2	3	-	-	2	-	2	2	1	3	2	2	1	3	1	2
CO3	3	1	2	2	3	2	-	1	3	3	2	1	3	-	3
CO4	3	-	-	2	1	2	-	2	3	3	3	-	3	-	3
CO5	3	3	2	3	3	2	2	-	3	2	3	1	2	1	1
CO6	3	1	-	-	3	-	-	-	3	3	1	-	1	-	1

D. Recommended Study Material:

Text Books:

1. Peurifoy, L., Schexnayder, C.J. and Shapira, A., Construction Planning, Equipment and Methods, McGraw Hill, New Delhi, 8th Edition, 2010.
2. Roy Chudley, Roger Geeno, "Advanced Construction Technology" Latest Edition
3. Deodhar, S.V. "Construction Equipment and Job Planning", Khanna Publishers Delhi, 2008

Reference Books:

1. Jha, Kumar Neeraj, "Construction Project Management: Theory and Practice", Pearson Education India, 2011

2. Seetharaman, S., Construction Engineering & Management, Umesh Publications, 2007.
3. Kotadia, A.S., "Construction Management and equipments", Mahajan Publishing House, 8th Edition, 2010
4. Sharma, S.C., Construction Equipment & Management, Khanna Publications, New Delhi, 1988.
5. Sankar, S.K. and Saraswati, S., Construction Technology, Oxford University Press, New Delhi, 2008.
6. Sahu, G.C. and Jena, Joygopal, "Building Materials and Construction", McGraw Hill Education, Latest Edition
7. Punmia, B. C., Jain, Ashok Kumar, "Soil Mechanics and Foundations", Laxmi Publication, 16th edition.

Web Materials:

1. <http://nptel.ac.in/courses/105103023/>
2. <https://www.youtube.com/watch?v=XEAcl1FHZpY>
3. <https://www.youtube.com/watch?v=pN2Th96poHQ>
4. <https://www.youtube.com/watch?v=D0DCtN0SXGQ>
5. <https://www.youtube.com/watch?v=pBJfKHRzDEM>

LIST OF PRACTICAL/TUTORIAL

Sr. No.	Name of Tutorials
1.	Estimating Earth Work For Construction Projects
2.	Machine Power
3.	Owning Operating Cost
4.	Output Analysis of Equipment
5.	Replacement Analysis of Equipment
6.	Excavating Equipment-Basic parts, selection and operation
7.	Handling and Hauling Equipment- Basic parts, selection and operation
8.	Conveying Equipment and Drilling Equipment- Basic parts, selection and operation
9.	Ground Water Control During Excavation
10.	Latest Building Material and Technology

**CL383: INSTRUMENTATION & SENSOR TECHNOLOGIES FOR CIVIL
ENGINEERING APPLICATIONS**
B TECH 6th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	03	02	05	04
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Fundamentals of Measurement, Sensing and Instrumentation	10
2	Sensor Installation and Operations	10
3	Data Analysis and Interpretation	12
4	Frequency Domain Signal Processing and Analysis	13

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | |
|---|-----------------|-------------|
| 1 Fundamentals of Measurement, Sensing and Instrumentation | 10 Hours | 22 % |
| 1.1 Definition of measurement and instrumentation, physical variables, common types of sensors. | | |
| 1.2 Describe the function of these sensors; | | |
| 1.3 Use appropriate terminology to discuss sensor applications; and qualitatively interpret signals from a known sensor type, types of instrumentation, Sensor Specifics, Permanent installations, Temporary installations; | | |
| 2 Sensor Installation and Operations | 10 Hours | 22 % |
| 2.1 Predict the response of sensors to various inputs;
Construct a conceptual instrumentation and monitoring | | |

	program		
2.2	Describe the order and methodology for sensor installation Differentiate between types of sensors and their modes of operation and measurement		
2.3	Approach to Planning Monitoring Programs, Define target, Sensor selection, Sensor siting, Sensor Installation & Configuration, Advanced topic, Sensor design, Measurement uncertainty		
3	Data Analysis and Interpretation	12 Hours	27 %
3.1	Fundamental statistical concepts, Data reduction and interpretation, Piezometer, Inclinator, Strain gauge, etc.		
3.2	Time domain signal processing, Discrete signals, Signals and noise		
3.3	Examples of statistical information to calculate are: Average value (mean), On average, how much each measurement deviates from the mean (standard deviation), Midpoint between the lowest and highest value of the set (median), Most frequently occurring value (mode), Span of values over which your data set occurs (range)		
4	Frequency Domain Signal Processing and Analysis	13 Hours	29 %
4.1	Explain the need for frequency domain analysis and its principles; Draw conclusions about physical processes based on analysis of sensor data;		
4.2	Combine signals in a meaningful way to gain deeper insight into physical phenomena, Basic concepts in frequency domain signal processing and analysis,		
4.3	Fourier Transform, FFT (Fast Fourier Transform), Example problems: Noise reduction with filters, Leakage, Frequency resolution		

C. Course Outcomes (COs):

On the completion of the course one should be able to

- CO1 Analyze the errors during measurements
- CO2 Specify the requirements in the calibration of sensors and instruments
- CO3 Describe the noise added during measurements and transmission
- CO4 Describe the measurement of electrical variables
- CO5 Describe the requirements during the transmission of measured signals and construct Instrumentation/Computer Networks
- CO6 Suggest proper sensor technologies for specific applications and design and set up measurement systems and do the studies

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	2	-	-	-	-	1	1	1	1	1
CO2	3	2	2	1	2	2	-	-	-	-	1	1	1	1-	1
CO3	3	2	2	1	2	2	-	-	-	-	1	1	1	1	1
CO4	3	2	2	1	2	2	-	-	-	-	1	1	1	1	1
CO5	3	2	2	1	2	2	-	-	-	-	1	1	1	1	1
CO6	3	2	2	1	2	2	-	-	-	-	1	1	1	1	1

D. Recommended Study Material:

Text Books:

- Alan S Morris (2001), Measurement and Instrumentation Principles, 3rd/e, Butterworth Hienemann
- David A. Bell (2007), Electronic Instrumentation and Measurements 2nd/e, Oxford Press
- S. Tumanski (2006), Principle of Electrical Measurement, Taylor & Francis
- Ilya Gertsbakh (2010), Measurement Theory for Engineers, Springer

List of Experiments

Experiment No.	Name of Experiment
1.	Instrumentation of typical civil engineering members/structures/structural elements
2.	Use of different sensors, strain gauges, inclinometers,

3.	Performance characteristics
4.	Errors during the measurement process
5.	Calibration of measuring sensors and instruments
6.	Measurement, noise and signal processing
7.	Analog Signal processing
8.	Digital Signal Processing
9.	Demonstration & use of sensor technologies



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering
Chandubhai S Patel Institute of Technology
M. S. Patel Department of Civil Engineering
Bachelor of Technology Programme
(Fourth Year Civil Engineering)

Effective From 2022-23



Vision

“To provide state of the art education in Civil Engineering guided by innovative research leading to centre of excellence in Civil Engineering education having recognition at national and international levels”

Mission

“Being a pioneering branch of Engineering, the department of Civil Engineering under the shelter of CHARUSAT is intended as a facilitator for creating a liaison between the brilliant student community and the next generation industrial needs”

Programme Educational Objectives (PEO's):

PEO 1: The graduate will possess foundation of engineering knowledge and exhibiting critical thinking and problem solving skills

PEO 2: The graduates will have trait of lifelong learning and be able to inculcate the capabilities to meet the diversified needs of industry, academia and research.

PEO 3: The graduate will exhibit the professional ethics and be supportive to the social needs

PEO 4: The graduates will possess comprehending, analyzing and designing capabilities to generate sustainable solutions

Programme Outcomes (PO's)

Engineering Graduates will be able to:

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSO's):

By the completion of Civil Engineering program, the student will attain:

PSO 1: The ability to serve the infrastructure sector with capabilities to plan, design, analyze and building civil engineering-based systems

PSO 2: The ability to adopt to the state-of-the-art practices in all sectors of Civil Engineering.

PSO 3: Employability skills with the cognizance of social and environmental necessity along with ethical responsibility to have a successful career and to become an entrepreneur.

CHARUSAT welcomes you for a Bright Future



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Accredited with Grade A by NAAC,
Accredited with Grade A by KCG

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Civil Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Academic Year – 2022-23

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Bachelor of Technology Programmes

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least of 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen subject of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate programme (B. Tech.)

Minimum 8 semesters (4 academic years)

Maximum 12 semesters (6 academic years)

3. Eligibility for Admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of Admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme Structure and Credits

As per Annexure – I attached

6. Attendance

All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

Student's attendance in a course should be 80%.

7. Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.2 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 7.1.3 Final examination by the University through modes such as; written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, is set to 70% of the marks for each the course.

7.2 Internal Evaluation

As per Annexure – I attached

7.3 University Examination

The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.4 In order to earn the credit in a course a student has to obtain grade other than FF.

7.5 Performance at Internal & University Examination

- 7.5.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows:

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

- 7.5.2 A student failing to score 40% in the final examination will get an FF grade.
- 7.5.3 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8. Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	<80 ≥73	<73 ≥66	<66 ≥60	<60 ≥55	<55 ≥50	<50 ≥45	<45
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Numerical point (Grade Point) corresponding to the letter grade	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where,
 C_i = Number of credits of course i
 G_i = Grade Point for the course i
 $i = 1$ to n
 n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where,
 C_i = Number of credits of course i
 G_i = Grade Point for the course i
 $i = 1$ to n
 n = number of courses of all semesters up to which CGPA is computed
- (iii) No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

9. Award of Degree

- 9.1 Every student of the programme who fulfills the following criteria will be eligible for the award of the degree:
- 9.1.1 He/ She should have earned minimum required credits as prescribed in course structure; and
 - 9.1.2 He/ She should have cleared all internal and external evaluation components in every course; and
 - 9.1.3 He/ She should have secured a minimum CGPA of 4.5 at the end of the programme;
 - 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.
- 9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction	:	CGPA ≥ 7.5 & ≤ 10.0
First class	:	CGPA ≥ 6.0 & < 7.5
Second Class	:	CGPA ≥ 5.0 & < 6.0
Pass	:	CGPA < 5.0

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)															
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CIVIL ENGINEERING (CBCS)															
Level	Course Code	Course Title	Teaching Scheme								Examination Scheme				
			Contact Hours				Credit				Theory		Practical		Total
			Theory	Practical	Tutorial	Total	Theory	Practical	Project	Total	Internal	External	Internal	External	
Level 4	CL451	Design of Reinforced Concrete Structures	3	2	1	6	3	1		4	30	70	25	25	150
	CL452	Design Of Steel Structures	3	2	1	6	3	1		4	30	70	25	25	150
	CL445	Summer Internship - II	0	3	0	3	0	0	3	3	0	0	75	75	150
	CL453	Construction Project Management	3	2	0	5	3	1		4	30	70	25	25	150
	CL454	Geotechnical Engineering - II	3	2	1	6	3	1		4	30	70	25	25	150
	CL455	Professional Practices	3		1	4	3	0		3	30	70	0	0	100
	CL470-496	Programme Elective - III	3	2	0	5	3	1		4	30	70	25	25	150
						35				26					1000
	CL458	Major Project	0	36	0	36	0	0	18	18	0	0	400	400	800

CONTENTS

CL451: DESIGN OF REINFORCED CONCRETE STRUCTURES	13
CL452: DESIGN OF STEEL STRUCTURES	17
CL445: SUMMER INTERNSHIP-II.....	21
CL453 CONSTRUCTION PROJECT MANAGEMENT	27
CL454: GEOTECHNICAL ENGINEERING – II.....	32
CL455: PROFESSIONAL PRACTICES	36
CL470: STRUCTURAL DYNAMICS & EARTHQUAKE ENGINEERING	40
CL471: ADVANCED STRUCTURAL ANALYSIS	44
CL477: FIELD APPLICATIONS OF GEOTECHNICAL ENGINEERING	47
CL478: TRANSPORTATION ENGINEERING- II.....	51
CL479: ENVIRONMENTAL ENGINEERING- II.....	57
CL480: WATER RESOURCES ENGINEERING II	61
CL481: DESIGN OF STRUCTURES USING INTERNATIONAL CODES	66
CL484: TRANSPORT PROJECT PLANNING AND EVALUATION	69
CL485: WATERSHED MANAGEMENT	74
CL487: ADVANCED CONSTRUCTION TECHNOLOGY	78
CL488: ADVANCED GEOTECHNICAL ENGINEERING	82
CL489: ENVIRONMENTAL POLLUTION AND CONTROL.....	87
CL492: GIS & REMOTE SENSING APPLICATIONS IN CIVIL ENGINEERING	91
CL493: COMPUTATIONAL METHODS IN WATER RESOURCES ENGINEERING	96
CL494: ADVANCED HIGHWAY & TRAFFIC ENGINEERING.....	100
CL496: GREEN BUILT ENVIRONMENT	104
CL458: MAJOR PROJECT	108

B. Tech. (Civil Engineering) Programme

SYLLABI (Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CL45I: DESIGN OF REINFORCED CONCRETE STRUCTURES
B TECH 7th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	1	2	6	4
Marks	100	-	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Load Assessment and Design Philosophy	02
2	Design of Beams	07
3	Design of Conventional Slabs	06
4	Design of Compression Members	07
5	Design of Footings	04
6	Design of Staircase	03
7	Design of Retaining Wall	06
8	Design of Flat Slab	05
9	Analysis & Design of Multi-storied Building	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | |
|--|----------|-----|
| 1 Load Assessment and Design Philosophy | 02 Hours | 4% |
| 1.1 Dead Load, Live Load, Wind Load, Earthquake Load and Load Combinations | | |
| 1.2 Basic Design Philosophy: Working Stress Method, Limit State Method | | |
| 1.3 Indian Standard Provisions, Failures of Structures | | |
| 2 Design of Beams | 07 Hours | 16% |
| 2.1 Methods of design of concrete structures | | |

2.2	Limit state of collapse: Flexure, compression, shear		
2.3	Characteristic design values & partial safety factors		
2.4	Limit state of serviceability: Deflection, cracking		
2.5	Basis of design, loads & forces		
2.6	Behavior of R.C.C. beams under design loads		
2.7	Design of singly reinforced sections		
2.8	Design of doubly reinforced sections		
2.9	Design of T- sections		
2.10	Detailing of Beams		
3	Design of Conventional Slabs	06 Hours	13%
3.1	Types of slabs		
3.2	Design of One-way slab (simply supported)		
3.3	Design of Two-way slab (restrained, simply supported)		
3.4	Detailing of Slab		
4	Design of Compression Members	07 Hours	16%
4.1	Analysis & design of axially loaded short column		
4.2	Design of axially loaded column with uni & bi-axial Bending		
4.3	Detailing of Column		
5	Design of Footings	04 Hours	9%
5.1	Pressure distribution under Footing		
5.2	Design of isolated footing (square & oblong)		
5.3	Design of Combined Footing		
5.4	Detailing of Footing		
6	Design of Staircase	03 Hours	7%
6.1	Classification of Stairs		
6.2	Design Requirements of Stairs		
6.3	Design of Dog-legged Staircase		
6.4	Detailing of Staircase		
7	Design of Retaining Wall	06 Hours	13%
7.1	Types of Retaining Wall		
7.2	Active & Passive Earth Pressure		
7.3	Design of Cantilever Retaining Wall		
7.4	Design of Counter fort Retaining Wall		

7.5 Detailing of Retaining Wall

8 Design of Flat Slab

05 Hours

11%

8.1 Types and Components of Flat Slab

8.2 Design of flat slab by direct method based on IS: 456

8.3 Detailing of Flat Slab

9 Analysis & Design of Multi-Storey Frame Buildings

05 Hours

11%

9.1 Structural Layout of Building

9.2 Design loads on buildings: Gravity, Wind and Earthquake loads

9.3 Member proportioning

9.4 Analysis and design of RCC framed buildings

9.5 Detailing of Members

C. Course Outcomes (CO):

On the successful completion of this course one should be able to

CO1 Analyse and design singly and doubly reinforced concrete beams under flexure and shear, including regular (rectangular shaped) and T-beams.

CO2 Analyse and design one way slab, two way slab, and reinforced concrete columns.

CO3 Analyse and design isolated and combined footings.

CO4 The students can identify the typical failure modes of staircase and retaining walls with detailing.

CO5 Students can understand the load transfer mechanism in multi-story building and also perform analysis and design of all structural members.

CO6 The students can able to understand the behaviour of flat slab under forces and application of design with detailing.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	1	2	-	1	-	-	-	-	-	-	3	3	3	3
CO2	3	3	2	1	1	-	-	-	-	-	1	3	3	3	3
CO3	3	2	2	-	1	-	-	1	-	1	1	2	3	3	3
CO4	1	2	1	1	-	-	-	1	-	1	2	2	3	2	3
CO5	2	1	3	-	1	-	-	1	-	2	1	2	1	2	1
CO6	1	2	1	1	-	-	-	1	-	1	2	2	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

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Page 15 of 110

Text Books:

1. Shah, H. J., Reinforced Concrete Vol-I & II, Charotar Publishing House.
2. Varghese, P. C., Limit State Design of Reinforced Concrete, Prentice – Hall of India.
3. S Unnikrishna Pillai & Devdas Menon, Reinforced Concrete Design, Second Edition, Tata McGraw Hill
4. Punamia, B. C., & Jain, A.K., R.C.C. Designs (Reinforced Concrete Structures), Laxmi Publications (P) Ltd.

Reference Books:

1. Shah, V.L., & Karve, S.R., Limit State Theory and Design of Reinforced Concrete, Structures Publications.
2. Ramchandra & Gehlot V., Limit State Design of Concrete Structures, Scientific Publishers, India.
3. Sinha, S.N., Handbook of Reinforced Concrete Design, Tata McGraw Hill Publishing Company Limited.

IS Specifications:

1. IS: 456, Plain and Reinforced Concrete-Code of Practice
2. IS: 875 (Part 1 to 5), Code of Practice for Design Loads
3. SP 16, Design Aids for Reinforced Concrete to IS: 456
4. SP 34, Handbook on Concrete Reinforcement and Detailing

Term Work:

1. Design of Flexure Members
2. Design of Conventional Slabs
3. Design of Compression Members
4. Design of Footings
5. Design of Staircase
6. Design of Retaining Wall
7. Design of Flat Slab
8. Analysis & Design of Multi-storied Building along with Detailing of Structural Elements in Drawing Sheet

CL452: DESIGN OF STEEL STRUCTURES
B TECH 7th SEMESTER (CIVIL ENGINEERING)

Credit and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	01	2	6	4
Marks	100	-	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Load Assessment and Design Philosophy	02
2	Connections Design	08
3	Design of Tension Members	07
4	Design of Compression Members	06
5	Design of Steel Beam	06
6	Design of Plate Girder	06
7	Design of Gantry Girder	05
8	Analysis & Design of Roof Truss	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | |
|---|-----------------|------------|
| 1 Load Assessment and Design Philosophy | 02 Hours | 4% |
| 1.1 Dead Load, Live Load, Wind Load and Load Combinations | | |
| 1.2 Basic Design Philosophy, Indian Standard Provisions, Failures of Structures | | |
| 2 Connections Design | 08 Hours | 17% |
| Bolted Connections | | |
| 2.1 Bolt as connection, types of bolts, types of bolted joints | | |
| 2.2 Load transfer mechanism, failure of bolted joints | | |

- 2.3 Bearing type connections
- 2.4 Tensile strength of plate
- 2.5 Strength and efficiency of the joint
- 2.6 Beam – Beam connection, Column – Beam connection, Unstiffened connection

Welded Connections

- 2.7 Welding processes, welding electrodes, advantages of welding
- 2.8 Types and properties of welds, types of Joints
- 2.9 Effective area of welds
- 2.10 Design of welds, simple joints
- 2.11 Moment resistant connections
- 2.12 Beam-to-column connections

3 Design of Tension Members

07 Hours 16%

- 3.1 Types of tension members
- 3.2 Net sectional area, effective net area, types of failures
- 3.3 Design strength of tension members, slenderness ratio
- 3.4 Design of tension member

4 Design of Compression Members

06 Hours 13%

- 4.1 Effective length, slenderness ratio, types of sections
- 4.2 Types of buckling
- 4.3 Column formula, design strength of angle sections, I-section and more
- 4.4 Design of axially loaded compression members
- 4.5 Introduction to Built-Up columns
- 4.6 Design of Lacing
- 4.7 Design of Battening

5 Design of Steel Beam

06 Hours 13%

- 5.1 Beam Type, Section Classification, Lateral Stability, Effective Length
- 5.2 Design of Laterally Supported Beams in Bending
- 5.3 Design of Laterally Un-Supported Beams
- 5.4 Shear Behaviour
- 5.5 Web Buckling and Web Crippling

6 Design of Plate Girder

06 Hours 13%

- 6.1 Introduction

6.2 Components of Plate Girder

6.3 Buckling of Plate Elements

6.4 Buckling of Web Plate

6.5 Requirements of Plate Girder Components

6.6 Design of Plate Girder

7 Design of Gantry Girder

05 Hours 12%

7.1 Introduction

7.2 Loads on Gantry Girder

7.3 Maximum Load Effects

7.4 Selection of Gantry Girder

7.5 Design of Gantry Girder

8 Analysis & Design of Roof Truss

05 Hours 12%

8.1 Selection of the Type of Truss, Spacing of Truss

8.2 Panel Layout of Truss

8.3 Loads on the Roof Truss, Load Combinations

8.4 Analysis of Roof Truss, Deflection of Truss

8.5 Selection of Sections, Connections

8.6 End Bearings, Bracing of Truss

8.7 Design of Roof Truss

C. Course Outcomes (CO):

On the successful completion of this course one should be able to

CO1 The students will be able to understand the designing of connections.

CO2 Students will be able to design bolted and welded connections for tension members, compression members and beams.

CO3 The students will be able to design the steel structure members which are either subjected to axial loading or transverse kind of loading.

CO4 Students able to design basic elements of steel structure like tension members, compression members, beams and beam-columns.

CO5 Students can give economical design of girder considering the effects of bending and shear with proper connections.

CO6 Students can be able to understand the force transfer mechanism of roof truss and economical design of members.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	3	3	1	1	-	-	1	1	2	2	3	3	2	3
CO2	3	3	3	1	1	-	-	1	1	2	2	3	3	2	3
CO3	2	3	3	1	1	-	-	1	1	2	2	3	3	2	3
CO4	2	3	3	1	1	-	-	1	1	2	2	3	3	2	3
CO5	1	1	3	1	-	-	-	1	1	2	2	3	3	2	3
CO6	1	1	3	1	2	-	-	1	1	2	2	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Duggal, S.K., Limit State Design of Steel Structures, McGraw Hill Education (P) Ltd, New Delhi
2. Subramanian, N., Steel Structures: Design and Practice, Oxford University Press.

Reference Books:

1. S.S. Bhavikatti, Design Of Steel Structures (By Limit State Method As Per IS: 800), I. K. International Pvt Ltd,.
2. V L Shah, Veena Gore, Limit State Design of Steel Structures, Structures publications.
3. Steel Table

IS Specifications:

1. IS: 800, General Construction in Steel- Code of Practice
2. IS: 875 (Part 1 to 5), Code of Practice for Design Loads
3. SP 6, Handbook for Structural Engineers

Term Work:

1. Design of Bolted Connections
2. Design of Welded Connections
3. Design of Tension Members
4. Design of Compression Members
5. Design of Steel Beam
6. Design of Plate Girder
7. Design of Gantry Girder
8. Design of Roof Truss

CL445: SUMMER INTERNSHIP-II
B TECH 7th SEMESTER (CIVIL ENGINEERING)

Credit and Hours:

Teaching Scheme	Project	Total	Credit
Hours / week	45	90	3
Marks	150	150	

A. Instructional Method and Pedagogy:

- Summer internship shall be at least of 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers are also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student is not able to attend the internship for a few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator on the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.

- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

B. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Title is to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be “Times New Roman” font with a font size of 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. A soft copy of the report shall be submitted in a CD and enclosed with the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the type of work, administrative structure, number of each category of employees, etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the company, and general information regarding the nature of work you carried out.

- **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
- **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

C. Learning outcomes:

After completion of the course students will able:

- CO1 To apply knowledge and skills gained in company/industry/organization to real-world problems and to solve engineering problems.
- CO2 To learn to work as a team and to work with teammates from other disciplines.
- CO3 To use experience related to professional and ethical issues in the actual work environment.
- CO4 To explain the impact of engineering solutions employed in a project, in a global, economic, environmental, and societal context.
- CO5 To find relevant sources (e.g., library, Internet, experts) and gathers information and to demonstrate knowledge of contemporary issues related with engineering in general and to use new tools and technologies.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	3	2	1	1	1	3	3	3	3	3	3
CO2	-	-	-	-	-	-	-	-	3	2	-	-	-	-	-
CO3	-	-	-	-	-	1	-	3	-	-	-	-	-	-	-
CO4	3	3	2	2	2	2	3	1	-	1	1	2	1	1	1
CO5	2	3	2	3	3	2	2	1	-	2	1	2	2	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Summer Internship Grade Form

Confidential

Name :

Supervisor Name :

Email :

Company name :

Department :

Duration :

Part-A: Work place

Average of the grades on the Summer Internship Evaluation Form filled by the employer : _____

Is the work done related to engineering? [Y/N] : _____

Does the supervisor has a similar engineering background? [Y/N] : _____

..... If all conditions in Part-A are satisfied, continue to Part-B, else mark Unsatisfactory in Overall Evaluation

Part-B: Report

Satisfactory ☐

Revision required ☐

If revision is required, changes needed must be stated on the report. The report is returned to the student until satisfactory.

Due date for resubmission:

...../...../20....

Student is given two weeks for each revision.

To be set by the department coordinator

..... If the report in Part-B is Satisfactory, continue to Part-C, else return it to the student for Revision

Part-C: Final version of the report

Based on the final version of the report, as evaluated on the back side of this form:

Sum of the Assessment/quality scores of Performance Criteria : _____
To be satisfactory, the sum must be at least 50.

The Assessment/quality score of Report Quality : _____
To be satisfactory, the score must be at least 7.

Overall Evaluation

Satisfactory ☐

Unsatisfactory ☐

Evaluator:

Name:.....

Signature

Date

...../...../20....

Evaluation of the Company/Department

- ☐ I strongly recommend this place for future students
- ☐ I am satisfied with this place
- ☐ I recommend this place not be allowed for future students.

Performance Criteria	On what page(s) of the report is the evidence of this found?	Assessment/quality score (from 0=missing to 15=full)
1. Able to apply knowledge and skills learned in school to real-world problems		
2. Able to solve engineering problems related to systems and applications		
3. Able to function as a team		
4. Able to work with teammates from other disciplines		
5. Aware of professional and ethical issues in the work environment		
6. Able to explain the impact of engineering solutions, employed in a project, in a global, economic, environmental, and societal context		
7. Find relevant sources (e.g. library, Internet, experts, seminars) and gathers information		
8. Demonstrate knowledge of contemporary issues related with engineering in general		
9. Able to use new tools and technologies		
Report Quality	On what page(s) of the report is the counter evidence of this found?	Assessment/quality score (from 0=missing to 15=full)
Able to prepare reports with high standards in terms of content, organization, style and language (the Summer Internship report itself is to be evaluated)		

¹If you think that a question does not apply to this particular summer internship, please write NA (not applicable).

CL453 CONSTRUCTION PROJECT MANAGEMENT
B TECH 7th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Construction Project Management	07
2	Project Feasibility Study	03
3	Management Techniques	05
4	Project management through networks	05
5	Critical Path Method	10
6	Program Evaluation and Review Technique (PERT)	05
7	Resource Allocation and Resource Scheduling	05
8	Contracts and Tender	05

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

B. Detailed Syllabus:

1	Introduction to Construction Project Management	07 Hours	15%
1.1	Introduction to Construction Management		
1.2	Necessity, objectives of construction management		
1.3	Unique features of Construction Project		
1.4	Stages in construction project management, Factors affecting various stages of construction management		
1.5	Construction Resources		
1.6	Function of Construction Management		
1.7	Causes of failures construction projects and its relevance with construction management		

2	Project Feasibility Study	03 Hours	6%
2.1	Need of Project Feasibility Study		
2.2	Technical Analysis		
2.3	Financial Analysis		
2.4	Economic Analysis		
2.5	Ecological Analysis		
3	Management Techniques	05 Hours	11%
3.1	Work Breakdown structure for various projects		
3.2	Gantt or Bar chart		
3.3	Mile stone Chart		
3.4	Line of Balance technique		
4	Project management through networks	05 Hours	11%
4.1	Definition and Objective of network techniques		
4.2	Types of network		
4.3	Elements of network like activity, dummy and event		
4.4	Interrelationship of Events, Interrelationship of Activity		
4.5	Network Rules and Fulkerson's rule for Numbering Events		
4.6	Advantages of Network techniques over conventional techniques		
5	Critical Path Method	10 Hours	24%
5.1	Introduction		
5.2	Activity Time estimate-EST, LST, EFT, LFT		
5.3	Event time estimate- T_E , T_L		
5.4	Critical activities, Critical path and Floats		
5.5	Cost optimization by crashing a Network		
5.6	Updating of CPM Network		
6	Program Evaluation and Review Technique (PERT)	05 Hours	11%
6.1	PERT Network : Introduction , Difference between PERT and CPM		
6.2	Expected time, frequency distribution, mean, variance and standard deviation		
6.3	PERT Network Analysis		
7	Resource Allocation and Resource Scheduling	05 Hours	11%
7.1	Introduction		
7.2	Recourse smoothing		
7.3	Resource levelling		

8 Contracts and Tender

05 Hours

11%

- 8.1 Contract- definition , Essentials of contract
- 8.2 Types of contract & Suitability for construction projects
- 8.3 Tender – definition, Tender Notice and Tender submission
- 8.4 Opening of tender, Scrutiny of tender, Acceptance of tender
- 8.5 Rejection of Tender

C. Course Outcomes:

On the successful completion of this course, the students will be able to:

- CO1 Explain various functions of construction management.
- CO2 Identify alternative methods of course of action and select best course of action through feasibility analysis of construction project.
- CO3 Apply basic principles of management and its applications in construction industry.
- CO4 Estimate, analyse and control project durations by using various management techniques like Bar Charts, PERT and CPM in construction industry
- CO5 Manage resources through cost time trade off, updating, rescheduling and compressing with applications in construction industry.
- CO6 Understand and apply knowledge of the tendering procedure and various construction contract systems.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	-	-	-	-	-	2	-	-	-	-	3	-	-	2	-
CO2	2	2	3	1	-	1	2	1	-	1	-	-	2	-	2
CO3	3	-	-	-	-	-	-	-	1	-	-	1	1	-	-
CO4	1	-	-	-	2	-	-	-	-	3	3	-	1	-	1
CO5	1	1	1	-	2	2	2	1	1	-	-	-	1	2	1
CO6	-		1	-		3	-	2	2	2	-	2	1	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Chitkara, K. K., Construction Project Management Planning, Scheduling and Controlling, Tata McGraw Hill, New Delhi.
2. Jha, K.N., Construction Project Management, Pearson Publications.
3. Seetharaman, S., Construction Engineering & Management, Umesh Publications, 2007.
4. Peurifoy, L., Schexnayder, C.J. and Shapira, A., Construction Planning, Equipment and Methods, McGraw Hill, New Delhi, 8th Edition, 2010.
5. Punamia, B.C. and Khandelwal, K.K., Project Planning and Control with PERT and CPM, Laxmi Publications, New Delhi, 2004.
6. Kotadia, A.S., Construction Management and Equipments, Mahajan Publishing House.

Reference Books:

1. Sharma, M.R., Fundamentals of Construction Planning and Management, S.K. Kataria & Son, New Delhi, 2012.
2. Srinath, L.S., PERT & CPM Principles and Applications, Tata McGraw Hill, New Delhi.
3. Gahlot, P.S. and Dhir, B.M., Construction Planning & Management, New Age International (P) Ltd., New Delhi.
4. Sharma, S.C., Construction Equipment & Management, Khanna Publications, New Delhi, 1988.
5. Sengupta and Guha, Construction Management and Planning, Tata McGraw Hill, New Delhi.

Web Materials:

1. http://nptel.iitm.ac.in/courses/IIT-MADRAS/Infrastructure_Planning_Management/index.php
2. http://www.deere.com/en_US/cfd/construction/deere_const/media/pdf/attachments.pdf
3. http://www.fta.dot.gov/documents/Construct_Proj_Mangmnt_CD.pdf
4. <http://www.netmba.com/operations/project/pert/>
5. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-%20Guwahati/cpm/index.html>
6. <http://www.youtube.com/watch?v=wJ8HZ7hqUs8>
7. <http://www.youtube.com/watch?v=IOnerkINAO>
8. <http://www.youtube.com/watch?v=2Ow8JUgRCIQ>
9. <http://www.youtube.com/watch?v=UEXrsZ3vkx0>

10. <http://www.youtube.com/watch?v=7cCaY3zBhcs>
11. <http://www.youtube.com/watch?v=HPC4lWTMjRM>
12. <http://www.youtube.com/watch?v=RYnUDLey-g4>

CL454: GEOTECHNICAL ENGINEERING – II
B TECH 7th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	1	2	6	4
Marks	100		50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Soil Exploration	06
2	Earth Pressure	10
3	Stability of Slopes	10
4	Shallow Foundation	10
5	Deep Foundation	14
6	Stress Distribution	08

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours (Tutorial): 15

Total Hours: 90

B. Detailed Syllabus:

- | | | |
|--|-----------------|-------------|
| 1 Soil Exploration | 06 Hours | 13% |
| 1.1 Introduction, Stages in sub-surface Exploration, Depth of Exploration, Location and number of pits and Borings | | |
| 1.2 Methods of boring/exploration | | |
| 1.3 Sampling of soils, types of soil samplers | | |
| 1.4 Standard penetration test, Cone penetration test: Static and Dynamic | | |
| 1.5 Field plate load test | | |
| 2 Earth Pressure | 07 Hours | 16 % |
| 2.1 Introduction, effects of wall movements on earth pressure, lateral | | |

earth pressure at rest		
2.2 Rankine's & Coulomb's theory for active and passive earth pressure conditions for cohesionless and cohesive backfill		
3 Stability of Slopes	09 Hours	20 %
3.1 Introduction, types of slope failure, factor of safety		
3.2 Slice method, friction circle method, Taylor's stability number & other methods of analysis		
3.3 Improving stability of slopes		
4 Shallow Foundation	08 Hours	18 %
4.1 Introduction, bearing capacity of soil, types of failure in soil, allowable bearing pressure		
4.2 Terzaghi's bearing capacity theory, factors affecting bearing capacity, depth of foundation		
4.3 Settlement-consideration & computation, effect of water-table		
4.4 IS code of practice for computing bearing capacity		
5 Deep Foundation	10 Hours	22 %
5.1 Introduction, types of piles, method of installation and load carrying behavior, necessity of pile foundation		
5.2 Static pile load formulae, pile load test, dynamic pile formulae		
5.3 Bearing capacity of single pile subjected to vertical load in sands and clays, pile subjected to uplift load		
5.4 Negative skin friction, group action of piles in Cohesionless & cohesive soil		
6 Stress Distribution	05 Hours	11 %
6.1 Stress strain parameters, geostatic stresses, concentrated force - Boussinesq's equations		
6.2 Pressure distribution diagram, vertical stress distribution on horizontal plane, vertical stress distribution on vertical plane, vertical pressure distribution under uniformly loaded circular area, line load, strip load		
6.3 Newmark's influence charts, Westergaard's analysis		
6.4 Contact pressure distribution		

C. Course Outcomes (COs):

- CO1 Student will have the understanding of various soil exploration techniques and also the various I.S. code criteria for SPT test and results.
- CO2 Student will understand the analysis of lateral earth pressure against retaining wall.
- CO3 Students will learn to analysis the slope stability using different methods.
- CO4 Students will gain the knowledge to determine the bearing capacity of shallow and deep foundation in sand and clay.
- CO5 Students will learn to determine the vertical pressure through different methods.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	1	1	-	-	2	2	3	1	1	-	1
CO2	3	2	2	1	2	-	-	-	2	1	1	1	1	-	-
CO3	3	2	2	1	2	-	-	-	2	1	1	1	1	-	-
CO4	3	3	2	1	2	-	-	-	2	1	1	1	1	1	1
CO5	3	1	1	1	1	-	-	-	2	1	1	1	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Arora, K.R., Soil Mechanics & Foundation Engineering, Standard Publication, New Delhi.
2. Punamia, B.C., Soil Mechanics & Foundation Engineering, Laxmi Publication Pvt. Ltd., New Delhi.
3. Murthy, V.N.S., Soil Mechanics & Foundation Engineering; SaiKripa Technical Consultants, Bangalore.
4. Shroff A. V., Shah D. L., “Soil Mechanics & Geotechnical Engineering”, Oxford-IBH New Delhi.

Reference Books:

1. Singh Alam, Soil Engineering, Vol. – I and II, Asia Publication House.
2. Fang and Einterkorn, Foundation Engineering Handbook.

3. Peck, Thomson and Thornburn, Foundation Engineering,
4. ShamsheerPrakash and GopalRanjan, Analysis and Design of Retaining Structures,
5. Sarita Publications.
6. Nayak, N.B., Foundation Engineering Manual.

Web Materials:

1. <http://edudel.nic.in>
2. <http://bis.org.in/other/quake.htm>
3. http://www.vastu-design.com/india_homes.htm
4. <http://www.thepeninsulaneighborhood.com/ThePlan.html>
5. http://www.historytution.com/indus_valley_civilization/town_planning.html
6. <http://nptel.ac.in/courses/105101083/>

LIST OF EXPERIMENTS

Experiment No.	Name of Experiment
1	Relative Density
2	Standard Proctor Compaction Test
3	Unconfined Compression Test
4	Free Swell Index
5	Laboratory Vane Shear Test
6	Standard Penetration Test
7	Laboratory Plate Load Test
8	Static Cone Penetration Test
9	CBR Test

CL455: PROFESSIONAL PRACTICES
B. TECH 7th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Total	Credit
Hours / Week	3	1	4	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Estimation	04
2.	Rules & Method of Measurement	03
3.	Detailed Quantity Estimation	22
4.	Specification	03
5.	Rate Analysis	05
6.	Valuation	08

Total Hours (Theory): 45

Total Hours (Lab): 0

Total Hours (Tutorial): 15

Total Hours: 60

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|----|
| 1. | Estimation | 04 Hours | 8% |
| 1.1 | General | | |
| 1.2 | Purpose of estimates | | |
| 1.3 | Types of estimates | | |
| 1.4 | Approximate estimate, general principle of approximate estimate, Construction cost index, approximate method of costing for buildings, water supply and sanitary works. | | |
| 1.5 | Detail estimate, data required for detail estimate, preparation of detail estimate, Standard measurement form, Abstract of | | |

measurement. Factors to be consider during preparation of a detailed estimate.

1.6 Procedure of estimate

1.7 Quantity estimate, Revised estimate, Supplementary estimate, complete estimate and Annual maintenance or repair estimate Data required for sanction of project, Administrative approval, expenditure sanction, Technical Sanction

2 Rules and Methods of Measurement **03 Hours** **6%**

2.1 General rules which are applicable during the calculation of quantity for different associated items

2.2 Principles in selecting units of measurement for items, various units and standard modes of measurement for different item of work and materials as per IS:1200 revised

2.3 Deduction criteria for various item of works

3. Detailed Quantity Estimation **22 Hours** **50%**

3.1 **Building Estimate:** General items of work for building estimate and their units of measurements, long wall short wall and centre line methods.

- Estimation of quantity of load bearing structure.
- Quantity analysis of R.C.C. component: footing, column, lintel, chajja, beam, slab, with schedule of bar bending, material estimate.
- Quantity analysis of steel roof truss
- Abstract preparation

3.2 Estimation of earthwork for roads.

4 Specifications **03 Hours** **7%**

4.1 Definition

4.2 Objective, importance, use, types of specification

4.3 General and special specification

4.4 Specification for material and workmanship

4.5 Design and principles of specification

4.6 Sources of information

4.7 Typical specification of various item of works

5	Rate Analysis	05 Hours	12%
5.1	Rate analysis and requirement of the rate analysis		
5.2	Factors affecting rate analysis		
5.3	Method of preparation of rate analysis of work		
5.4	Quantity of materials per unit rate of work		
5.5	Estimating Labor		
5.6	Cost of equipment or tools and plants		
5.7	Overhead expenses		
5.8	Contractor Profit		
5.9	Task of work		
5.10	Load of trucks		
5.11	Rate analysis of all typical items of works for building construction		
6	Valuation	08 Hours	17%
6.1	Valuation, value, price and cost, purpose and principle of valuation		
6.2	Different form of value		
6.3	Mortgage and lease		
6.4	Freehold and Leasehold property		
6.5	Sinking Fund, depreciation, types of depreciation		
6.6	Year of purchase		
6.7	Outgoings		
6.8	Methods of Valuation		
6.9	Rent Fixation		

C. Course Outcomes:

On the successful completion of this course

- CO1 The students will get a diverse knowledge of estimating, costing and professional practice, which will be use full in tackling real life problems.
- CO2 The students will be able to understand the procedure to carry out the estimation and steps to prepare reports of the construction works.
- CO3 The students will learn the purpose and importance of valuation

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	1	-	-	1	1	-	-	2	-	2	-	1	1	1
CO2	2	-	-	-	1	1	-	-	1	1	-	-	1	-	-
CO3	2	-	-	-	1	1	-	-	1	1	-	-	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Dutta, B.N., Estimating & Costing in Civil Engineering Theory and Practice, UBS Publishers & Distributors Limited, New Delhi, 1997.
2. Chakraborti, M., Estimating, Costing, Specification and Valuation on Civil Engineering, M. Chakraborty Publication, 2007.
3. Roshan Namavati, Theory & Practice Of Valuation, (Land & Buildings) for Architects, Engineers, Surveyors, Advocates, & Income Tax Practitioners, Universal Book Corp

Reference Books:

1. Patil, B.S., Civil Engineering Contracts, Vol. – I, Orient Longman Publication, 1998.
2. Rangwala, S.C., Elements of Estimating and Costing, Professional practice, Charotar Publishing House, Anand.
3. Aggarwal, A., Upadhyay, A.K., Civil Estimating, Costing & Valuation, S.K Kataria & Sons, New Delhi.
4. Chandola, S.P. and Vazirani, Estimating and Costing, Khanna Publication.

Web Materials:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT%20Guwahati/cpm/index.html>

CL470: STRUCTURAL DYNAMICS & EARTHQUAKE ENGINEERING
B TECH 7th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE-III

Credits and Hours:

Teaching Scheme	Theory	Practical/ Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Theory of vibrations	15
2	Earthquake Basics & Design Philosophy	08
3	Lateral Load Analysis of Building	10
4	Ductile Detailing	06
5	Special Topics	06

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|----------|---|-----------------|------------|
| 1 | Theory of vibrations | 15 Hours | 34% |
| 1.1 | Introduction, Difference between Static and Dynamic load, Basic Terminologies, | | |
| 1.2 | Simplified single degree of freedom system, mathematical modeling of buildings, natural frequency, resonance v/s increased response | | |
| 1.2 | Responses of buildings to different types of vibrations like free and forced, damped and un-damped | | |
| 1.3 | Response to multi degree (maximum three) of freedom systems, mode shapes | | |
| 2 | Earthquake Basics & Design Philosophy | 08 Hours | 18% |

2.1	Introduction of Earthquake, Definitions of basic terms,		
2.2	Causes of earthquake and their characteristics, Seismographs, Seismic Zoning practices in India, Seismic Codes		
2.3	Earthquake parameters, Characterization of ground motion		
2.4	Earthquake Intensity & Magnitude, Recording Instruments		
2.5	Philosophy of earthquake resistant design, earthquake proof v/s earthquake resistant design		
2.6	Four virtues of earthquake resistant (strength, stiffness, ductility and configuration), Seismic structural configuration		
2.7	Damages caused during past earthquakes (worldwide)		
3	Lateral Load Analysis of Building	10 Hours	22%
3.1	Lateral analysis of the building systems, Lateral load distribution,		
3.2	Torsionally coupled and un-coupled Systems		
3.3	Design Lateral Loads for RC Building, Earthquake resistant design of masonry structure as per IS 4326, Concept of Response Spectra, Design Response Spectrum		
3.4	Equivalent lateral load concept, Rigid floor diaphragm, Codal Provisions		
4	Ductile Detailing	06 Hours	13%
4.1	Importance of ductility - Methods of introducing ductility into RC structures, Ductile detailing of various structural elements, Design Methodology IS 1893, IS 13920 and IS 4326 - Codal provisions - Design as per the codes		
5	Special Topics	06 Hours	13%
5.1	Introduction of different control systems: Passive control: base isolation and active control: bracing system, TMD and some latest invention		
5.2	Soil liquefaction, causes and its remedial measure		
5.3	Introduction to Disaster Management: Types of Disaster, Phases of disaster management, Disaster rescue, psychology and plan of rescue operations		

C. Course Outcomes

On the successful completion of this course, the students will be able to

- CO1 Apply the concepts of theory of vibration to SDOF and MDOF and understand the response of structures.
- CO2 Understand basic earthquake mechanisms, tectonics, types of ground motion, and propagation and interpret the earthquake ground motion data, and also understand causes types and measurement of an earthquake.
- CO3 Analyze and design of various lateral load resisting systems to develop response spectrum for earthquake ground motion and estimate lateral load on the structures as per codal stipulations.
- CO4 Apply the concepts of ductility and related codal specification for earthquake resistant design.
- CO5 To Know the importance of energy dissipation devices, Effect of Soil liquefaction and role of disaster Management during Earthquake.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	-	2	-	-	-	-	-	1	-	-	2	-	-
CO2	2	2	1	-	-	-	-	2	-	-	-	1	2	2	-
CO3	3	2	3	1	1	-	-	2	-	1	-	1	3	2	2
CO4	3	2	3	2	1	-	-	2	-	1	-	1	3	1	1
CO5	1	1	1	1	-	-	-	1	-	-	-	1	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Chopra, A.K., Dynamics of Structures, PrenticeHall, N.J.
2. MarioPaz, StructuralDynamics, McGraw-Hill
3. Manish Shrikhande & PankajAgrawal; Earthquake resistant design of structures, PHI Publication, New Delhi
4. Duggal, S.K., Earthquake resistance design of structures ;Oxford University Press, New Delhi.

Reference Books:

1. Park & Pauly, Behavior of RC Structure
2. Clough & Penzin, Dynamics of structures, McGraw-Hill.
3. IITK-GSDMAEQ27-V-3.0, Design Example of a Six Storey Building
4. Murthy, C.V.R., Earthquake Tips, NICEE
5. Jain A. K., Dynamics of Structures with MATLAB, Pearson

Web Materials:

1. http://en.wikipedia.org/wiki/Earthquake_engineering
2. <http://www.curee.org>
3. <http://www.earthquakeengineering.com/>
4. <http://www.nicee.org/>
5. <http://www.earthquakeinfo.org/>
6. opensees.berkeley.edu/
7. <http://nptel.ac.in/>
8. <http://ocw.mit.edu/courses/civil-and-environmental-engineering/>
9. www.eeri.org/

Other Materials:

1. IS:875, Code of Practice for Design Loads
2. IS:1893(Part-I), Criteria for Earthquake Resistant Design
3. IS:4326, Earthquake Resistant Design and Construction of Buildings- Code of Practice
4. IS:4327, Earthquake Resisting Design & Construction Building
5. IS:13920, Ductile Detailing of RC Structures
6. IITK-GSDMA Guidelines

CL471: ADVANCED STRUCTURAL ANALYSIS
B TECH 7th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE-III

Credits and Hours:

Teaching Scheme	Theory	Tutorial / Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	02
2	Direct Stiffness Method	28
3	Finite Element Method	15

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction	02 Hours	04 %
1.1	Basic concepts of Analysis		
2	Direct Stiffness Method	28 Hours	62%
2.1	Introduction: Stiffness Method		
2.2	Overview of different stiffness & rotation-transformation matrices		
2.3	Analysis of beam, plane truss, plane frames, grid, space truss and space frame by stiffness method member approach		
2.4	Analysis of beam, plane truss and plane frame under various secondary effects like support sinking, prestraining and temperature effect		
2.5	Symmetry and Anti-symmetry		
2.6	Oblique supports and elastic supports		

- 3 **Finite Element Method** 15 Hours 34%
- 3.1 Introduction to FEM, Advantages and disadvantages of FEM
- 3.2 Types of problems, stresses & equilibrium, strain-displacement relations, stress-strain relations
- 3.3 Application of FEM to one dimensional (for bar & beam) & two dimensional problems using constant strain triangles
- 3.4 Two dimensional iso-parametric elements: Four node quadrilateral elements, numerical integration, higher order elements
- 3.5 Application of FEM to two dimensional truss element

C. Course Outcomes:

On the completion of the course the students will be able:

- CO1 To solve the structural problems using Finite element method.
- CO2 Use Direct stiffness method for the analysis of Frame structures
- CO3 To recognize the importance of structural analysis and the different tools used to determine the response of a structural system to external loads.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	1	1	-	-	-	-	-	-	2	-	1
CO2	3	3	2	2	1	-	-	-	-	-	-	-	1	-	-
CO3	2	2	1	1	2	-	-	-	-	-	1	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Weaver William and Gere James, Matrix Analysis of Framed Structures, CBS Publishers
2. Dawe, D.J., Matrix and Finite Element Displacement Analysis of Structures, Clarendon Press.
3. Menon Devdas, Advanced Structural Analysis, Narosa Publishing House
4. Desai & Ables, Introduction to the Finite Element Method, CBS

5. Bhavikatti S.S., Finite Element Analysis, New age international limited, publishers

Reference Books:

1. Krishnamoorthy C.S., Finite Element Analysis, Tata McGraw-Hill
2. Cook, R.D., Concepts & Applications of Finite Element Analysis, John Wiley & Sons
3. Wang, C.K., Intermediate Structural Analysis, Tata McGraw Hill

CL477: FIELD APPLICATIONS OF GEOTECHNICAL ENGINEERING
B TECH 7th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE-III

Credits and Hours:

Teaching Scheme	Theory	Tutorial/Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Reinforced Soil Structures	10
2	Ground Improvement Techniques	12
3	Foundation of Marine Structures	09
4	Geosynthetics in Civil Engineering	06
5	Stabilization of soils for road constructions	08

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Reinforced Soil Structures	10 Hours	22%
1.1	Concept of soil reinforcement		
1.2	Reinforcing materials and their properties		
1.3	Design of soil reinforcement for stability		
1.4	Durability of reinforcement materials		
1.5	Applications of Reinforced earth structures		
2	Ground Improvement Techniques	12 Hours	27%
2.1	Principles of ground improvement		
2.2	Properties of compacted soil, Compaction control tests.		
2.3	Mechanical Stabilization: Principle of mechanical modification,		

- Dynamic Stabilization, Vibro-floatation, Precompression and Compaction piles.
- 2.4 Hydraulic modification, Dewatering systems, Preloading and vertical drains, Sand Drain, Electro-kinetic dewatering
- 2.5 Chemical modification, Modification by admixtures
- 2.6 Bearing capacity & Slope stability improvement
- 3 Foundation of Marine Structures 09 Hours 20%**
- 3.1 Origin, Nature and distribution of marine soils and their engineering properties
- 3.2 Sampling and sample disturbance, In-situ testing of marine soil
- 3.3 Offshore platforms
- 3.4 Shallow foundations on marine soil
- 3.5 Deep foundations on marine soil
- 4 Geosynthetics in Civil Engineering 06 Hours 13%**
- 4.1 Introduction: An overview on the development and applications various geosynthetics - geotextiles, geogrids, geonets, geomembranes and geocomposites.
- 4.2 Designing with geotextiles: Geotextile properties and test methods, functions, designing for separation, reinforcement, stabilization, filtration, drainage.
- 4.3 Different test on geosynthetics
- 5 Stabilization of Soils for Road Construction 08 Hours 18%**
- 5.1 The need for a stabilized soil, choice of stabilizers
- 5.2 Testing and field control
- 5.3 Stabilization in India for rural roads
- 5.4 Use of geo-fabrics in unpaved road construction.
- 5.5 Case studies.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1 Learn the concept and designing of reinforced soil structure.

CO2 Understand the principle of ground improvement technique and various

different methods to improve the soil properties in economical way.

CO3 Identify important aspects of foundation design for marine structures.

CO4 Identify which type of geosynthetics is required for given engineering problems.

CO5 Apply fundamental knowledge of soil quality improvement in design & construction of road.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	3	3	2	1	2	2	1	1	2	2	3	3	2	3
CO2	3	2	2	2	1	2	2	2	1	2	2	2	3	3	3
CO3	3	3	3	3	2	2	3	2	2	2	3	3	3	3	3
CO4	2	2	2	3	1	2	2	1	2	2	2	3	2	2	2
CO5	3	3	3	2	2	2	3	2	2	2	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Koerner, R.M., Designing with Geosynthetics, (Third Edition), Prentice Hall, 1997.
2. Reinforced Soil and Geo-textiles- J. N. Mandal, proceedings FIGC- 1988, Oxford and IBH publishing company private Ltd., New Delhi.
3. Khanna S.K and Justo C.E.G., Highway Engineering, New Chand and Brothers, Roorkee, 1998.
4. H.G.Poulos, Marine Geotechniques, Unwin Hyman, London.
5. Swamisaran, Analysis and Design of Substructures, Oxford & IBH Publishing Company Private Ltd., Delhi.
6. Dr. Purushothama Raj, Ground Improvement Techniques, Laxmi Publications, New Delhi.

Reference Books:

1. Jewell, R.A., Soil Reinforcement with Geotextile, CIRIA, London, 1996.
2. Remote Sensing in Civil Engineering, by Kennie, T.J.M. and Matthews M.C. Surrey University Press, Glasgow. Taylor,

3. Hausmann, M.R., Engineering Principles of Ground Modification, McGraw Hill, 1990.
4. Jones, C.J.F.P., Earth Reinforcement and Soil Structures, Butterworth Publications, 1996.

Web Materials:

1. <http://edudel.nic.in>
2. <http://bis.org.in/other/quake.htm>
3. <http://www.thepeninsulaneighborhood.com/ThePlan.html>

LIST OF TUTORIALS

Tutorial No.	Name of Tutorial
1	Reinforced Soil Structures
2	Ground Improvement Techniques
3	Foundation of Marine Structures
4	Geosynthetics in Civil Engineering
5	Stabilization of Soils for Road Construction

Credits and Hours:

A. Outline of the Course:

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Railway Engineering	12 Hours	26 %
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1.1 Introduction

Permanent way and railway track components, Ideal requirement of permanent way, Capacity of railway track, Gauge

Rails-function and types of rails, rail sections, defects in rails, creep of rails, rail joints and welding of rails

Sleepers - function, types, spacing and density

Ballast- Function and Types of Ballast material, Sub-ballast

	Rail fixtures and Fastenings Subgrade and embankment.		
1.2	Geometric design of Railway track Gradients, Speed of the trains on curves, Super elevation, Curves, Radius of the curves, Widening on curves.		
1.3	Crossing, Stations, Yards, Signaling and Interlocking Points and crossings - turnouts, switches, crossings. Track junctions - types and salient features. Railway stations - requirements, facilities, classifications, platforms, loops, sidings, Railway yards - types, required equipment in yards. Signaling and control system - Necessity, objectives, classification. Interlocking of signals and points, Mechanical Interlocking devices.		
1.4	Railway Track Railway track - construction, drainage, maintenance. Recent developments in railways - high speed trains, modernization in track for high speed, Metro rails, Monorail, Maglev Rails, Tube Rails, Automation in operation and control.		
2	Bridge Engineering	05 Hours	11 %
2.1	Introduction: Selection of site, Data collection, Stages of investigation		
2.2	Classification, types of substructures, flooring joints, bridge bearings, movable bridges, temporary bridges, Construction methods and Maintenance of Bridge.		
3	Tunnel Engineering	06 Hours	14 %
3.1	Introduction: Necessity/Advantage of a tunnel, Classification of Tunnels, Size and shape of a tunnel, Alignment of a Tunnel, Portals and Shafts		
3.2	Lighting, Ventilation and Dust control: Tunnel Lighting, Ventilation of Tunnel, Methods of Ventilation, Dust control, Drainage system of Tunnel and Safety		

4	Airport Engineering	14 Hours	31 %
4.1	General: Introduction to Air transport authorities, Air crafts and its characteristics, Airport classifications as per ICAO.		
4.2	Airport Planning: Regional planning - concepts and advantages, location and planning of airport as per ICAO and F.A.A. recommendations, airport Elements - airfield, terminal area, obstructions, approach zone, zoning laws, airport capacity, airport size and site selection.		
4.3	Run Way Design: Wind rose and orientation of runway, wind coverage and crosswind component, factors affecting runway length, basic runway length, and corrections to runway length, runway geometrics and runway patterns (configurations). Runway marking, threshold limits cross section of runway		
4.4	Taxiway Design: Controlling factors, taxiway geometric elements, layout, exit taxiway, location and geometrics, holding apron, turnaround facility. Aprons - locations, size, gate positions, aircraft parking configurations and parking systems, Fuel storage area, blast pads, wind direction indicator		
4.5	Air Traffic Control and Visual Aids: Air traffic control objectives, control system, control network - visual aids - landing information system, airport markings and lighting.		
5	Harbour Engineering	08 Hours	18 %
5.1	Harbour Planning: Harbour components, characteristics of good harbour and principles of harbour planning, size of harbour, site selection criteria and layout of harbours. Surveys to be carried out for harbour planning.		
5.2	Marine Structures: General design aspects, breakwaters - function, types general design principles, wharves, quays, jetties, piers, pier heads, dolphin, fenders, mooring accessories - function, types, suitability, design and construction features.		
5.3	Docks and Locks: Tidal basin, wet docks - purpose, operation of lock gates and passage, repair docks - graving docks, floating		

docks.

- 5.4 Port Amenities and Navigational Aids: Ferry, transfer bridges, floating landing stages, transit sheds, warehouses, cold storage, aprons, cargo handling equipment, purpose and general description, Channel and entrance demarcation, buoys, beacons, light house, electronic communication devices.

C. Students Learning Outcomes:

The course content should be taught and learning imparted with the aim to develop required knowledge and skills so that they are able to acquire following competency:

- CO1 The students will gain an experience in the implementation of Railway, Bridge, Tunnel, Airport and Harbour Engineering on engineering concepts which are applied in field of Transportation Engineering.
- CO2 The students will get a diverse knowledge of Railway, Bridge, Tunnel, Airport and Harbour engineering practices applied to real life problems.
- CO3 The students will learn to understand the theoretical and practical aspects of Railway, Bridge, Tunnel, Airport and Harbour engineering along with the design and management applications.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	-	2	-	1	1	1	-	3	2	-	1	2	1	1
CO2	1	-	2	1	2	1	-	-	2	3	2	2	2	1	1
CO3	1	-	2	-	3	-	-	-	2	2	-	1	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Satish Chandra and M.M. Agrawal, Railway Engineering, Oxford University Press, New Delhi
2. S.C. Rangwala, K.S. Rangwala and P.S. Rangwala, Principles of Railway Engineering, Charotar Publishing House, Anand.

3. S.P.Bindra, Principles and Practice of Bridge Engineering, Dhanpat Rai & Sons, New Delhi
4. S.C. Saxena, Tunnel Engineering, Dhanpat Rai & Sons, New Delhi
5. Dr. S. K. Khanna, M.G.Arora and S.S. Jain, Airport Planning & Design, Nem Chand & Bros., Roorkee
6. Airport Engineering, Charotar Publishing House Pvt. Ltd, Anand
7. R. Srinivasan and S. C. Rangwala, Harbour, Dock and Tunnel Engineering, 1995, Charotar Pub.House, Anand

Reference Books:

1. S.C.Saxena And S.P. Arora, A Text Book of Railway Engineering, Dhanpat Rai Sons, New Delhi
2. D.J. Victor, Essential of Bridge Engineering, Oxford & IBH Pub. Co. Ltd. Mumbai
3. G.V. Rao Airport Engineering, Tata McGraw Hill Pub. Co., New Delhi
4. S. P. Bindra, A Course in Docks and Harbour Engineering, 1992, Dhanpat Rai & Sons, New Delhi
5. Alonzo Def. Quinn, Design and Construction of Ports and Marine Structure, McGraw - Hill Book Company, New York

Web Materials:

1. <http://www.cphbooks.com/html/40ae.htm>
2. <http://as.wiley.com/WileyCDA/WileyTitle/productCd-0471527556.html>
3. <http://cphbooks.com/html/38re.htm>

LIST OF TUTORIALS

Tutorial No.	Name of Tutorials
1	Introduction, Railway Track Gauge, Alignment of Railway lines
2	Track and Track stresses, Rails, Sleeper
3	Ballast, Track fittings, Geometric design of Track
4	Points and crossings, Railway Stations and Yards
5	Signaling and Interlocking
6	Bridge –General, Classification of Bridge Construction methods, Maintenance

7	Tunnel Classification and Alignment
8	Lighting, Ventilation, Drainage and Safety
9	Layout of Airport, components of airport, authorities of airport
10	Factors affecting runway design , Wind rose diagram, Design of runway
11	Design of Taxiway
12	Navigational aids of Airport
13	Components of harbor, factors affecting the site selection
14	Harbour infrastructure

CL479: ENVIRONMENTAL ENGINEERING- II
B TECH 7th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE-III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Solid Waste Management	08
2	Waste: Collection and Transportation	10
3	Waste Processing Techniques	09
4	Disposal of Solid Waste	10
5	Noise: Source, generation, Effects & Control	08

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | |
|---|----------------------|
| 1 Introduction to Solid Waste Management | 08 Hours 18 % |
| 1.1 Municipal Solid Waste Management - Sources, nature and characteristics; | |
| 1.2 Quantitative and qualitative aspects; Engineering principles, assessment and management. | |
| 1.3 Solid waste problems - Industrial, Mining, Agricultural, Domestic (urban) wastes, Biomedical waste, E-waste, Plastic Waste and Construction Waste, Management of lead acid battery. | |
| 2 Waste: Collection and Transportation | 10 Hours 22 % |
| 2.1 Collection and storage of municipal solid waste; Methods of | |

- collection - House to House collection
- 2.2 Type of Vehicles-Manpower requirement collection routes; on site storage methods-materials used for containers Reduction of solid waste at source-on site segregation of solid waste
- 2.3 Recycling and Reuse Need for transfer and transport; transfer station selection of location, operation and maintenance; transportation Methods manual, Mechanical methods with or without compaction, economy in transportation of waste optimization of transportation routes.
- 3 Waste Processing Techniques** **09 Hours 20 %**
- 3.1 Processing techniques-biological and chemical conversion technologies – composting and its methods
- 3.2 Vermicomposting, mechanical composting, In vessel composting
- 3.3 Incineration, pyrolysis, gasification.
- 4 Disposal of Solid Waste** **10 Hours 22%**
- 4.1 Segregation, reduction at source, recovery and recycle; dumping of solid waste-sanitary waste
- 4.2 Sanitary landfills-site selection-design and operation of sanitary landfill-secure landfills-landfill bioreactors-leachate and landfill gas management
- 4.3 Landfill closure and environmental monitoring-landfill remediation
- 4.4 Municipal solid waste in Indian conditions, legal aspects of solid waste disposal
- 5 Noise: Source, generation, Effects & Control** **08 Hours 18 %**
- 5.1 Fundamentals of Noise: Basics of Acoustics: Sound power, Sound intensity and Sound pressure levels
- 5.2 Plane, Point and Line sources, Multiple sources Outdoor and indoor noise propagation;
- 5.3 Effects of noise – noise induced deafness, presbycusis, acoustic trauma, other physiological and psychological effects
- 5.4 Special noise environments – infrasound, ultrasound, impulsive

sound and sonic boom.

5.5 Noise standards and indices.

5.6 Noise monitoring: Occupational, ambient and road traffic noise monitoring

C. Course Outcomes:

On the completion of the course one should be able to

- CO1 Evaluate the subject from the technical, legal and economical points by learning of all terms related to general solid waste management
- CO2 Examine the technical points that are required to set up a solid waste management system
- CO3 Apply the legal legislation related to solid waste management.
- CO4 Plan a solid waste management system for decision makers
- CO5 Link cause and effect of noise pollution
- CO6 Develop noise pollution mitigation/abatement strategies

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	1	-	-	-	2	1	-	-	-	-	-	-	1	-
CO2	-	3	1	3	-	1	2	-	1	2	1	-	-	1	2
CO3	1	1	1	1	-	3	2	3	1	2	-	-	-	-	2
CO4	-	1	3	1	2	2	2	2	2	2	3	3	3	3	3
CO5	3	1	-	3	-	1	1	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Garg, S.K., Environmental Engg. Vol. – II, Khanna Publications.
2. Peavy, Rowe and Tchobanoglous, Environmental Engg, Tata Mcgraw Hill.

Reference Books:

1. Solid Waste: Engineering Principles & Management Issues- G. Tchobanoglous, GH. Theisen & R. Eliassen, McGraw Hill Int. Ed. Singapore, 1977.

2. Hazardous Waste Management (2nd ed) - Lagrega, MD, P L Buchingham & JC Evans, McGraw Hill, NY, 2001.
3. Bioremediation Principles – Eweis, JB, Ergas SJ, Chang DYP and Schroeder ED, McGraw-Hill, Singapore, 1998.

Web Materials:

1. <http://nptel.iitm.ac.in>
2. <http://www.epa.gov>
3. <http://www.nesc.wvu.edu>
4. <http://www.indiaenvironmentportal.org.in>
5. <http://www.filtersource.com>
6. <https://dgserver.dgsnd.gov.in>

List of Experiments

Experiment No.	Name of Experiment
1.	Sample preparation; sampling techniques; conning and quartering method; overburden and other wastes sampling.
2.	Proximate Analysis and Ultimate Analysis
3.	Determination of Calorific Value
4.	Determination of pH & buffered pH
5.	Determination of EC & CEC
6.	Determination of exchangeable Na & K
7.	Determination of ESP and SAR
8.	Determination of organic matter and organic carbon C:N ratio;
9.	Determination of plant available P and total P
10.	Leachate Analysis
11.	Ambient and Source Noise Monitoring

CL480: WATER RESOURCES ENGINEERING II
B TECH 7th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE-III

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum Number of Hours
1.	Runoff and Hydrograph	08
2.	Floods	10
3.	Discharge Measurement	08
4.	Groundwater Exploration	06
5.	Sediment Transport in Alluvial Rivers and Channels	10
6.	Hydroelectric Power	03

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|-----|-------------------------------------|-----------------|------------|
| 1. | Runoff and Hydrograph | 08 Hours | 18% |
| 1.1 | Estimation of Runoff | | |
| 1.2 | Dependable Flow | | |
| 1.3 | Runoff Co – efficient, Runoff Cycle | | |
| 1.4 | Synthetic Unit Hydrograph | | |
| 1.5 | Distribution Graph | | |
| 1.6 | Triangular Unit Hydrograph | | |
| 1.7 | Instantaneous Unit Hydrograph | | |

2. Floods and Flood Routing	10 Hours	22%
2.1 Definition, Factors Affecting Flood, Classification of Floods		
2.2 Estimation of Floods		
2.3 Gumbel's Distribution		
2.4 Log-Pearson Type III Distribution		
2.5 Risk		
2.6 Flood Routing: Definition, Types of Flood Routing		
2.7 Flood Routing Through Channel		
2.8 Flood Routing Through Reservoir		
3 Discharge Measurement	08 Hours	18%
3.1 Definition, The stage of a River		
3.2 Measurement of Discharge: Area - Slope Method, Area - Velocity Method, Salt Titration Method		
3.3 Discharge Measurement by Hydraulic Structures, Hydraulic Model Method, Ultrasonic Method, Electromagnetic Induction Method, Moving - Boat Technique		
3.4 Stage Discharge Relation		
3.5 Stream Gauging Network		
4 Groundwater Exploration	06 Hours	13 %
4.1 Geophysical Investigations: Electrical Resistivity Method, Seismic Refraction and Reflection Method, Other Surveying Methods		
4.2 Borehole Geophysical Techniques: Electrical Logging, Radioactive Logging, Induction Logging, Sonic Logging, Fluid Logging, Well Construction Measurements, Downhole photography		
4.3 Water Well Design, Well Diameter, Depth, Design of Well Screen		
4.4 Open Well versus Borewells		
4.5 Infiltration Gallery		
4.6 Water Well Construction: Installation of Well Screens, Recovering Well Screens, Well Maintenance		

5	Sediment Transport in Channels	10 Hours	22 %
5.1	Importance of Sediment Transport		
5.2	Sediment Load, Bed Formation		
5.3	Mechanics of Sediment Transport		
5.4	Shield's Entrainment Method for Design of Non – Scouring Stable Channels having Protected Side Slope in Alluvium		
5.5	Stability of Channel Slopes (Design of Non – Scouring Channels with Unprotected Side Slope)		
5.6	Estimation of Transported Sediment in a channel: Suspended Load and Its Measurement, Bed Load and Its Measurement		
5.7	Lining of Irrigation Canals: Types and Economics of Lining		
6	Hydroelectric Power	03 Hours	7%
6.1	Thermal and Hydropower		
6.2	Classification of Hydel Plants		
6.3	Important Terms and Definitions Connected with Hydropower		
6.4	Principal Components of a Hydro – Electric Scheme		
6.5	Comparison of Hydropower and Thermal Power with Reference to Indian Conditions, Hydropower Potentials of India		
6.6	Electricity Generation from various Sources in the World		

C. Course Outcomes:

At the end of the course the students will be able to

- CO1 Perform engineering runoff and hydrograph computations.
- CO2 Estimate flood and perform flood routing.
- CO3 Implement different methods for discharge calculations in river.
- CO4 Know the procedure of ground water exploration, well construction and management.
- CO5 Calculate sediment load in a channel.
- CO6 Demonstrate fundamentals of hydropower generation.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	3	3	3	1	1	3	1	1	3	2	2	3	2	3
CO2	2	3	3	3	1	3	3	1	1	3	2	2	3	2	3
CO3	2	3	1	2	1	1	1	1	-	3	2	2	1	2	3
CO4	2	3	3	1	2	1	1	-	1	-	3	1	1	1	2
CO5	2	3	1	2	1	1	1	1	-	3	2	2	1	2	3
CO6	2	-	2	1	-	1	1	-	-	-	3	1	3	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Subramanya, K., Engineering Hydrology, Tata McGraw Hill, New Delhi.
2. Raghunath, H.M., Groundwater, 1987, Wiley Eastern Ltd., New Delhi.
3. Garg, S.K., Irrigation Engineering and Hydraulic Structures, Khanna Publishers, New Delhi.

Reference Books:

1. Garg, S.P., Groundwater and Tube Wells, 1993, Oxford & IBH Publishing Co.
2. Modi, P.N., Irrigation Water Resources and Water Power Engineering, Standard Book House, New Delhi.
3. Raghunath, H.M., Hydrology – Principles, Analysis and Design, 1986, Wiley Eastern Ltd.
4. Todd, D.K., Groundwater Hydrology, 1993 John Wiley & Sons.
5. M. J. Deodhar, Elementary Engineering Hydrology, Pearson India.
6. Punmia and Pande, Lal, B.B., Irrigation and Water Engineering, Standard Publishers Distributors, New Delhi.

LIST OF TUTORIALS

Tutorial No.	Name of Tutorial
1	Runoff
2	Hydrograph
3	Floods

4	Flood Routing
5	Discharge Measurement
6	Groundwater Exploration
6	Sediment Transport in Channels
8	Hydroelectric Power

CL481: DESIGN OF STRUCTURES USING INTERNATIONAL CODES**B. TECH 8th SEMESTER (CIVIL ENGINEERING)****PROGRAMME ELECTIVE - IV**

Credits and Hours:

Teaching Scheme	Theory	Practical / Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Overview of ACI 318	15
2.	Euro code 2 – Design of concrete Structures	30

Total Hours (Theory): 45**Total Hours (Lab): 30****Total Hours: 75****B. Detailed Syllabus:**

- | | | |
|---|-----------------|------------|
| 1 Overview of ACI 318 | 15 Hours | 33% |
| 1.1 Standards for tests and materials | | |
| 1.2 Construction Requirements: Durability, Concrete quality, mixing and placing, Formwork, Embedded pipes and construction joints | | |
| Details of Reinforcement | | |
| 1.3 General Requirements : Analysis and design- general considerations, Strength and serviceability requirements, Flexural and axial loads, Shear and Torsion | | |
| 1.4 Structural Systems or Elements: two way Slab systems, Walls, Footings, Composite concrete flexural members | | |
| 1.5 Informatives: Appendix C (Alternative load and strength reduction factor), Appendix E (Steel reinforcement information) | | |

2	Euro code 2 – Design of Concrete Structures	30 Hours	67%
2.1	Scope of Euro code 2		
2.2	Materials : Concrete, Reinforcing steel, Design compressive and tensile strengths, Ductility characteristics		
2.3	Durability and cover to reinforcement: Requirements for durability, Methods of verifications, Minimum cover		
2.4	Ultimate Limit States (ULS): Bending with or without axial force, Shear, Members not requiring design shear reinforcement, Members requiring design shear reinforcement, Shear between web and flanges of T-sections, Shear at the interface between concrete cast at different times, Shear and transverse bending, Torsion, Design procedure		
2.5	Serviceability Limit States (SLS): Stresses, Crack control, General considerations, Minimum reinforcement areas, Control of cracking without direct calculation, Calculation of crack widths, Deflection control, General considerations, Cases where calculations may be omitted.		
2.6	Detailing of members and particular rules: General, Beams, Shear reinforcement, Columns, Transverse reinforcement, Deep beams, Foundations,		
2.7	Informatives: Annexure A (Modification of partial factors for materials), Annexure B (Creep and shrinkage strain), Annexure C (Properties of reinforcement suitable for use with this Euro code), Annexure E (Indicative strength classes for durability), Annexure F (Tension reinforcement expressions for in-plane stress conditions), Annexure J (Detailing rules for particular situations)		

C. Course Outcomes:

On the completion of the course the students will be able to:

- CO1 Understand and design reinforced concrete structures by using the concepts of Euro code.
- CO2 Understand and design reinforced concrete structures by using the concepts of ACI code.
- CO3 Deal with design procedures with different fundamentals.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	3	2	2	1	1	1	1	1	1	1	3	1	1
CO2	3	2	3	2	2	1	1	1	1	1	1	1	3	1	1
CO3	2	2	3	-	-	1	-	1	1	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Reference Books:

1. Prab Bhatt, T.J. MacGinley, & Ban Seng Choo, Reinforced Concrete Design to Eurocodes: Design Theory and Examples, Fourth Edition, CRC Press, United States
2. W.M.C. McKenzie, Design of Structural Elements, Palgrave Macmillan
3. W.H. Mosley, Reinforced Concrete Design: to Eurocode 2, Palgrave Macmillan

Web Materials:

1. <http://www.phd.eng.br/wp-content/uploads/2015/12/en.1992.2.2005.pdf>
2. http://www.eurocodes.fi/1992/paasivu1992/sahkoinen1992/1110_WS_EC2.pdf
3. <http://publications.lib.chalmers.se/records/fulltext/188834/188834.pdf>
4. <https://archive.org/stream/gov.law.aci.318.1995/aci.318.1995#page/n11/mode/2up>

CL484: TRANSPORT PROJECT PLANNING AND EVALUATION**B. TECH 8th SEMESTER (CIVIL ENGINEERING)****PROGRAMME ELECTIVE - IV**

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	-	2	5	4
Marks	100	-	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	2 Hours
2	Travel Survey and Traffic Impact Assessment	4 Hours
3	Transportation Planning Process and Modeling	15 Hours
4	Public Transportation Systems	7 Hours
5	Transportation System Management	4 Hours
6	Travel Demand, Elastic Model and Demand Management	4 Hours
7	Project Formulation and Evaluation Techniques	6 Hours
8	Introduction to Transportation Engineering Software's	3 Hours

Total Hours (Theory): 45**Total Hours (Lab): 30****Total Hours: 75****B. Detailed Syllabus:**

1	Introduction	2 Hours	4%
1.1	Introduction to planning		
1.2	Levels of planning, Objectives & goals of planning		
1.3	Road Infrastructure planning and evaluation of projects		
1.4	Planning Morphology		
2	Travel Survey and Traffic Impact Assessment	4 Hours	8%

2.1	Travel Survey: Travel Survey Process		
2.2	Travel Survey: Data Processing and Interpretation, Future Methodologies		
2.3	Traffic Impact Assessment: Content of TIA		
2.4	Traffic Impact Assessment: TIA and Transportation Planning		
3	Transportation Planning Process and Modeling	15 Hours	33%
3.1	Conventional and Four Stage Modeling Process		
3.2	Trip Generation Models		
3.3	Trip Distribution Models and Calibration		
3.4	Trip Assignment Models		
3.5	Mode Choice and Modal Split Models.		
4	Public Transportation Systems	7 Hours	20%
4.1	Introduction to public transportation systems		
4.2	Transit Classification and Right of way		
4.3	Transit System Performance: Transit Capacity, Frequency and Headway, Quality of Service		
4.4	Bus Rapid Transit: System, Technology and Operation		
4.5	Rail Rapid Transit: System, Technology and Operation		
4.6	Multi-modal Transportation System		
5	Transportation System Management	4 Hours	8%
5.1	Introduction to TSM		
5.2	Need, scope and Strategies for TSM		
5.3	TSM planning & classification		
6	Travel Demand, Elastic Model and Demand Management	4 Hours	8%
6.1	Demand Management		
6.2	Elastic Model: Types of elasticity, Sensitivity & elasticity		
6.3	Sensitivity of travel demand, Factors affecting Elasticity		
6.4	Demand Model- Kraft Demand Model		
6.5	Traffic Management Measures		
6.6	Advantages and Disadvantages		
7	Project Formulation and Evaluation Techniques	6 Hours	13%

- 7.1 Surveys and Investigations- Traffic Surveys, Conventional Ground Survey, Drainage Study, Soil Investigation, Pavement Design Investigation
- 7.2 Design Drawing, Estimate and Project Report- Design, Drawing, Estimate, Earthwork Quantities, Project Reports, Stages in project preparation
- 7.3 Transport Pricing- Methods of road pricing, Transport costs, Elasticity of demand, Average cost and Marginal cost pricing, Pricing Policy, Congestion Pricing, Public and Private Transport Pricing
- 8 Introduction to Transportation Engineering Software's 3 Hours 6%
- 8.1 Introduction
- 8.2 Mx Road, TransCAD
- 8.3 IIT Pave, VISSIM, VISUM
- 8.4 HDM4

C. Students Learning Outcomes:

The course content should be taught and learning imparted with the aim to develop required knowledge and skills so that they are able to acquire following competency:

- CO1 The students would be able to understand and evaluate current scenarios of traffic management and improve it and they will gain the skill for collecting data about travel behaviour and analysing the data for use in transport planning.
- CO2 Basic understanding of what transportation planning is, its theoretical backgrounds and applications
- CO3 Students would be aware of the Principles and Planning of Transportation Infrastructure and would also be able to correlate economy and growth of transportation sector along with basic knowledge to quantify the benefits from well-developed transportation facilities.
- CO4 The students would have knowledge on planning of various transit systems like bus and rail, their scheduling and management strategies.
- CO5 Students would be equipped with the economic principles in dealing with transport supply and demand and would have gained knowledge on various transportation software tools and their application in solving transportation

problems on a real time basis.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	1	1	2	2	-	-	1	2	1	1	1	2	2	2
CO2	2	2	1	3	1	1	-	-	2	1	1	2	3	2	2
CO3	1	1	2	1	1	1	-	-	1	2	1	1	1	1	1
CO4	-	-	2	2	2	1	1	-	1	2	1	2	1	1	2
CO5	1	1	1	1	1	-	1	1	1	1	2	1	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Khanna, S.K. & Justo, C.E.G., Highway Engineering, NemChand & Bros, Roorkee (U.A).
2. Kadiyali, L.R., Traffic Engineering & Transport Planning, Khanna Publishers, New Delhi.
3. Kadiyali, L.R. & Lal, N.B., Principles & Practices of Highway Engineering, Khanna Publishers, New Delhi.
4. Jotin Khisty, S.C. and Kent Lall, B., Transportation Engineering – An Introduction, Pearson Prentice-Hall, NJ
5. Salter, R J., Highway Traffic Analysis and Design, ELBS

Reference Books:

1. Sharma, S.K., Principles, Practice and Design of Highway Engineering, S. Chand & Co., New Delhi.
2. Hutchinson, B.G., Principles of Urban Transport Systems Planning, Scripta, McGraw-Hill, NewYork, 1974.
3. Hutchison, B.G., Introduction to Transportation Engineering, & Planning, McGraw Hill Book Co.
4. John W. Dickey, Metropolitan Transportation Planning, Tata McGraw Hill Pub. Co.
5. Vukan R. Vuchic, Urban Public Transportation System & Technology, Prentice Hall, Inc.

6. Papacostas C.S., Prevedouros, "Transportation Engineering and Planning, 3rd Edition, Prentice Hall of India, New Delhi, 2002
7. Rajaraman, V., Computer Oriented Numerical Methods, Prentice – Hall of India, 1995
8. Chapra S.C., and Canale R.P., Numerical Methods for Engineers, McGraw – Hill, 2004
9. Michael J.Bruton , "An Introduction to Transportation Planning", Hutchinson,1985
10. Michael D.Meyer and Eric J.Miller , "Urban Transportation Planning – A Decision Oriented Approach", McGraw Hill Book Company, New York,1984
11. F.D.Hobbs, "Traffic Planning and Design", Poargam on Oress
12. John W.Dickey, "Metropolitan Transportation Planning" – Tata McGraw Hill Publishing Company Limited, New Delhi, 1980.
13. Paul H.Wright, "Transportation Engineering – Planning and Design", John Wiley and Sons, New York, 1989.

LIST OF TUTORIALS

Tutorial No.	Name of Tutorial
1	Introduction
2	Travel Survey and Traffic Impact Assessment
3	Transportation Planning Process and Modeling
4	Public Transportation Systems
5	Transportation System Management
6	Travel Demand, Elastic Model and Demand Management
7	Project Formulation and Evaluation Techniques
8	Introduction to Transportation Engineering Software's

CL485: WATERSHED MANAGEMENT
B. TECH 8th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - IV

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the course:

Sr. No.	Title of the unit	Min. number of hours
1.	Introduction and basic concepts	04
2.	Watershed Modeling	07
3.	Soil Erosion Modeling	05
4.	Management of Water Quality	06
5.	Storm Water and Flood and Drought Management	06
6.	Integrated Watershed Management	07
7.	Urban Storm-water Management	05
8.	Use of modern techniques in watershed management	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|----------|--|-----------------|------------|
| 1 | Introduction and basic concepts | 04 Hours | 09% |
| 1.1 | What is Watershed, Watershed management, historical look at watershed management | | |
| 1.2 | Basic hydrology, occurrence and movement of water | | |
| 1.3 | Human interventions to manage water flow or quality | | |
| 2 | Watershed modeling | 07 Hours | 16% |
| 2.1 | Standard modeling approaches and classifications | | |

2.2	System concept for watershed modeling		
2.3	Modeling of rainfall-runoff process, subsurface flows and groundwater flow		
3	Soil Erosion Modeling	05 Hours	11%
3.1	Soil Erosion, Soil water Relationship		
3.2	Types and causes of soil erosion, Estimation of soil erosion		
3.3	Different methods to control soil erosion		
4	Management of Water Quality	06 Hours	13%
4.1	Water quality and pollution, types and sources of pollution		
4.2	Water quality modeling		
4.3	Environmental guidelines for water quality		
5	Storm Water and Flood and Drought Management	06 Hours	13%
5.1	Storm water management, design of drainage system,		
5.2	Estimation of design flood and design droughts in a watershed		
5.3	Flood routing through channels and reservoir, flood control and reservoir operation		
5.4	Drought assessment and classification, drought analysis techniques		
5.5	Drought mitigation planning		
6	Integrated Watershed Management	07 Hours	16%
6.1	Introduction to integrated approach, conjunctive use of water resources		
6.2	Rainwater harvesting, Different methods of water harvesting		
6.3	Proposed water harvesting in India through interlinking of rivers		
7	Urban Stormwater Management	05 Hours	11%
7.1	Issues pertaining to watershed management in urban environments		
7.2	Methods to control pollution from urban stormwater		
7.3	Measures to control urban stormwater pollution		

- 8 Use of modern techniques in watershed management 05 Hours 11%
- 8.1 Applications of Geographical Information System(GIS) and Remote Sensing(RS) in Watershed Management
- 8.2 Role of RS and GIS in Watershed Management and Decision Support System

C. Course Outcomes (COs):

On the completion of the course one should be able to:

- CO1 Understand the fundamental principles and concepts of watershed management.
- CO2 Apply various approaches towards storm water management, rainwater harvesting, and flood and drought management.
- CO3 Apprehend soil erosion, watershed modelling, and water quality problems.
- CO4 Attest the concept of integrated watershed management.
- CO5 Corroborate for development and planning of watershed.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	1	1	1	-	1	1	-	-	-	1	2	1	1	-
CO2	3	2	1	1	-	2	1	-	-	-	1	2	1	1	-
CO3	3	2	2	2	2	2	2	-	-	-	2	2	1	2	1
CO4	3	3	3	2	2	3	3	-	1	-	3	2	2	1	2
CO5	3	2	3	3	2	3	3	1	1	1	3	3	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Watershed Management by J.V.S. Murthy New Age Publishers
2. Watershed Management by Madan Mohan Das, PHI Publication

Reference Books:

1. Watershed management: Guidelines for Indian Conditions by E.M. Tideman, Omega Scientific Publishers.
2. Hydrology and Soil Conservation Engineering by Ghanshyam Das, Prentice Hall India.

3. Watershed Planning & Management by Dr. Rajvir Singh, Yash Publishing House.
4. Watersheds - Processes, Assessment and Management by Pau A. Debarry, John Wiley & Sons.

Web Materials:

1. <http://www.epa.gov>
2. <http://www.indiaenvironmentportal.org.in>
3. <http://nptel.ac.in>

LIST OF EXPERIMENTS

Experiment No.	Name of Experiment
1	Introduction to Sampling in water and wastewater analysis
2	Determination of BOD/COD ratio
3	Monitoring of Noise
4	Determination of PM10 and PM2.5 in ambient air
5	Determination of gaseous pollutants in ambient air
6	Determination of various parameters in solid waste
7	Treatability studies of industrial wastewater

CL487: ADVANCED CONSTRUCTION TECHNOLOGY**B. TECH 8th SEMESTER (CIVIL ENGINEERING)****PROGRAMME ELECTIVE - IV**

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Substructure Construction	15
2	Superstructure Construction	08
3	Concreting Process & Equipments	06
4	Advanced Concrete Technology	08
5	Sustainable Construction Technologies	08

Total Hours (Theory): 45**Total Hours (Lab): 30****Total Hours: 75****B. Detailed Syllabus:**

- | | | | |
|----------|---|-----------------|------------|
| 1 | Substructure Construction | 15 Hours | 33% |
| 1.1 | Box jacking, Pipe jacking | | |
| 1.2 | Under water construction of diaphragm walls and basement | | |
| 1.3 | Tunnelling techniques | | |
| 1.4 | Piling techniques and load testing, Battered piles | | |
| 1.5 | Driving well and Caissons, | | |
| 1.6 | Cofferdams: types, design considerations, construction sequence, sinking of cofferdam | | |
| 1.7 | Cable anchoring and grouting | | |
| 1.8 | Sheet piles: laying operations for built up offshore | | |

system			
1.9	Shoring for deep cutting		
2	Superstructure Construction	08 Hours	18%
2.1	Techniques of construction for continuous concreting operation in tall buildings		
2.2	Large span structures – segmental bridge construction techniques		
2.3	In-situ pre-stressing in high rise structures		
2.4	Post tensioning techniques		
3	Concreting Process & Equipments	06 Hours	14%
3.1	Aggregate crushers, feeders and screens, aggregate handling equipments		
3.2	Concrete batching, mixing and pumping equipments		
3.3	Ready mix concrete equipment		
3.4	Tools and plants for hot weather concreting		
3.5	Underwater concreting		
4	Advanced Concrete Technology	08 Hours	18%
4.1	Properties and applications of High strength and high performance concrete		
4.2	Reactive powder concrete		
4.3	Lightweight, heavyweight, and mass concrete		
4.4	Fibre reinforced concrete		
4.5	Self-compacting concrete, Self-healing concrete		
4.6	Polymer Concrete, Epoxy resins and screeds		
5	Sustainable Construction Technologies	08 Hours	17%
5.1	Necessity and importance		
5.2	Criteria for defining as sustainable materials		
5.3	Recycled building products from industrial, agricultural and urban waste stream materials		
5.4	Recent developments in sustainable building materials and technologies		
5.5	Smart buildings		

C. Course Outcomes (COs):

On the successful completion of this course, the students will be able to:

- CO1 Understand and apply operational details of important equipment, techniques and methods employed in substructure construction.
- CO2 Understand the systems and techniques used in heavy constructions works and apply the same in construction sector.
- CO3 Understand the application of different equipments, techniques and methods of producing and placing of concrete.
- CO4 Have in-depth knowledge of different types of advanced concrete and its application.
- CO5 Apply different materials and methods used for sustainable building construction.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	-	-	-	1	1	-	-	-	-	-	-	2	1	1
CO2	1	-	-	-	1	1	-	-	-	-	-	-	-	1	1
CO3	1	-	-	-	2	1	-	-	-	-	-	-	-	1	-
CO4	-	1	2	1	2	1	2	-	-	-	-	-	2	1	2
CO5	-	-	-	-	-	3	3	3	2	1	1	2	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Energy Conservation Building Code, Bureau of Energy Efficiency, New Delhi, 2007.
2. J.R Waters, ‘Energy Conversation in Building: A Guide to part of the building regulations’, Black well publishing, 2003.
3. Peurifoy R. L., C. J. Schexnayder, A. Shapira and R. Schmitt, ‘Construction planning, equipment, and methods’, 8th ed., McGraw Hill, New York, 2010.
4. Jerry Irvine, ‘Advanced Construction Techniques’, CA Rocketr, 1984.
5. Shetty M.S., ‘Concrete Technology’, S.Chand and Company Ltd. Delhi, 2003.

6. Ben C. Gerwick, Construction of Pre-stressed Concrete, Wiley-Interscience; 2 edition, 1997.

Reference Books:

1. Ross Spiegel and Dru Meadows, 'Green Building Materials: A guide to Product Selection and Specification', John Wiley & Sons, 2012
2. Roy Chudley and Roger Greeno, 'Construction Technology', Prentice Hall, 2005.
3. Sankar, S.K. and Saraswati, S., 'Construction Technology', Oxford University Press, New Delhi
4. Gambhir, M. L., 'Concrete Technology', McGraw Hill Education, 2006.
5. Krishnaraju, N., 'Advanced Concrete Technology', CBS Publishers.
6. Krishna Raju N, Prestressed Concrete, Tata McGraw Hill Publishing, 2006.
7. Tyagi, A.K. (Ed). Handbook on Energy Audits and Management Tata Energy Research Institute, 2000.

Web Materials:

1. http://www.pipejacking.org/about_pipe_jacking
2. <http://www.jackedstructures.com/home.html>
3. <https://www.youtube.com/watch?v=wUlQyiHfex0>
4. <http://www.deepexcavation.com/en/Diaphragm+wall+construction+methods>
5. <http://www.pilingcontractors.com.au/processes/diaphragm-walls>
6. <http://nptel.ac.in/courses/105104034/1>
7. <https://www.deepexcavation.com/en/shoring-engineering>
8. https://www.researchgate.net/publication/281174992_A_SHORT_STUDY_ON_LAUNCHING_TECHNIQUES
9. <http://www.post-tensioning.org/Uploads/Conference/2012%20Convention/Segmental%20Bridge%20Constructin%20Techniques.pdf>
10. <https://www.concretenetwork.com/post-tension/basics.html>
11. <http://www.teriin.org/>

CL488: ADVANCED GEOTECHNICAL ENGINEERING**B. TECH 8th SEMESTER (CIVIL ENGINEERING)****PROGRAMME ELECTIVE - IV**

Credits and Hours:

Teaching Scheme	Theory	Tutorial/Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	01
2	Effective Stress Principle	04
3	Introduction to Rock Mechanics	08
4	Foundation on Expansive Soils	10
5	Foundation on Collapsible Soils	05
6	Environmental Waste Effects On Soils	07
7	Machine Foundations	10

Total Hours (Theory): 45**Total Hours (Lab): 30****Total Hours: 75****B. Detailed Syllabus:**

1	Introduction	01 Hour	02%
1.1	Course review		
1.2	Overview of Soil mechanics and Rock mechanic		
1.3	Overview of foundation engineering		
2	Effective Stress Principal	03 Hours	07%
2.1	Effective stresses in saturated soil		
2.2	Effective stresses in unsaturated soil		
3	Introduction to Rock Mechanics	08 Hours	18%
3.1	Introduction		

3.2	Geological classification of rocks		
3.3	Index properties of rocks		
3.4	Classification of rocks for engineering purpose		
3.5	Rock strength and failure criteria		
3.6	Modes of failure of rocks		
4	Foundation on Expansive Soils	10 Hours	22%
4.1	Identification of expansive soils		
4.2	Parameters of expansive soils		
4.3	Causes of moisture changes		
4.4	Preventive measures for expansive soils		
4.5	Techniques For Controlling Swelling: Horizontal Moisture Barriers, Vertical Moisture Barriers, Surface And Subsurface Drainage, Prewetting, Soil Replacement Sand Cushion Techniques, CNS Layer Technique.		
4.6	Modification of Swelling Characteristics: Lime Stabilization, Mechanisms, Limitations, Lime Injection, Lime Columns, Mixing, Chemical Stabilization,		
4.7	Design of Foundation on Expansive soils: Drilled Pier, Belled Drilled Pier, Under-Reamed Piles.		
5	Foundation on Collapsible Soils	05 Hours	11%
5.1	Types of collapsible soils		
5.2	Parameters of collapsible soils		
5.3	Treatment Methods for Collapsible Soil: Induce Collapse Prior to Construction, Decrease Foundation Pressures, Solidify the Soil, Densify Collapsible Soil, Other Methods		
5.4	Design of foundation on un-wetted collapsible soil, Soil subjected to wetting.		
6	Environmental Waste Effects on Soils	08 Hours	17%
6.1	Physico-chemical properties of soil		
6.2	Man made changes in geotechnical environment - mining, embankments, pumping, reservoir, landfills and reclamation effect and control.		

- 6.3 Control of contamination with use of clay barriers, geosynthetics, cut-off walls, leachate collection systems.
- 6.4 stabilization of contaminated soils: materials and techniques in control of ground pollution and treatment

7 Machine Foundations **10 Hours 23%**

- 7.1 Introduction of Soil dynamics, basic definitions
- 7.2 Types of machine foundation
- 7.3 General criteria for design of machine foundation
- 7.4 Vibration analysis of a machine foundation
- 7.5 Examples based on topic
- 7.6 Indian Standard on Design and Construction of Foundation for
- 7.7 Reciprocating Machines
- 7.8 Liquefaction

C. Course Outcomes (COs):

At the end of the course, the students will be able to

- CO1 Learn the various theoretical and practical aspects of geotechnical engineering and apply a diverse knowledge that is applicable to real life problems and can conduct independent work along with the design.
- CO2 Understand the concept of effective stress and its influence on soil behavior, and able to understand of the influence of water flow on the engineering behavior of soils.
- CO3 To be able to analyze and to determine mechanical and engineering properties of rocks for engineering application, critically review rock mechanics principles and methods and their applications to engineering practices. And analyses rock slope stability and foundations on rock.
- CO4 Design and analyze the foundation on problematic soils.
- CO5 The student will learn to assess the quality of soil and extent of soil pollution and soil degradation and suitable remediation.
- CO6 Understand soil response when subjected to dynamic actions, fundamentals of wave propagation and seismology and the knowledge of in situ and laboratory tests for soil dynamic characterization.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	2	3	3	1	1	-	1	1	2	1	3	2	-	3
CO2	1	1	2	1	-	-	-	-	-	1	1	2	2	1	1
CO3	3	2	3	1	-	-	1	1	1	2	-	3	3	-	2
CO4	3	2	3	2	-	1	1	1	-	2	-	2	2	1	2
CO5	1	2	1	3	2	2	2	-	-	1	-	2	-	1	1
CO6	3	3	3	2	1	2	1	-	1	1	-	2	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Soil Mechanics & Foundation Engineering, K.R Arora, Standard Publication, New Delhi.
2. Soil Mechanics & Foundation Engineering, B.C Punamia, Laxmi Publication Pvt Ltd. New Delhi
3. Soil Mechanics & Foundation Engineering by Murthy, V.N.S, Sai Kripa Technical Consultants, Bangalore.
4. Geo-environmental Engineering: Site Remediation, Waste Containment, and Emerging Waste Management Technologies by Hari D Sharma and Krishna R Reddy, John Wiley & Sons.

Reference Books:

1. Singh Alam, Soil Engineering, Vol. I and II, Asia Publication House.
2. Basics and Soil Mechanics by Gopal Ranjan and A S R Rao, New Age International pvt ltd., New Delhi.
3. Design Aids in Soil Mechanics and Foundation Engineering by Kaniraj, McGraw Hill Education.
4. Principles of Foundation Engineering by B.M Das, Cengage, US.
5. Fang and Einterkorn, Foundation Engineering Handbook, Springer.
6. Foundation Engineering by Peck, Thomson and Thornburn, Wiley, India.

7. Foundation Engineering Manual by Nayak, N.B., Dhanpat Rai Publications, New Delhi.
8. Handbook of Machine Foundation by Sribivasula and Vaidyanathan, Tata McGraw Hill Book Co., New Delhi.
9. Geo-Environmental Engineering – Principles and Applications by Lakshmi N. Reddy, Hilary. I. Inyang, Makcel Dekker Ink, 2000
10. Geotechnical and Geo-environmental Engineering Handbook by R. Kerry Rowe, Springer, US.
11. Geotechnical Practice for Waste Disposal by D.E .Daniel, Chapman & Hall, London.

LIST OF TUTORIALS

Tutorial No.	Name of Tutorial
1	Effective Stress Principle
2	Introduction to Rock Mechanics
3	Foundation on Expansive Soils
4	Foundation on Collapsible Soils
5	Environmental Waste Effects On Soils
6	Machine Foundations

CL489: ENVIRONMENTAL POLLUTION AND CONTROL
B. TECH 8th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - IV

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of the Course	02
2.	Industrial Wastewater Management	18
3.	Hazardous Waste Management	12
4.	Air Pollution Control	10
5.	Environmental Policies for Pollution Prevention	03

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | |
|--|------------------------|
| 1. Overview of the Course: | 02 Hours 04% |
| 1.1 Concept of pollution prevention and cleaner production, environmental management hierarchy | |
| 1.2 Industrialization and sustainable development | |
| 2. Industrial Wastewater Management: | 18 Hours 40% |
| 2.1 Effects of industrial wastes on sewerage system and receiving water bodies | |
| 2.2 Industrial waste survey: Process flow charts, condition of waste stream | |
| 2.3 Pre-treatment of industrial wastewater: Volume reduction, strength | |

reduction, neutralization, equalization and proportion, removal of organic and inorganic dissolved solids

- 2.4 Wastewater treatment in specific industries: Distillery, sugar, pulp and paper, cement, textile, dairy, fertilizer, pesticides, pharmaceutical, etc.

3. Hazardous Waste Management: 12 Hours 27%

- 3.1 Introduction, sources, classification, regulations for hazardous waste management, hazardous waste characterization
- 3.2 Waste minimization and resource recovery: Approaches, development of a waste tracking system
- 3.3 Transportation of hazardous waste: Requirements, regulations
- 3.4 Physico-chemical, chemical and biological treatment of hazardous waste
- 3.5 Sanitary landfill: Design approach, leachate and gaseous collection system, facility siting and process selection for treatment, storage, disposal facility (TSDF)
- 3.6 Recent developments in hazardous wastes management
- 3.7 Biomedical waste management: Sources, treatment and disposal

4. Air Pollution Control: 10 Hours 22%

- 4.1 Meteorology: Composition and structure of the atmosphere, wind circulation, solar radiation, lapse rates, atmospheric stability conditions, wind velocity profile, Maximum Mixing Depth (MMD), temperature inversions, windrose diagram
- 4.2 Monitoring of particulate matter and gaseous pollutants: Respirable, non-respirable and nano - particulate matter, CO, CO₂, Hydrocarbons (HC), SOX and NOX, photochemical oxidants
- 4.3 Pollutants dispersion models: Description and application of point, line and areal sources
- 4.4 Air pollution control equipment for particulate matter & gaseous pollutants: Gravity settling chambers, centrifugal collectors, wet collectors, fabric filters, electrostatic precipitator (ESP), adsorption, absorption, scrubbers, condensation and combustion

5. Environmental Policies for Pollution Prevention

03 Hours 07%

- 5.1 Introduction to environmental legislations
- 5.2 Environmental auditing: Financial and managerial opportunities

C. Course Outcomes:

At the end of the course the students will be able to

- CO1 Discover correctly the knowledge related to environmental pollution
- CO2 Understand the theories and practical aspects of pollution control along with the design and management applications.
- CO3 Propose ideas to control environmental pollution with respects to professionalism and ethics.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	-	-	-	1	-	-	-	2	-	2	-	-	2
CO2	3	3	3	3	3	3	3	2	3	3	3	3	3	3	3
CO3	-	1	3	1	3	3	3	3	3	3	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Mahajan, S.P., Pollution Control in Process Industries, Tata McGraw Hill Publishing Company Ltd., New Delhi.
2. Washington, D.C., Eckenfelder, Industrial Water Pollution Control, McGraw hill Company, New Delhi, American Chemical Society, USA, 2000.
3. Rao, C.S., Environmental Pollution Control Engineering, Wiley Eastern Ltd., New Age International Ltd.
4. Rao, M.N. and Rao, H.V.N., Air Pollution, Tata McGraw-Hill

Reference Books:

1. Wark, K., Warner, C.F. and Davis, W.T., Air Pollution - Its Origin and Control, Harper & Row Publishers, New York, 1998.
2. Perkins, H.C., Air Pollution, McGraw Hill, 1974.
3. Stern, A.C., Air Pollution, Vol. - I, II, III.

4. Nemerow, N.N., Liquid Waste of Industry Theories, Practices and Treatment. Addison Willey, New York, 1971.
5. Ross, R.D., Industrial Waste Disposal, Reinhold Environmental Series - New York, 1978.
6. Tchobanoglous, G., Theissen, H. and Eliassen, R., Solid Waste Engineering - Principles and Management Issues, McGraw Hill, New York, 1991.

Web Materials:

1. <http://www.epa.gov>
2. <http://www.indiaenvironmentportal.org.in>
3. <http://nptel.ac.in>

LIST OF EXPERIMENTS

Experiment No.	Name of Experiment
1	Introduction to Sampling in wastewater analysis
2	Determination of BOD/COD ratio
3	Determination of PM10 and PM2.5 in ambient air
4	Determination of gaseous pollutants in ambient air
5	Determination of pH and conductivity for hazardous waste
6	Determination of loss on drying for hazardous waste
7	Treatability studies of industrial wastewater
8	Leachate Analysis

CL492: GIS & REMOTE SENSING APPLICATIONS IN CIVIL ENGINEERING
B. TECH 8th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - IV

Credits and Hours:

Teaching Scheme	Theory	Practical / Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Remote Sensing	03
2	Remote sensors and platforms	07
3	Satellite data Analysis	05
4	Digital Image processing	07
5	Introduction to Geographic Information System (GIS)	05
6	GIS data processing	06
7	Introduction to navigational systems	06
8	Integration of GIS and Remote Sensing, Applications in Civil Engineering	06

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction to Remote Sensing	03 Hours	7 %
1.1	History and development of remote sensing		
1.2	Electromagnetic radiation and its interaction with matter		
1.3	Atmospheric effects and corrections		
2	Remote sensors and platforms	07 Hours	16 %
2.1	Overview of Remote sensors		

2.2	Remote sensing satellite platforms and orbits		
2.3	Optical sensor systems		
2.4	Microwave sensor systems		
2.5	Indian Satellite Program		
2.6	Recent trends in Remote Sensing techniques: Microwave remote sensing, Lasers and radars, Hyper spectral remote sensing		
3	Satellite data Analysis	05 Hours	11 %
3.1	Methods of acquisition of digital images from conventional and space-borne scanners		
3.2	Satellite data products		
3.3	Visual image interpretation		
3.4	Image Processing Systems		
4	Digital Image processing	07 Hours	16%
4.1	Digital Image, Media for digital data recording, Storage, Data formats		
4.2	Distortions in Space-borne Digital images and Restoration		
4.3	Matching and Registration of satellite images		
4.4	Image enhancement through Point Operations : Neighborhood Operations on Satellite Images,		
4.5	Image Transforms : Fourier, Wavelet, Hough, color and PCT Arithmetic Operations,		
4.6	Digital Image Analysis Through Supervised and Unsupervised Classification,		
4.7	Post-classification Analysis: statistical accuracy estimation.		
4.8	Image processing softwares.		
5	Introduction to Geographic Information System (GIS)	05 Hours	11 %
5.1	Introduction to GIS, Concept of GIS and Functions and advantages		
5.2	Spatial data concepts, map reference systems		
5.3	Spatial data - sources, models, structures, Data quality		
5.4	Analysis and interpolation		
6	GIS data processing	06 Hours	13 %
6.1	Spatial data model, Geospatial analysis		
6.2	Attribute management and metadata concept		

6.3	GIS Data sources, Organizing data for analysis		
6.4	Linking spatial and attribute data		
6.5	Spatial decision support systems,		
6.6	GIS Softwares		
7	Introduction to navigational systems	06 Hours	13 %
7.1	Overview of GPS and GNSS		
7.2	Current navigational system, components, Functioning and Uses		
8	Integration of GIS and Remote Sensing and their Applications in Civil Engineering	06 Hours	13 %
	Case studies in the following areas:		
8.1	Environmental engineering		
8.2	Water Resources		
8.3	Intelligent Transport Systems		
8.4	Land-use – Land cover – Urban planning		
8.5	Disaster Management		

C. Course Outcomes:

On the successful completion of this course:

- CO1 Ability to develop thematic maps using Remote Sensing Images and carry out multilayer analysis using GIS technology
- CO2 The students will be in a position to use geo-information science and earth observation technology to generate, integrate, analyze and visualize spatial data
- CO3 The students be able to apply practical skills to carry out an independent final assignment (a case study project).
- CO4 The students may undertake GPS based surveys and geographical positioning assignments.
- CO5 The students can be able to pursue careers in the field of remote sensing and GIS academia, governmental agencies, businesses and consultancies around the world.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	-	1	1	3	2	1	-	-	-	1	2	1	1	1
CO2	3	-	1	1	3	2	1	-	-	-	1	2	1	1	1
CO3	3	1	1	1	3	2	1	-	-	-	1	2	1	1	1
CO4	3	1	1	1	3	2	1	-	-	-	1	2	1	1	1
CO5	3	1	1	1	3	2	1	-	-	-	1	2	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. B. Bhatta, Remote Sensing and GIS, 2nd Edition, Oxford University Press, New Delhi
2. George Joseph, Fundamentals of Remote Sensing, Universities Press, India, 2005
3. A.M. Chandra and S.K. Ghosh, Remote Sensing and Geographical Information System, Narosa Publishing House, New Delhi.

Reference Books:

1. Gopi, Advanced Surveying: Total Station, GIS and Remote Sensing, Pearson Education India, 2006, (ISBN 8131743012, 9788131743010)
2. J.R. Jensen, Introductory Digital Image Processing, Prentice-Hall, New Jersey
3. J.R. Jensen, Remote Sensing of Environment: An Earth Perspective, Pearson Education, Delhi, 2004
4. P.A. Burrough and R.A. McDonnell, Principles of Geographical Information Systems, 2nd ed. Oxford, England, Oxford University Press.
5. T.M. Lillesand, R.W. Kiefer and J.W. Chipman, Remote Sensing and Image Interpretation, 5th edition, John Wiley and Sons, India
6. Kali Charan Sahoo, Textbook of Remote Sensing and Geographical Information System, Atlantic Publishers.

LIST OF EXPERIMENTS

Experiment No.	Name of Tutorial / Experiment
1	Visual Image Interpretation: Photographic Products
2	Digital Interpretation of True Color Composite and False Color Composite
3	Image Registration and Geo-referencing
4	Image Enhancement
5	Image Transformation
6	Multiband Image Operations
7	Supervised Image Classification
8	Unsupervised Image Classification
9	Accuracy Assessment of image classification
10	Introduction to GIS, Spatial Data Generation
11	Linking of Spatial and Attribute Data
12	Linking of Spatial and Attribute Data
13	Integration of GIS and RS, civil engineering application

CL493: COMPUTATIONAL METHODS IN WATER RESOURCES ENGINEERING**B. TECH 8th SEMESTER (CIVIL ENGINEERING)****PROGRAMME ELECTIVE - IV**

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of the Course	03
2.	Probabilistic analysis	06
3.	Regression Analysis	08
4.	Ordinary Differential Equations	10
5.	Partial Differential Equations	10
6.	Infinite Series	08

Total Hours (Theory): 45**Total Hours (Lab): 30****Total Hours: 75****B. Detailed Syllabus:**

1. Overview of the Course: 03 Hours 7%
 - 1.1 Need for computational and statistical methods, overview of the applications in Civil Engineering in general and Water Resources Engineering in particular.
 - 1.2 Review of numerical techniques for finding roots of non-linear equations
2. Probabilistic analysis: 06 Hours 13%
 - 2.1 Review of basic concepts of probability and probability distributions
 - 2.2 Probabilistic analysis and treatment of hydro-meteorological and

water quality data		
2.3 Flood frequency analysis		
3. Regression Analysis:	08 Hours	18%
3.1 Simple linear and multiple linear regression, evaluation of regression, confidence limits		
3.2 Applications – rainfall-runoff analysis, rating curves, water quality modeling		
4. Ordinary Differential Equations:	10 Hours	22%
4.1 Nature of problems, boundary and initial equations		
4.2 Euler's method, modified Euler's method, Predictor-Corrector method, Runge-Kutta method, Boundary value problems		
4.3 Applications for reservoir routing, gradually varied flow problems, pipe networks.		
5. Partial differential equations:	10 Hours	22%
5.1 Classification and nature of problems, Concepts of finite difference method, Solution of parabolic equations, pollutant transport, Solutions of elliptical equation, Laplace equation and Poisson equation.		
5.2 Flow through porous media, Solution of hyperbolic equation		
5.3 Unsteady flow through open channels, propagation of waves		
5.4 Concept of finite volume method		
6. Infinite Series:	08 Hours	18%
6.1 Basics of Finite Element Method, FEM vs FDM, Element shapes, shape functions, development of shape functions for linear elements		
6.2 Formulation of FEM for stress analysis problems		
6.3 Flow through porous media, Galerkin's method and Variational method		

C. Course Outcomes:

At the end of the course the students will be able to

- CO1 Apply numerical methods for solution of differential equations in Water Resources Engineering
- CO2 Apply finite difference schemes for solution of hydraulic and hydrologic models.
- CO3 Formulate finite element model for solution of flow through porous media.
- CO4 Perform statistical analysis of water resources systems.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	2	3	3	2	1	2	1	-	2	3	3	2
CO2	3	3	3	2	3	3	1	1	2	1	-	2	3	3	2
CO3	3	3	3	2	3	3	1	1	2	1	-	2	3	3	2
CO4	3	3	3	3	3	3	2	3	2	2	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Hoffman, J.D., (2011), “*Numerical Methods for Engineers and Scientists*”, CRC Press, Special Indian Edition
2. Numerical Methods by Chapra and Canale.
3. Schilling, R.J., and S.L. Harris, (2007), “*Applied Numerical Methods for Engineering*”, CENGAGE Learning, India Edition

Reference Books:

1. Kotteguda, N.T. and Renzo Resso, (1998), “*Statistics, Probability and Reliability for Civil and Environmental Engineers*”, McGraw Hill Companies Inc., New York
2. Neural Network Fundamentals with Graphs, Algorithms, Applications-Bose N.K. & Liang P. McGraw Hill N.Y.
3. Computational Fluid Dynamics by Anderson.
4. Abbot, M.A. and Verver (1996), “*Computational Hydraulics*”, Elsevier Publications

LIST OF TUTORIALS

Tutorial No.	Name of Tutorial
1	Probabilistic analysis and treatment of hydro-meteorological and water quality data
3	Flood frequency analysis
4	rainfall-runoff analysis, rating curves, water quality modelling
5	Euler's method, modified Euler's method, Predictor-Corrector method, Runge-Kutta method, Boundary value problems
6	Applications for reservoir routing, gradually varied flow problems, pipe networks.
7	Flow through porous media, Solution of hyperbolic equation
8	Unsteady flow through open channels, propagation of waves
9	Applications of infinite series

CL494: ADVANCED HIGHWAY & TRAFFIC ENGINEERING
B. TECH 8th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE - IV

Credits and Hours:

Teaching Scheme	Theory	Practical/ Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Pavement Materials Characterization	10
2	Pavement Evaluation	08
3	Road Safety & Environmental Impact Assessment	07
4	Design of Intersections and Parking Analysis	05
5	Traffic Flow Theory	10
6	Road Making Machinery & Modern Highway System	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|----------|---|-----------------|------------|
| 1 | Pavement Materials Characterization | 10 Hours | 22% |
| 1.1 | Requirements of paving concrete, Design of mixes – IRC, Absolute volume, Vibrated Concrete mix design | | |
| 1.2 | Composites, Plastics and Geo-synthetics: Plastics and polymerization process, properties, Durability and chemical composition, Reinforced Polymer Composites, Geo-synthetics, Dry Powdered Polymers- Enzymes. | | |
| 1.3 | Reclaimed/Recycled Waste Products: Reclaimed Materials – waste products in civil engineering applications, Effect of waste | | |

products on materials, Structure and properties, Self-healing and Smart materials.

2	Pavement Evaluation	08 Hours	18%
2.1	Pavement Evaluation: Methods of Pavement Evaluation, Visual Rating Pavement Serviceability Index (PSI).		
2.2	Roughness Measurements: Importance of smooth riding surface, measurement of road roughness, Towed Fifth Wheel Bump Integrator, Skid Resistance: Measurement, Importance and Governing Factors Pavement Deterioration Research.		
3	Road Safety & Environmental Impact Assessment	07 Hours	16%
3.1	Introduction, Collision & condition diagram, Causes of accidents, Accident studies & records, Analysis of accident studies, Road Safety Improvement Strategies.		
3.2	Road Safety Audit Process, Black Spot, Black Route, Area Identification		
3.3	Transport related different types of pollution& their sources, Effects of Weather Conditions, Vehicular emission parameters, Pollution standards, Measurement and analysis of vehicular emission, Mitigative Measures.		
3.4	Urban and non-urban traffic noise sources, Noise level factors, Effects of traffic noise, Propagation and measurement of traffic noise, Prediction and control measures, Noise studies, Noise standards.		
3.5	EIA requirements of highway projects, EIA procedures, guidelines, EIA practices in India.		
4	Design of Intersection and Parking Analysis	05 Hours	11%
4.1	Principles of design, Channelization, Roundabouts, Staggered intersections, Design of rotary intersection.		
4.2	Introduction to Parking Studies ,Types of parking facilities, Modes of parking , Parking studies and analysis, Evaluation of parking parameters, Ill effects of parking		
5	Traffic Flow Theory	10 Hours	22%
5.1	Uninterrupted traffic Flow Theory :		

Fundamentals of Traffic flow theory, Uninterrupted Traffic flow including Macroscopic and Microscopic Traffic flow models

5.2 Interrupted traffic Flow Theory :

Fundamentals of Interrupted Traffic Flow, Shockwave Analysis, Car following theory, Queuing Theory, Vehicle arrival: Gap and Gap acceptance

6 Road Making Machinery & Modern Highway System 05 Hours 11%

6.1 Role of Labour vs. machinery in road construction, Earth work machinery, volume changes, clearing and grubbing, Tractors, Tractor Dozers, scrapers, Graders.

6.2 Rock excavation machinery, Transporting Equipment, Compaction Equipment , Production of aggregates

6.3 Bituminous concrete road equipment, Cement Concrete road making Equipment, Equipment Usage charges

6.4 Automated Highway System , ITS

6.5 Green Highway, Smart Highway, Electric Highway & Solar Highway

C. Course Outcomes:

The course content should be taught and learning imparted with the aim to develop required knowledge and skills so that they are able to acquire following competency:

CO1 The student will develop an understanding of the basic concepts of different paving material and their characteristics.

CO2 The student will be able to identify the distress in pavement and will be able to decide the evaluation and maintenance strategies for the pavement.

CO3 The student will able to understand about the road safety audit process and EIA.

CO4 The student will able to design the parking facilities and at grade intersection.

CO5 The student will capable to understand the traffic flow behavior on the basis of flow theories.

CO6 The student will learn about the modern equipment and highway system.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	2	2	2	2	-	2	-	1	1	-	2	1	2	2
CO2	1	2	2	2	3	1	-	-	2	2	1	2	3	1	3
CO3	1	3	3	1	3	2	3	1	2	3	1	3	2	1	3
CO4	2	3	1	2	3	1	1	-	2	2	-	2	3	1	2
CO5	2	2	1	2	3	1	-	-	2	2	-	2	2	1	2
CO6	1	-	1	-	2	-	-	-	-	1	-	1	-	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Dr. L. R. Kadiyali and Dr. N. B. Lal, Principles & practices of Highway Engineering, Khanna Publication Delhi
2. Dr. L. R. Kadiyali, Traffic Engineering & Transport Planning, Khanna Publication Delhi.
3. S. K Khanna & C. E.G Justo, Highway Engineering, Nem Chand & Bros, Roorkee.
4. Elvik, R., and Vaa, T., The Handbook of Road Safety Measures, Elsevier, 2004.
5. Chakroborty & Das, Principles of Transportation Engineering, PHI Learning Pvt. Ltd. Delhi

LIST OF TUTORIALS

Tutorial No.	Name of Tutorials
1	Pavement Materials Characterization
2	Pavement Evaluation
3	Road Safety Audit Process
4	Environmental Impact Assessment
5	Numerical on Design of Intersection
6	Numerical on Parking Analysis
7	Traffic Flow Theory
8	Road Making Machinery & Modern Highway System

CL496: GREEN BUILT ENVIRONMENT
B. TECH 7th SEMESTER (CIVIL ENGINEERING)
PROGRAMME ELECTIVE-III

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Sustainable Architecture & Sites	09
2	Water Management	09
3	Energy Management	09
4	Materials & Resources	09
5	Indoor Environmental Quality	09

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|----------|---|-----------------|------------|
| 1 | Sustainable Architecture & Sites | 09 Hours | 20% |
| 1.1 | Integrated Approach for Green Building design | | |
| 1.2 | Soil erosion & pollution control measures | | |
| 1.3 | Storm water management | | |
| 1.4 | Reduction of heat island effect | | |
| 1.5 | Minimizing night sky pollution | | |
| 2 | Water Management | 09 Hours | 20% |
| 2.1 | Water Balance and approach for water efficiency | | |
| 2.2 | Water efficient plumbing fixtures | | |
| 2.3 | Effective irrigation techniques | | |
| 2.4 | Waste-water treatment & reuse | | |

- 2.5 Rainwater harvesting
- 3 Energy Management** 09 Hours 20%
- 3.1 Energy Performance Evaluation
- 3.2 Approach for Enhancing Energy Efficiency
- 3.3 Performance Improvement Opportunities; Building Envelop, Air Conditioning, Lighting d. Appliances and Equipment
- 3.4 Renewable Energy systems
- 4 Materials & Resources** 09 Hours 20%
- 4.1 Green Products and Materials for enhancing performance
- 4.2 Waste management – During construction and post-occupancy
- 4.3 Suggested approach - 3 R concept (Reduce, Reuse & Recycle)
- 4.4 Case studies
- 5 Indoor Environmental Quality** 09 Hours 20%
- 5.1 Indoor Air quality – Codes and Standards
- 5.2 Approach for improving Indoor air quality
- 5.3 Enhancing occupants' Comfort, Health and Wellbeing; Thermal comfort, Visual Comfort, Acoustics, Olfactory comfort, Ergonomics

C. Course Outcomes:

On the successful completion of this course, the students will be able to:

- CO1 Understand and apply trends & technologies of Green concepts in construction Industry.
- CO2 Have in-depth knowledge and exposure to the Green concepts in design, construction and operation of buildings and built environment.
- CO3 Apply different materials and methods used for sustainable building construction.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	-	2	1	2	2	3	1	2	1	1	2	2	1	2
CO2	2	-	2	2	2	2	2	-	1	-	-	1	1	1	1
CO3	1	-	1	-	1	1	2	-	1	-	-	-	-	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Books:

1. Energy Conservation Building Code, Bureau of Energy Efficiency, New Delhi, 2007.
2. Ross Spiegel and Dru Meadows, 'Green Building Materials: A guide to Product Selection and Specification', John Wiley & Sons, 2012
3. Karthik Karuppu and NVICO, 'Green Building Guidance: The Ultimate Guide for IGBC Accredited Professional Examination', Notion Press, 2019.
4. A. K. Jain, 'The Idea Of Green Building', Khanna Publishers, 2014
5. David Johnston, 'Toward a Zero Energy Home: A Complete Guide to Energy Self-Sufficiency at Home', Taunton Press Inc, 2010
6. Francis D. K. Ching and Ian M. Shapiro, 'Green Building Illustrated', Wiley, 2014

Web Materials:

1. <https://igbc.in/igbc/redirectHtml.htm?redVal=showResourcesnosign>
2. <https://igbc.in/igbc/redirectHtml.htm?redVal=showGreenNewBuildingsnosign#overview>
3. <https://igbc.in/igbc/redirectHtml.htm?redVal=showGovtIncentivesnosign>
4. <http://cdn.ifma.org/sfcdn/membership-documents/green-rating-systems-htg-final.pdf>

B. Tech. (Civil Engineering) Programme

SYLLABI (Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CL458: MAJOR PROJECT
B.TECH 8th SEMESTER (CIVIL ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	36	36	29
Marks	0	800	800	

A. Outline of the Course:

- Students have to select their domain of work and company/ industry/ organization well in advance before the commencement of the project.
- Students are advised to carry out project in such a way that it is complete in all aspects and report should reflect comprehensive domain knowledge
- Students have to take approval of company/ industry/ organization from concerned faculty.
- Students have to take NOC from the concerned faculty, if needed and then they have to submit their confirmation letter from company/ industry/ organization before the commencement of project.
- Students must meet and report to their assigned faculty mentor about outline of the project work which they are going to carry out in the whole semester in 15 days after the commencement of the project.
- Students have to keep the record of activities carried out during their project tenure in the form of daily progress report and weekly progress report. Report weekly to your faculty mentor/ internal guide with filled weekly report as per the guidelines will be issued by concerned faculty.
- Project will be evaluated internally twice during the semester along with final evaluation at the end of the semester as a part of continuous evaluation.
- Students are required to prepare a report as per the guidance and directions from their assigned faculty mentor.
- After the completion of the project work, students must have to take completion certificate from the concerned company/ industry/ organization and submit the certificate to the concerned faculty.

- Students have to submit hard binding report with CD.
- Content of reports should be unique such that, Plagiarism check of project report of individual student shall be less than the university norms. Also, students have to attach completion certificate in their project report.

B. Instructional Method and Pedagogy:

- Type of Project will be assigned to an individual/ group of students based on their inclination/ willingness/ interest.
- The project may include a site visit/ software training/ research project/ real life problem as per the project type, where individual/ group of students can avail an opportunity to build an appreciation for the concepts to be utilized in understanding the actual scenario.
- 2 internal evaluations & 1 final evaluation are required. Each internal evaluation carries 100 Marks, 100 marks for report and 100 marks from internal guide & External evaluation carries 400 marks
- At the end of the semester, students have to submit the final project report. The project report should consist of at least 40 - 50 pages.
- At the end of semester, a presentation of the project is required to be done in group or individually on scheduled date for at least 15 minutes.

C. Course Outcomes:

On the successful completion of this course

- CO1 The students will be able to hold ideas about projects and understand the significance of projects. They will be able to demonstrate a sound technical knowledge, skills and attributes of a professional engineer.
- CO2 The students will also gain a better approach towards design and performance of various infrastructure related projects. They will be able to provide design engineering solutions to complex problems utilizing a systems approach.
- CO3 Students will be able to communicate with engineers and community at large in written as well as in oral forms. Students will also be able to learn about the technical writing which is adopted at organization.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	3	2	2	-	1	1	2	1	-	2	1	2	2	1
CO2	2	3	3	2	1	2	1	-	-	-	1	1	3	3	-
CO3	-	-	-	-	3	-	-	-	-	3	-	-	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Project related study is to be carried out by each student/group of students.

1. Reading Materials, web materials, blogs, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic
4. Various codes related to civil engineering

Web Links:

1. <https://www.asce.org/>
2. <https://cecr.in/>
3. <https://www.sciencedirect.com/>
4. <https://www.elsevier.com/en-in>
5. <https://bis.gov.in/>